

Algebraic Techniques for Rational Expressions

Factoring, Simplification, Polynomial Division, and Partial-Fraction
Decomposition

A Recognition and Mastery Workbook

Purpose

This is an algebra-only boot camp for students preparing to study calculus. It develops the ability to identify the shape of a polynomial or rational-expression problem, select the appropriate algebraic tool or sequence of tools, carry out the work correctly, preserve restrictions, and verify the result. No limits, derivatives, or integrals are used.

Contents

How to Use This Book	vii
Diagnostic Vocabulary	viii
I Part I: Factoring Toolbox	1
Factoring Strategy	2
1 Greatest Common Factor and Sign Extraction	3
1.1 Recognize the shape	3
1.2 Procedure	3
1.3 Mastery exercises	3
2 Difference of Squares	5
2.1 Recognize the shape	5
2.2 Procedure	5
2.3 Mastery exercises	5
3 Perfect-Square Trinomials	7
3.1 Recognize the shape	7
3.2 Procedure	7
3.3 Mastery exercises	7
4 Sum and Difference of Cubes	9
4.1 Recognize the shape	9
4.2 Procedure	9
4.3 Mastery exercises	9
5 Monic Quadratic Trinomials	11
5.1 Recognize the shape	11
5.2 Procedure	11
5.3 Mastery exercises	11
6 General Quadratic Trinomials	13
6.1 Recognize the shape	13

6.2	Procedure	13
6.3	Mastery exercises	13
7	Factoring by Grouping	15
7.1	Recognize the shape	15
7.2	Procedure	15
7.3	Mastery exercises	15
8	Substitution and Higher-Power Patterns	17
8.1	Recognize the shape	17
8.2	Procedure	17
8.3	Mastery exercises	17
9	Multi-Stage Complete Factorization	19
9.1	Recognize the shape	19
9.2	Procedure	19
9.3	Mastery exercises	19
II	Part II: Rational-Expression Operations	21
	Rational-Expression Discipline	22
10	Simplification by Factoring and Cancellation	23
10.1	Recognize the shape	23
10.2	Procedure	23
10.3	Mastery exercises	23
11	Multiplication of Rational Expressions	26
11.1	Recognize the shape	26
11.2	Procedure	26
11.3	Mastery exercises	26
12	Division of Rational Expressions	29
12.1	Recognize the shape	29
12.2	Procedure	29
12.3	Mastery exercises	29
13	Addition and Subtraction of Rational Expressions	33

13.1 Recognize the shape	33
13.2 Procedure	33
13.3 Mastery exercises	33
14 Complex Rational Expressions	37
14.1 Recognize the shape	37
14.2 Procedure	37
14.3 Mastery exercises	37
 III Part III: Polynomial Division	 41
Polynomial Division as an Algorithm	42
 15 Polynomial Long Division	 43
15.1 Recognize the shape	43
15.2 Procedure	43
15.3 Mastery exercises	43
 16 Synthetic Division	 46
16.1 Recognize the shape	46
16.2 Procedure	46
16.3 Mastery exercises	46
 17 Choosing Long or Synthetic Division	 49
17.1 Recognize the shape	49
17.2 Procedure	49
17.3 Mastery exercises	49
 IV Part IV: Partial-Fraction Decomposition	 52
Partial Fractions: The Master Setup	53
 18 Distinct Linear Factors	 54
18.1 Recognize the shape	54
18.2 Procedure	54
18.3 Mastery exercises	54
 19 Heaviside Cover-Up Method	 58

19.1 Recognize the shape	58
19.2 Procedure	58
19.3 Mastery exercises	58
20 Strategic Substitution	62
20.1 Recognize the shape	62
20.2 Procedure	62
20.3 Mastery exercises	62
21 Equating Coefficients	66
21.1 Recognize the shape	66
21.2 Procedure	66
21.3 Mastery exercises	66
22 Repeated Linear Factors	70
22.1 Recognize the shape	70
22.2 Procedure	70
22.3 Mastery exercises	70
23 Distinct Irreducible Quadratic Factors	74
23.1 Recognize the shape	74
23.2 Procedure	74
23.3 Mastery exercises	74
24 Repeated Irreducible Quadratic Factors	78
24.1 Recognize the shape	78
24.2 Procedure	78
24.3 Mastery exercises	78
25 Mixed Denominator Types	82
25.1 Recognize the shape	82
25.2 Procedure	82
25.3 Mastery exercises	82
26 Improper Rational Expressions and Partial Fractions	86
26.1 Recognize the shape	86
26.2 Procedure	86
26.3 Mastery exercises	86

27 Hybrid Coefficient Methods	90
27.1 Recognize the shape	90
27.2 Procedure	90
27.3 Mastery exercises	90
28 Verifying Decompositions	94
28.1 Recognize the shape	94
28.2 Procedure	94
28.3 Mastery exercises	94
V Part V: Tool Selection Capstone	98
29 Systematic Decision Process	99
30 Capstone: Identify, Select, Execute, Verify	100
VI Part VI: Reference and Answer Key	105
31 Formula and Procedure Reference	106
32 Answer Key	107
A. Greatest Common Factor and Sign Extraction	107
B. Difference of Squares	108
C. Perfect-Square Trinomials	108
D. Sum and Difference of Cubes	109
E. Monic Quadratic Trinomials	110
F. General Quadratic Trinomials	111
G. Factoring by Grouping	112
H. Substitution and Higher-Power Patterns	112
I. Multi-Stage Complete Factorization	113
J. Simplification by Factoring and Cancellation	114
K. Multiplication of Rational Expressions	115
L. Division of Rational Expressions	116
M. Addition and Subtraction of Rational Expressions	119
N. Complex Rational Expressions	121
O. Polynomial Long Division	123
P. Synthetic Division	125

Q. Choosing Long or Synthetic Division	126
R. Distinct Linear Factors	128
S. Heaviside Cover-Up Method	130
T. Strategic Substitution	132
U. Equating Coefficients	135
V. Repeated Linear Factors	137
W. Distinct Irreducible Quadratic Factors	139
X. Repeated Irreducible Quadratic Factors	142
Y. Mixed Denominator Types	144
Z. Improper Rational Expressions and Partial Fractions	146
AA. Hybrid Coefficient Methods	149
AB. Verifying Decompositions	151
CAP. Capstone	153
Before Declaring the Algebra Complete	161

How to Use This Book

This workbook is deliberately organized by tool before ending with an unlabeled capstone. While working inside a chapter, first learn the recognition cues and procedure. In the capstone, no chapter heading tells you what to do: you must diagnose the expression yourself.

recognize the shape $\longrightarrow$ select the tool(s) $\longrightarrow$ execute $\longrightarrow$ verify

Each technique chapter contains fifty problems in five levels. Problems 1–10 are direct. Problems 11–20 add routine variations. Problems 21–30 disguise the pattern or require preparation. Problems 31–40 combine stages. Problems 41–50 remove procedural hints.

Standing rules

1. Factor completely before cancelling or writing a partial-fraction template.
2. Cancel factors, never individual terms connected by addition or subtraction.
3. A rational expression is defined only where every original denominator is nonzero.
4. Cancelling a factor does not restore a value excluded by the original expression.
5. A partial-fraction decomposition begins only after the fraction is proper and the denominator is completely factored.
6. Verification is part of the procedure, not an optional extra.

Diagnostic Vocabulary

Term	Meaning
Term	A piece separated from another piece by addition or subtraction.
Factor	A piece multiplied by another piece. Only factors can be cancelled.
Proper rational expression	$P(x)/Q(x)$ with $\deg P < \deg Q$.
Improper rational expression	$P(x)/Q(x)$ with $\deg P \geq \deg Q$. Polynomial division is required before partial fractions.
Distinct factor	A factor that occurs once.
Repeated factor	A factor occurring to a power greater than one.
Irreducible quadratic	A real quadratic with negative discriminant; it has no real linear factors.
Complete factorization	A factorization in which no allowed factoring tool applies to any remaining factor over the stated number system.
Restriction	A value excluded because an original denominator is zero or, in division, because the divisor is zero.
Identity	An equation true for every value for which both sides are defined.

Part I

Part I: Factoring Toolbox

Factoring Strategy

Factoring reverses multiplication. Begin with the greatest common factor because it simplifies every later pattern. Then count terms: two terms suggest squares or cubes; three terms suggest a perfect square or quadratic trinomial; four or more terms may suggest grouping. Higher powers may hide a quadratic pattern.

Recommended order

GCF $\longrightarrow$ special products $\longrightarrow$ trinomials $\longrightarrow$ grouping or substitution $\longrightarrow$ inspect every new factor again.

Chapter 1

Greatest Common Factor and Sign Extraction

Learning target

Factor out the largest numerical and variable factor shared by every term.

1.1 Recognize the shape

Look for a common coefficient, a lowest shared power of each variable, or a leading negative sign.

1.2 Procedure

First remove the GCF. Factoring out -1 is often useful when it makes the leading term positive.

Common failure

Removing only part of the common factor, or forgetting that extracting a negative changes every sign inside the parentheses.

Worked example

Problem. $42x^3 + 66x^2$

Diagnosis. The shape is common numerical/monomial factor. The appropriate tool is: extract the greatest common factor.

Result. $6x^2(7x + 11)$

Check. Multiply the GCF by the remaining polynomial.

1.3 Mastery exercises

Level I: direct recognition

1. $24x^3 + 60x^2$

2. $6x^2 + 15x$

3. $42x^3 + 66x^2$

4. $-35x^3 + 50x$

5. $40x^3 - 88x^2$

6. $36x^3 + 16x^2$

7. $-12x^4 - 54x^2$

8. $-84x^4 + 48x^2$

9. $-8x^2 - 88x$

10. $-12x^3 + 84x^2$

Level II: routine variations

- | | |
|---------------------|---------------------|
| 1. $12x^5 - 96x^3$ | 6. $-60x^3 - 84x$ |
| 2. $-24x^3 + 36x$ | 7. $-18x^4 - 21x^3$ |
| 3. $12x^2 + 32x$ | 8. $16x^4 + 44x^3$ |
| 4. $-45x^4 + 9x^3$ | 9. $-30x^4 - 5x^2$ |
| 5. $108x^4 + 48x^2$ | 10. $70x^4 - 40x^3$ |

Level III: disguised or prepared forms

- | | |
|-----------------------------|------------------------------|
| 1. $-35x^8 - 40x^6 + 35x^4$ | 6. $48x^8 - 8x^6 + 48x^3$ |
| 2. $-96x^4 + 12x^3 - 72x$ | 7. $12x^7 - 36x^6 - 6x^3$ |
| 3. $-25x^8 + 10x^6 - 10x^5$ | 8. $-50x^5 + 50x^4 - 20x^3$ |
| 4. $20x^7 - 30x^6 + 25x^2$ | 9. $-72x^8 - 45x^7 + 27x^6$ |
| 5. $-20x^8 + 30x^6 - 60x^4$ | 10. $-64x^8 - 40x^7 + 56x^5$ |

Level IV: multi-stage execution

- | | |
|--------------------------------|---------------------------------|
| 1. $-24x^{10} + 12x^9 + 72x^8$ | 6. $-40x^7 + 10x^5 - 10x^4$ |
| 2. $30x^7 - 36x^6 + 36x^4$ | 7. $12x^{10} + 48x^6 + 18x^5$ |
| 3. $-8x^{10} + 64x^6 - 40x^4$ | 8. $-54x^9 + 63x^8 - 18x^4$ |
| 4. $24x^5 + 48x^4 + 42x^2$ | 9. $20x^9 + 12x^8 - 24x^7$ |
| 5. $72x^9 - 63x^8 - 36x^7$ | 10. $-21x^{10} - 15x^7 + 18x^6$ |

Level V: independent mastery

- | | |
|------------------------------|------------------------------|
| 1. $56x^3 - 576x^2 + 648x$ | 6. $40x^5 + 216x^4 + 80x^3$ |
| 2. $4x^4 + 44x^3 + 72x^2$ | 7. $3x^4 + 15x^3 + 18x^2$ |
| 3. $-5x^3 + 5x^2 + 30x$ | 8. $84x^7 + 660x^6 - 864x^5$ |
| 4. $18x^6 - 114x^5 + 36x^4$ | 9. $36x^4 - 76x^3 - 96x^2$ |
| 5. $-48x^4 + 282x^3 + 36x^2$ | 10. $4x^5 - 24x^4 - 108x^3$ |

Chapter 2

Difference of Squares

Learning target

Recognize and factor a subtraction of two perfect squares.

2.1 Recognize the shape

Exactly two terms; subtraction; both terms are squares.

2.2 Procedure

Use $A^2 - B^2 = (A - B)(A + B)$, then inspect the new factors again.

Common failure

Applying the formula to a sum of squares. Over the real numbers, $A^2 + B^2$ does not factor by the difference-of-squares identity.

Worked example

Problem. $x^2 - 9$

Diagnosis. The shape is difference of two perfect squares. The appropriate tool is: apply $A^2 - B^2 = (A - B)(A + B)$ repeatedly.

Result. $(x - 3)(x + 3)$

Check. Expand the two conjugate factors.

2.3 Mastery exercises

Level I: direct recognition

1. $4x^2 - 1$

2. $25x^2 - 4$

3. $x^2 - 9$

4. $36x^2 - 81$

5. $25x^2 - 81$

6. $25x^2 - 4$

7. $36x^2 - 1$

8. $16x^2 - 4$

9. $49x^2 - 49$

10. $4x^2 - 81$

Level II: routine variations

1. $9x^4 - 36$

2. $4x^4 - 9$

3. $25x^4 - 1$

4. $4x^6 - 25$

5. $25x^4 - 49$

6. $x^4 - 36$

7. $x^4 - 1$

8. $16x^8 - 49$

9. $x^4 - 9$

10. $4x^6 - 25$

Level III: disguised or prepared forms

1. $16x^2 + 24x + 5$

2. $9x^2 + 24x + 7$

3. $9x^2 + 36x$

4. $x^2 + 6x + 8$

5. $x^2 + 12x + 20$

6. $4x^2 + 16x + 12$

7. $25x^2 + 60x + 20$

8. $4x^2 + 20x + 24$

9. $4x^2 + 20x + 9$

10. $x^2 + 4x + 3$

Level IV: multi-stage execution

1. $6x^4 - 96x^2$

2. $20x^3 - 5x$

3. $18x^4 - 2x^2$

4. $6x^5 - 24x^3$

5. $5x^4 - 5x^2$

6. $4x^4 - 196x^2$

7. $2x^4 - 32x^2$

8. $64x^4 - 16x^2$

9. $8x^3 - 8x$

10. $32x^3 - 128x$

Level V: independent mastery

1. $16x^4 - 1296$

2. $256x^4 - 81$

3. $16x^4 - 256$

4. $x^4 - 2401$

5. $16x^4 - 625$

6. $16x^4 - 256$

7. $81x^4 - 625$

8. $81x^4 - 256$

9. $81x^4 - 81$

10. $256x^4 - 1$

Chapter 3

Perfect-Square Trinomials

Learning target

Recognize a trinomial produced by squaring a binomial.

3.1 Recognize the shape

The first and last terms are squares, and the middle term is $\pm 2AB$.

3.2 Procedure

Use $A^2 \pm 2AB + B^2 = (A \pm B)^2$.

Common failure

Checking only the first and last terms. The middle term must be exactly $\pm 2AB$.

Worked example

Problem. $16x^2 - 16x + 4$

Diagnosis. The shape is three-term perfect-square pattern. The appropriate tool is: match $A^2 \pm 2AB + B^2$.

Result. $4(2x - 1)^2$

Check. Square the binomial.

3.3 Mastery exercises

Level I: direct recognition

1. $25x^2 - 80x + 64$

2. $x^2 + 2x + 1$

3. $16x^2 - 16x + 4$

4. $9x^2 + 18x + 9$

5. $25x^2 + 60x + 36$

6. $x^2 + 10x + 25$

7. $16x^2 - 8x + 1$

8. $9x^2 + 24x + 16$

9. $9x^2 + 24x + 16$

10. $x^2 + 8x + 16$

Level II: routine variations

1. $x^2 - 10x + 25$

2. $16x^2 - 64x + 64$

3. $x^2 + 14x + 49$

4. $x^2 + 12x + 36$

5. $9x^2 - 30x + 25$

6. $36x^2 + 12x + 1$

7. $x^2 - 10x + 25$

8. $36x^2 + 48x + 16$

9. $9x^2 + 24x + 16$

10. $36x^2 + 24x + 4$

Level III: disguised or prepared forms

1. $16x^2 + 32x + 16$

2. $16x^2 + 48x + 36$

3. $9x^4 - 42x^2 + 49$

4. $49x^4 - 70x^2 + 25$

5. $16x^2 - 40x + 25$

6. $x^2 + 2x + 1$

7. $16x^4 + 24x^2 + 9$

8. $x^2 - 14x + 49$

9. $x^4 - 12x^2 + 36$

10. $4x^2 + 20x + 25$

Level IV: multi-stage execution

1. $245x^4 - 350x^3 + 125x^2$

2. $128x^3 + 192x^2 + 72x$

3. $72x^3 - 216x^2 + 162x$

4. $8x^5 + 32x^3 + 32x$

5. $45x^7 + 90x^4 + 45x$

6. $2x^6 - 28x^4 + 98x^2$

7. $196x^3 - 392x^2 + 196x$

8. $384x^3 + 192x^2 + 24x$

9. $216x^6 - 216x^4 + 54x^2$

10. $45x^3 + 270x^2 + 405x$

Level V: independent mastery

1. $54x^3 + 144x^2 + 96x$

2. $75x^3 - 270x^2 + 243x$

3. $180x^6 + 240x^4 + 80x^2$

4. $36x^4 - 120x^3 + 100x^2$

5. $36x^2 + 192x + 256$

6. $320x^{10} - 240x^6 + 45x^2$

7. $243x^6 + 54x^3 + 3$

8. $125x^4 + 250x^3 + 125x^2$

9. $256x^5 - 128x^3 + 16x$

10. $8x^9 - 8x^5 + 2x$

Chapter 4

Sum and Difference of Cubes

Learning target

Factor sums and differences of perfect cubes.

4.1 Recognize the shape

Exactly two terms; each term is a cube; the sign may be plus or minus.

4.2 Procedure

Use $A^3 - B^3 = (A - B)(A^2 + AB + B^2)$ and $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$.

Common failure

Losing the sign pattern in the quadratic factor: the first sign matches the binomial and the middle sign is the opposite.

Worked example

Problem. $64x^3 + 8$

Diagnosis. The shape is sum or difference of cubes. The appropriate tool is: apply the cube identity, after any GCF.

Result. $8(2x + 1)(4x^2 - 2x + 1)$

Check. Multiply the binomial and quadratic factors.

4.3 Mastery exercises

Level I: direct recognition

1. $x^3 - 216$

2. $8x^3 - 8$

3. $64x^3 + 8$

4. $8x^3 + 27$

5. $x^3 + 216$

6. $x^3 + 27$

7. $64x^3 + 125$

8. $27x^3 + 216$

9. $8x^3 + 8$

10. $8x^3 + 27$

Level II: routine variations

1. $27x^3 + 27$

2. $x^3 + 8$

3. $8x^3 + 1$

4. $x^3 + 216$

5. $64x^3 + 125$

6. $8x^3 + 1$

7. $x^3 + 64$

8. $x^3 - 1$

9. $64x^3 + 64$

10. $x^3 - 216$

Level III: disguised or prepared forms

1. $64x^6 + 27$

2. $8x^6 + 1$

3. $x^6 + 216$

4. $27x^6 - 125$

5. $27x^6 - 125$

6. $27x^3 - 125$

7. $x^6 + 125$

8. $64x^6 - 1$

9. $27x^3 - 64$

10. $64x^3 - 27$

Level IV: multi-stage execution

1. $54x^4 + 128x$

2. $40x^8 - 40x^2$

3. $24x^3 - 375$

4. $2x^3 - 250$

5. $135x^7 - 625x$

6. $40x^6 + 40$

7. $135x^5 + 135x^2$

8. $135x^3 + 320$

9. $5x^8 - 40x^2$

10. $2x^5 + 16x^2$

Level V: independent mastery

1. $16x^8 + 16x^2$

2. $3x^9 + 3$

3. $24x^4 - 24x$

4. $3x^5 - 3x^2$

5. $24x^4 + 648x$

6. $54x^4 + 250x$

7. $5x^{11} + 1080x^2$

8. $5x^8 + 1080x^2$

9. $2x^5 + 54x^2$

10. $2x^7 - 2x$

Chapter 5

Monic Quadratic Trinomials

Learning target

Factor $x^2 + bx + c$ over the integers when possible.

5.1 Recognize the shape

A quadratic trinomial whose leading coefficient is 1.

5.2 Procedure

Find integers r, s with $r + s = b$ and $rs = c$; then $x^2 + bx + c = (x + r)(x + s)$.

Common failure

Choosing numbers with the correct product but the wrong sum.

Worked example

Problem. $x^2 - 6x + 8$

Diagnosis. The shape is monic quadratic trinomial. The appropriate tool is: find two integers with sum $-b$ and product c .

Result. $(x - 4)(x - 2)$

Check. Multiply the two binomials.

5.3 Mastery exercises

Level I: direct recognition

1. $x^2 - 9x + 8$

2. $x^2 - 9x + 20$

3. $x^2 - 6x + 8$

4. $x^2 - 8x + 12$

5. $x^2 - 9x + 14$

6. $x^2 - 3x + 2$

7. $x^2 - 11x + 28$

8. $x^2 - 16x + 63$

9. $x^2 - 11x + 24$

10. $x^2 - 11x + 28$

Level II: routine variations

1. $x^2 + 14x + 48$

2. $x^2 + 16x + 63$

3. $x^2 + 10x + 21$

4. $x^2 + 8x + 12$

5. $x^2 + 12x + 32$

6. $x^2 + 2x + 1$

7. $x^2 + 5x + 6$

8. $x^2 + 16x + 63$

9. $x^2 + 14x + 48$

10. $x^2 + 12x + 20$

Level III: disguised or prepared forms

1. $x^2 - x - 110$

2. $x^2 + 14x + 45$

3. $x^2 - 10x + 24$

4. $x^2 - 4x - 12$

5. $x^2 - 11x + 24$

6. $x^2 - 10x - 11$

7. $x^2 - 14x + 40$

8. $x^2 + 3x - 70$

9. $x^2 - 5x - 50$

10. $x^2 - 25$

Level IV: multi-stage execution

1. $x^2 - 2x - 15$

2. $x^2 + x - 72$

3. $x^2 + 10x + 21$

4. $x^2 - 19x + 90$

5. $x^2 + 14x + 45$

6. $x^2 + 12x + 27$

7. $x^2 + 7x$

8. $x^2 - 144$

9. $x^2 + 12x + 32$

10. $x^2 + 12x + 35$

Level V: independent mastery

1. $x^2 - 6x + 8$

2. $x^2 + 10x - 11$

3. $x^2 - 3x - 10$

4. $x^2 + x - 12$

5. $x^2 + 8x + 15$

6. $x^2 - 8x + 15$

7. $x^2 - 7x - 30$

8. $x^2 - 13x + 40$

9. $x^2 + 12x + 11$

10. $x^2 - 5x - 36$

Chapter 6

General Quadratic Trinomials

Learning target

Factor $ax^2 + bx + c$ when $a \neq 1$.

6.1 Recognize the shape

A quadratic trinomial with a nonunit leading coefficient.

6.2 Procedure

Find a middle-term split whose product is ac and whose sum is b , then factor by grouping.

Common failure

Finding a middle-term split but failing to factor the resulting four terms by grouping.

Worked example

Problem. $6x^2 + 23x + 21$

Diagnosis. The shape is nonmonic quadratic trinomial. The appropriate tool is: split the middle term or use an ac method, then group.

Result. $(2x + 3)(3x + 7)$

Check. Multiply the two binomials and compare all three coefficients.

6.3 Mastery exercises

Level I: direct recognition

1. $2x^2 + 13x + 6$

2. $2x^2 + 11x - 40$

3. $6x^2 + 23x + 21$

4. $4x^2 - 8x - 5$

5. $2x^2 - 3x - 20$

6. $3x^2 + 26x + 48$

7. $3x^2 - 26x + 16$

8. $6x^2 + 13x - 8$

9. $2x^2 + 13x - 24$

10. $2x^2 + 13x - 24$

Level II: routine variations

1. $16x^2 - 16x - 5$

2. $6x^2 + 31x + 35$

3. $12x^2 + 35x + 25$

4. $12x^2 - 11x + 2$

5. $2x^2 - 13x + 21$

6. $9x^2 + 18x - 7$

7. $2x^2 + 9x + 10$

8. $9x^2 - 36x + 35$

9. $2x^2 - 11x - 21$

10. $12x^2 + 23x + 5$

Level III: disguised or prepared forms

1. $15x^2 - 46x + 35$

2. $20x^2 + 17x - 24$

3. $2x^2 - 13x + 20$

4. $6x^2 - 5x - 6$

5. $3x^2 - 4x - 15$

6. $6x^2 + 11x + 3$

7. $4x^2 + 20x + 21$

8. $4x^2 - 3x - 1$

9. $15x^2 + 26x + 8$

10. $9x^2 + 6x - 35$

Level IV: multi-stage execution

1. $30x^2 - x - 42$

2. $36x^2 + 36x + 5$

3. $10x^2 + 21x - 49$

4. $3x^2 - 20x + 12$

5. $10x^2 + 31x + 24$

6. $9x^2 - 3x - 20$

7. $10x^2 + 19x - 15$

8. $6x^2 + 13x + 6$

9. $2x^2 + 11x + 15$

10. $12x^2 + 7x - 10$

Level V: independent mastery

1. $5x^2 + 32x + 12$

2. $14x^2 - 31x - 10$

3. $6x^2 + 5x - 25$

4. $3x^2 + 16x - 35$

5. $9x^2 + 9x - 28$

6. $35x^2 - 9x - 56$

7. $42x^2 - 79x + 35$

8. $25x^2 - 70x + 48$

9. $10x^2 + 7x - 12$

10. $7x^2 - 15x + 2$

Chapter 7

Factoring by Grouping

Learning target

Create and extract a repeated binomial factor.

7.1 Recognize the shape

Usually four terms that can be paired so each pair produces the same binomial.

7.2 Procedure

Group, factor each group, then factor out the common binomial.

Common failure

Factoring each group correctly but producing opposite binomials; factor out -1 from one group when necessary.

Worked example

Problem. $-2x^3 + 12x^2 - 15x - 4$

Diagnosis. The shape is four or more terms with a repeated binomial after grouping. The appropriate tool is: group terms and extract the common binomial.

Result. $-(x - 4)(2x^2 - 4x - 1)$

Check. Distribute the common binomial.

7.3 Mastery exercises

Level I: direct recognition

1. $20x^3 + 26x^2 + 4x - 2$

2. $5x^3 - 11x^2 - 17x - 4$

3. $-2x^3 + 12x^2 - 15x - 4$

4. $6x^3 - 5x^2 - 7x - 1$

5. $24x^3 - 8x^2 + 12x - 4$

6. $30x^3 - 50x^2 + 26x - 4$

7. $20x^3 + 20x^2 + 5x + 5$

8. $-10x^3 - 14x^2 + 14x + 4$

9. $2x^3 - 8x^2 - 11x + 5$

10. $15x^3 + 27x^2 - 11x + 1$

Level II: routine variations

1. $-8x^3 + 34x^2 - 31x - 7$

2. $-30x^3 - 35x^2 - 24x - 28$

3. $-5x^3 - 15x^2 - 7x - 21$

4. $-6x^3 - 42x^2 + 6x + 42$

5. $2x^3 - x^2 - 29x + 4$

6. $2x^3 - 13x^2 + 4x + 1$

7. $-24x^3 + 42x^2 + 16x - 28$

8. $-3x^3 - 5x^2 - 3x - 5$

9. $-8x^3 - 2x^2 + 28x + 7$

10. $5x^3 - x^2 - 35x + 7$

Level III: disguised or prepared forms

1. $8x^4 - 20x^3 + 6x - 15$

2. $-5x^3 - 37x^2 - 19x - 2$

3. $-15x^3 + 14x^2 + 5x + 4$

4. $2x^3 + 5x^2 - 44x + 7$

5. $-36x^4 + 12x^3 + 30x - 10$

6. $-10x^3 - 3x^2 + 6x - 1$

7. $-2x^3 + 12x^2 - 5x + 30$

8. $18x^4 + 18x^3 - 3x - 3$

9. $8x^3 - 12x^2 - 12x - 2$

10. $-25x^3 + 30x^2 + 30x - 36$

Level IV: multi-stage execution

1. $-18x^4 + 30x^3 - 9x + 15$

2. $18x^4 - 18x^3 + 9x - 9$

3. $18x^3 - 6x^2 - 21x - 4$

4. $25x^3 + 25x^2 - 10x - 10$

5. $-24x^4 + 18x^3 - 16x + 12$

6. $30x^3 + 14x^2 - 14x + 2$

7. $2x^3 - 2x^2 - 8x + 8$

8. $2x^3 + 8x^2 - 2x - 8$

9. $6x^3 - 7x + 1$

10. $12x^4 + 20x^3 + 9x + 15$

Level V: independent mastery

1. $-36x^3 - 6x^2 + 6x + 1$

2. $12x^3 + 9x^2 - 26x - 5$

3. $25x^3 + 5x^2 - 7x - 3$

4. $-30x^4 - 15x^3 - 24x - 12$

5. $16x^3 - 20x^2 + 4x - 5$

6. $24x^3 + 18x^2 - 19x - 5$

7. $-36x^3 - 24x^2 + 24x + 16$

8. $-36x^5 + 30x^4 - 24x + 20$

9. $12x^3 + x^2 - 3x - 2$

10. $9x^4 + 15x^3 - 3x - 5$

Chapter 8

Substitution and Higher-Power Patterns

Learning target

Treat expressions such as $x^4 - 5x^2 + 4$ as quadratics in a power of x .

8.1 Recognize the shape

Only powers $x^{2k}, x^k, 1$ occur for one fixed k .

8.2 Procedure

Set $u = x^k$, factor in u , replace u , and continue factoring.

Common failure

Factoring in the temporary variable u but forgetting to replace u by the original power and continue factoring.

Worked example

Problem. $x^4 - 3x^2 + 2$

Diagnosis. The shape is quadratic pattern in x^2 . The appropriate tool is: substitute $u = x^2$, factor in u , then replace u .

Result. $(x - 1)(x + 1)(x^2 - 2)$

Check. Expand after replacing the temporary variable.

8.3 Mastery exercises

Level I: direct recognition

1. $x^4 - 5x^2 + 6$

2. $x^4 + 8x^2 + 7$

3. $x^4 - 3x^2 + 2$

4. $x^4 - 13x^2 + 42$

5. $x^4 + 5x^2 + 6$

6. $x^4 - 11x^2 + 30$

7. $x^4 + x^2 - 12$

8. $x^4 - 9x^2 + 20$

9. $x^4 - x^2 - 6$

10. $x^4 + 11x^2 + 30$

Level II: routine variations

1. $x^4 - x^2 - 12$
2. $x^4 - 3x^2 - 28$
3. $x^4 + 9x^2 + 20$
4. $x^4 + 9x^2 + 14$
5. $x^4 + 13x^2 + 42$

6. $x^6 - 1$
7. $x^6 - 6x^3 + 8$
8. $x^4 + 5x^2 + 6$
9. $x^4 - 3x^2 - 10$
10. $x^4 - 5x^2 - 14$

Level III: disguised or prepared forms

1. $x^6 - 4x^3 + 3$
2. $x^6 + 6x^3 + 5$
3. $x^4 + 7x^2 + 12$
4. $x^4 + 2x^2 - 35$
5. $x^6 - 3x^3 - 28$

6. $x^4 + 8x^2 + 15$
7. $x^4 + 3x^2 + 2$
8. $x^4 - 2x^2 - 3$
9. $x^6 - 2x^3 - 24$
10. $x^4 - 4x^2 - 5$

Level IV: multi-stage execution

1. $2x^8 + 2x^5 - 4x^2$
2. $3x^6 + 9x^4 - 12x^2$
3. $2x^5 - 16x^3 + 14x$
4. $2x^5 - 24x^3 + 70x$
5. $3x^5 + 9x^3 + 6x$

6. $3x^7 - 6x^4 - 105x$
7. $x^4 + 11x^2 + 28$
8. $x^4 - 6x^2 + 8$
9. $x^4 + 6x^2 + 5$
10. $x^4 + 6x^2 + 5$

Level V: independent mastery

1. $3x^6 - 27x^4 + 60x^2$
2. $2x^8 - 18x^5 + 40x^2$
3. $3x^5 - 12x^3 - 15x$
4. $x^6 + 2x^3 - 24$
5. $x^4 - 10x^2 + 24$

6. $x^6 - 6x^3 + 5$
7. $x^6 - 11x^3 + 30$
8. $3x^{10} + 15x^6 + 18x^2$
9. $x^4 - 3x^2 - 28$
10. $5x^5 + 30x^3 + 40x$

Chapter 9

Multi-Stage Complete Factorization

Learning target

Combine several factoring tools in the correct order.

9.1 Recognize the shape

The first factorization reveals another recognizable shape.

9.2 Procedure

Use GCF first, then identities, trinomials, grouping, or substitution until no listed factorization applies.

Common failure

Stopping after the first successful step instead of checking every new factor.

Worked example

Problem. $5x^5 + 15x^4 - 15x^3 + 15x^2 - 20x$

Diagnosis. The shape is expression containing more than one factoring pattern. The appropriate tool is: GCF first, then repeatedly factor until no listed tool applies.

Result. $5x(x-1)(x+4)(x^2+1)$

Check. Multiply all final factors.

9.3 Mastery exercises

Level I: direct recognition

1. $5x^4 - 25x^2 + 20$

2. $4x^7 - 4x^5 - 4x^4 + 4x^2$

3. $5x^5 + 15x^4 - 15x^3 + 15x^2 - 20x$

4. $20x^6 + 60x^5 - 85x^4 - 375x^3 - 250x^2$

5. $2x^6 - 8x^4 - 16x^3 + 64x$

6. $4x^6 - 20x^4 + 16x^2$

7. $-3x^5 + 12x^3 + 24x^2 - 96$

8. $3x^5 - 6x^4 - 42x^3 - 6x^2 - 45x$

9. $3x^4 + 9x^3 - 69x^2 - 225x - 150$

10. $5x^7 - 45x^5 - 135x^4 + 1215x^2$

Level II: routine variations

1. $-6x^4 + 30x^2 - 24$
2. $2x^6 - 8x^4 - 16x^3 + 64x$
3. $2x^5 - 10x^4 - 10x^3 - 10x^2 - 12x$
4. $27x^5 + 81x^4 + 42x^3 - 36x^2 - 24x$
5. $5x^6 - 5x^4 - 5x^3 + 5x$
6. $5x^5 - 25x^3 + 20x$
7. $4x^7 - 100x^5 - 500x^4 + 12500x^2$
8. $-4x^5 - 4x^4 + 20x^3 - 4x^2 + 24x$
9. $54x^5 + 162x^4 + 102x^3 - 18x^2 - 12x$
10. $4x^5 - 16x^3 - 32x^2 + 128$

Level III: disguised or prepared forms

1. $5x^4 - 25x^2 + 20$
2. $-3x^6 + 3x^4 + 3x^3 - 3x$
3. $-4x^7 - 32x^6 - 52x^5 - 32x^4 - 48x^3$
4. $8x^6 + 24x^5 - 2x^4 - 54x^3 - 36x^2$
5. $-6x^8 + 24x^6 + 48x^5 - 192x^3$
6. $-6x^7 + 30x^5 - 24x^3$
7. $5x^6 - 5x^4 - 5x^3 + 5x$
8. $6x^7 + 18x^6 - 54x^5 + 18x^4 - 60x^3$
9. $20x^4 + 60x^3 + 20x^2 - 60x - 40$
10. $4x^5 - 16x^3 - 32x^2 + 128$

Level IV: multi-stage execution

1. $2x^4 - 10x^2 + 8$
2. $-6x^6 + 6x^4 + 6x^3 - 6x$
3. $-3x^6 - 9x^5 + 51x^4 - 9x^3 + 54x^2$
4. $45x^7 + 135x^6 + 70x^5 - 60x^4 - 40x^3$
5. $-3x^6 + 3x^4 + 3x^3 - 3x$
6. $3x^7 - 15x^5 + 12x^3$
7. $2x^6 - 18x^4 - 54x^3 + 486x$
8. $-4x^6 + 24x^5 - 36x^4 + 24x^3 - 32x^2$
9. $36x^4 + 108x^3 + 68x^2 - 12x - 8$
10. $2x^6 - 32x^4 - 128x^3 + 2048x$

Level V: independent mastery

1. $2x^6 - 10x^4 + 8x^2$
2. $-4x^5 + 100x^3 + 500x^2 - 12500$
3. $-3x^6 + 6x^5 + 42x^4 + 6x^3 + 45x^2$
4. $-6x^4 - 18x^3 + 84x^2 + 288x + 192$
5. $-4x^8 + 16x^6 + 32x^5 - 128x^3$
6. $4x^6 - 20x^4 + 16x^2$
7. $4x^9 - 16x^7 - 32x^6 + 128x^4$
8. $-6x^8 - 18x^7 + 18x^6 - 18x^5 + 24x^4$
9. $-16x^6 - 48x^5 + 4x^4 + 108x^3 + 72x^2$
10. $-4x^5 + 16x^3 + 32x^2 - 128$

Part II

Part II: Rational-Expression Operations

Rational-Expression Discipline

A rational expression is a fraction built from polynomials. Its algebra is ordinary fraction algebra, but factoring controls cancellation and denominator construction. Record restrictions from the original expression before simplifying; those restrictions survive even when a factor later cancels.

Terms are not factors

In $x + 3$, neither x nor 3 is a factor of the entire expression. In $x(x + 3)$, both x and $x + 3$ are factors. Cancellation applies only to a common nonzero factor of an entire numerator and denominator.

Chapter 10

Simplification by Factoring and Cancellation

Learning target

Simplify a rational expression without losing domain restrictions.

10.1 Recognize the shape

Numerator and denominator may share factors after factoring.

10.2 Procedure

Record original restrictions, factor completely, cancel identical nonzero factors only, and retain every original excluded value.

Illegal cancellation

$\frac{x+3}{x} \neq 3$ and $\frac{x^2+4}{x} \neq x+4$. Addition does not create a common factor. Factor first; cancel only an entire common factor.

Common failure

Cancelling terms rather than factors, or deleting an original excluded value after cancellation.

Worked example

Problem. $\frac{x^2 - 7x + 12}{x^2 - 3x - 4}$

Diagnosis. The shape is quotient with common polynomial factors. The appropriate tool is: factor both parts, cancel factors only, preserve original restrictions.

Result. $\frac{x-3}{x+1}$, with $x \neq -1, 4$.

Check. Multiply back or test an allowed value; recheck the restriction list.

10.3 Mastery exercises

Level I: direct recognition

1. $\frac{x^2 - 36}{x^2 - 11x + 30}$

2. $\frac{x^2 - 3x - 28}{x^2 - 4x - 21}$

$$3. \frac{x^2 - 7x + 12}{x^2 - 3x - 4}$$

$$4. \frac{x^2 + 7x + 10}{x^2 + 5x}$$

$$5. \frac{x^2 + 4x - 12}{x^2 + 11x + 30}$$

$$6. \frac{x^2 + 3x - 28}{x^2 + x - 42}$$

$$7. \frac{x^2 - 6x + 5}{x^2 - 4x + 3}$$

$$8. \frac{x^2 + 7x + 12}{x^2 + 8x + 15}$$

$$9. \frac{x^2 - 7x + 10}{x^2 + 2x - 35}$$

$$10. \frac{x^2 + 12x + 35}{x^2 + 9x + 14}$$

Level II: routine variations

$$1. \frac{x^2 + 7x + 10}{x^2 - 4x - 12}$$

$$2. \frac{x^2 - 9}{x^2 - x - 6}$$

$$3. \frac{x^2 - x - 12}{x^2 + 7x + 12}$$

$$4. \frac{x^2 - 6x}{x^2 - 9x + 18}$$

$$5. \frac{x^2 + 6x + 5}{x^2 + 12x + 35}$$

$$6. \frac{x^2 - 11x + 28}{x^2 - 10x + 21}$$

$$7. \frac{x^2 + 8x + 7}{x^2 + x - 42}$$

$$8. \frac{x^2 + 4x - 21}{x^2 + 9x + 14}$$

$$9. \frac{x^2 + 3x}{x^2 + 4x}$$

$$10. \frac{x^2 + 2x - 35}{x^2 + 6x - 7}$$

Level III: disguised or prepared forms

$$1. \frac{x^3 - 31x + 30}{x^3 - 4x^2 - 7x + 10}$$

$$2. \frac{x^3 - 2x^2 - x + 2}{x^3 - 4x^2 + x + 6}$$

$$3. \frac{x^3 - 21x - 20}{x^3 + x^2 - 25x - 25}$$

$$4. \frac{x^3 + 2x^2 - 33x - 90}{x^3 - 6x^2 - 9x + 54}$$

$$5. \frac{x^3 - 8x^2 + x + 42}{x^3 - 6x^2 - x + 30}$$

$$6. \frac{x^3 + 2x^2 - 15x}{x^3 - 3x^2 - 25x + 75}$$

$$7. \frac{x^3 - 43x + 42}{x^3 - 3x^2 - 46x + 168}$$

$$8. \frac{x^3 - 8x^2 + 9x + 18}{x^3 - 10x^2 + 19x + 30}$$

$$9. \frac{x^3 - 12x^2 + 47x - 60}{x^3 - 8x^2 + 15x}$$

$$10. \frac{x^3 - 2x^2 - 23x + 60}{x^3 - 3x^2 - 25x + 75}$$

Level IV: multi-stage execution

$$1. \frac{x^5 + 5x^4 - 11x^3 - 31x^2 - 12x - 36}{x^3 + 11x^2 + 36x + 36}$$

$$2. \frac{x^5 - 7x^4 - x + 7}{x^3 - 6x^2 - x + 6}$$

$$3. \frac{x^5 - 5x^4 + 3x^3 + 3x^2 + 2x + 8}{x^3 + 3x^2 - 6x - 8}$$

$$4. \frac{x^5 + x^4 - 13x^3 - 23x^2 - 14x - 24}{x^3 + 2x^2 - 16x - 32}$$

$$5. \frac{x^5 + 3x^4 - 5x^3 - 5x^2 - 6x - 8}{x^3 - x^2 - 2x}$$

$$6. \frac{x^5 + 5x^4 - 11x^3 - 31x^2 - 12x - 36}{x^3 + 4x^2 - 15x - 18}$$

$$7. \frac{x^5 - x^4 - 21x^3 + 39x^2 - 22x + 40}{x^3 + 4x^2 - 7x - 10}$$

$$8. \frac{x^5 + 2x^4 - 4x^3 - 4x^2 - 5x - 6}{x^3 + 4x^2 - 7x - 10}$$

$$9. \frac{x^5 + 4x^4 - 24x^3 - 96x^2 - 25x - 100}{x^3 - 21x - 20}$$

$$10. \frac{x^5 - 16x^4 + 82x^3 - 142x^2 + 81x - 126}{x^3 - 6x^2 - 19x + 84}$$

Level V: independent mastery

$$1. \frac{x^5 + 9x^4 + 24x^3 + 24x^2 + 23x + 15}{x^3 + x^2 - 9x - 9}$$

$$2. \frac{x^5 - 7x^4 - 8x^3 + 56x^2 - 9x + 63}{x^3 - 9x^2 + 11x + 21}$$

$$3. \frac{x^5 + 6x^4 - 15x^3 - 90x^2 - 16x - 96}{x^3 + 7x^2 - 6x - 72}$$

$$4. \frac{x^5 - 4x^4 - 46x^3 + 206x^2 - 47x + 210}{x^3 - 5x^2 - 49x + 245}$$

$$5. \frac{x^5 + 7x^4 - 13x^3 - 113x^2 - 14x - 120}{x^3 - 21x + 20}$$

$$6. \frac{x^5 + 6x^4 - 18x^3 - 78x^2 - 19x - 84}{x^3 + 3x^2 - 16x - 48}$$

$$7. \frac{x^5 + 9x^4 + 24x^3 + 24x^2 + 23x + 15}{x^3 + 4x^2 + 3x}$$

$$8. \frac{x^5 + 4x^4 - 46x^3 - 206x^2 - 47x - 210}{x^3 - 2x^2 - 41x + 42}$$

$$9. \frac{x^5 + 5x^4 - 37x^3 - 163x^2 - 38x - 168}{x^3 + 6x^2 - 27x - 140}$$

$$10. \frac{x^5 + 4x^4 - 26x^3 - 86x^2 - 27x - 90}{x^3 - 2x^2 - 15x}$$

Chapter 11

Multiplication of Rational Expressions

Learning target

Multiply rational expressions efficiently.

11.1 Recognize the shape

A product of fractions whose factors may cancel across numerators and denominators.

11.2 Procedure

Record restrictions, factor, cancel common factors across the entire product, then multiply what remains.

Common failure

Multiplying expanded polynomials before looking for cancellable factors.

Worked example

Problem. $\frac{(x-1)(x+3)}{(x-5)(x+1)} \cdot \frac{x(x+1)}{(x+3)(x+4)}$

Diagnosis. The shape is product of rational expressions. The appropriate tool is: factor, cancel across the product, then multiply.

Result. $\frac{x^2 - x}{x^2 - x - 20}$, with $x \neq -4, -3, -1, 5$.

Check. Multiply the simplified factors and compare at an allowed value.

11.3 Mastery exercises

Level I: direct recognition

$$1. \frac{(x-7)(x-2)}{(x-8)(x+2)} \cdot \frac{(x-8)(x-4)}{x(x-2)}$$

$$2. \frac{(x-5)(x+5)}{(x-8)(x+4)} \cdot \frac{(x-2)(x+4)}{(x+5)(x+7)}$$

$$3. \frac{(x-1)(x+3)}{(x-5)(x+1)} \cdot \frac{x(x+1)}{(x+3)(x+4)}$$

$$4. \frac{(x-3)(x+1)}{(x-4)(x-1)} \cdot \frac{(x-1)(x+7)}{(x-3)(x+2)}$$

$$5. \frac{(x-6)(x+5)}{(x+1)(x+2)} \cdot \frac{(x-2)(x+2)}{(x+5)(x+6)}$$

$$6. \frac{x(x+6)}{(x-3)(x+5)} \cdot \frac{(x-3)(x-1)}{(x+1)(x+6)}$$

$$7. \frac{(x+3)(x+8)}{(x-1)(x+7)} \cdot \frac{(x-4)(x-1)}{(x+3)(x+6)}$$

$$8. \frac{x(x+1)}{(x-6)(x-2)} \cdot \frac{(x-4)(x-2)}{(x-8)(x+1)}$$

$$9. \frac{(x-7)(x+6)}{x(x+1)} \cdot \frac{x(x+7)}{(x-7)(x+4)}$$

$$10. \frac{(x-7)(x+8)}{(x-8)(x-6)} \cdot \frac{(x-6)(x+4)}{(x-5)(x+8)}$$

Level II: routine variations

$$1. \frac{(x-8)(x+7)}{(x-7)(x-2)} \cdot \frac{(x-2)(x+6)}{(x-1)(x+7)}$$

$$2. \frac{(x-2)(x+6)}{(x+3)(x+8)} \cdot \frac{(x+3)(x+5)}{(x-4)(x+6)}$$

$$3. \frac{(x-4)(x-1)}{(x+2)(x+7)} \cdot \frac{(x-8)(x+2)}{(x-6)(x-1)}$$

$$4. \frac{(x+5)(x+8)}{x(x-4)} \cdot \frac{x(x+2)}{(x+3)(x+8)}$$

$$5. \frac{(x-8)(x-4)}{(x-1)(x+1)} \cdot \frac{(x+1)(x+2)}{(x-8)(x-6)}$$

$$6. \frac{x(x+5)}{(x-3)(x-1)} \cdot \frac{(x-4)(x-1)}{(x+3)(x+5)}$$

$$7. \frac{(x-6)(x+5)}{(x-7)(x+4)} \cdot \frac{x(x-7)}{(x-6)(x+8)}$$

$$8. \frac{(x+6)(x+8)}{(x-8)(x-7)} \cdot \frac{(x-7)(x+2)}{(x-1)(x+6)}$$

$$9. \frac{(x-2)(x+8)}{(x-4)(x+5)} \cdot \frac{(x+1)(x+5)}{x(x+8)}$$

$$10. \frac{(x-7)(x+8)}{(x-5)(x-1)} \cdot \frac{x(x-1)}{(x-7)(x-6)}$$

Level III: disguised or prepared forms

$$1. \frac{x^2+4x-5}{x^2-2x-8} \cdot \frac{(x-4)(x+8)}{(x+3)(x+5)}$$

$$2. \frac{x^2+11x+30}{x^2-x} \cdot \frac{(x-1)(x+8)}{(x-6)(x+6)}$$

$$3. \frac{x^2+7x+12}{x^2+5x} \cdot \frac{(x-7)(x+5)}{(x-1)(x+4)}$$

$$4. \frac{x^2+9x+18}{x^2+x-2} \cdot \frac{(x-3)(x+2)}{(x+3)(x+8)}$$

$$5. \frac{x^2-6x}{x^2-11x+24} \cdot \frac{(x-3)(x-1)}{x(x+1)}$$

$$6. \frac{x^2-15x+56}{x^2+x-12} \cdot \frac{(x-3)(x-2)}{(x-8)(x-6)}$$

$$7. \frac{x^2+3x-10}{x^2+12x+32} \cdot \frac{(x+1)(x+8)}{(x+3)(x+5)}$$

$$8. \frac{x^2-4}{x^2-4x} \cdot \frac{(x-7)(x-4)}{(x+2)(x+5)}$$

$$9. \frac{x^2+12x+35}{x^2+5x+4} \cdot \frac{(x-2)(x+1)}{x(x+7)}$$

$$10. \frac{x^2+2x-24}{x^2+6x-7} \cdot \frac{(x+7)(x+8)}{(x-4)(x-3)}$$

Level IV: multi-stage execution

$$1. \frac{x^2+7x+10}{x^2+3x-40} \cdot \frac{x^2+x-30}{x^2-x-30}$$

$$2. \frac{x^2+11x+24}{x^2-x-6} \cdot \frac{x^2+4x-21}{x^2-2x-15}$$

$$3. \frac{x^2-10x+24}{x^2-4x-32} \cdot \frac{x^2+7x+12}{x^2-13x+42}$$

$$4. \frac{x^2-8x+15}{x^2+x-56} \cdot \frac{x^2-11x+28}{x^2-11x+24}$$

$$5. \frac{x^2+2x-24}{x^2+11x+24} \cdot \frac{x^2-2x-15}{x^2+13x+42}$$

$$6. \frac{x^2-2x-3}{x^2+12x+32} \cdot \frac{x^2-3x-28}{x^2+7x+6}$$

$$7. \frac{x^2+6x+8}{x^2+15x+56} \cdot \frac{x^2+11x+24}{x^2+3x+2}$$

$$8. \frac{x^2+10x+16}{x^2+x-42} \cdot \frac{x^2+7x}{x^2+5x-24}$$

$$9. \frac{x^2-8x}{x^2+6x-7} \cdot \frac{x^2+x-2}{x^2-64}$$

$$10. \frac{x^2+11x+24}{x^2+x-20} \cdot \frac{x^2-25}{x^2+15x+56}$$

Level V: independent mastery

$$1. \frac{x^2 + 2x - 3}{x^2 - 4} \cdot \frac{x^2 - 4x - 12}{x^2 - 1}$$

$$2. \frac{x^2 + 6x + 8}{x^2 - 64} \cdot \frac{x^2 + x - 56}{x^2 - 2x - 24}$$

$$3. \frac{x^2 - x - 42}{x^2 - 11x + 30} \cdot \frac{x^2 - 6x + 5}{x^2 - 6x - 7}$$

$$4. \frac{x^2 + 9x + 8}{x^2 - 8x} \cdot \frac{x^2 + 3x}{x^2 + 6x - 16}$$

$$5. \frac{x^2 + x - 2}{x^2 - 2x - 24} \cdot \frac{x^2 - 6x}{x^2 + 7x - 8}$$

$$6. \frac{x^2 + 3x + 2}{x^2 + 10x + 24} \cdot \frac{x^2 + 9x + 20}{x^2 + 2x}$$

$$7. \frac{x^2 + 5x + 6}{x^2 - 5x} \cdot \frac{x^2 + 3x - 40}{x^2 + 8x + 15}$$

$$8. \frac{x^2 + x - 2}{x^2 + 4x - 32} \cdot \frac{x^2 + 3x - 28}{x^2 - 4x + 3}$$

$$9. \frac{x^2 - 10x + 21}{x^2 - 64} \cdot \frac{x^2 - 12x + 32}{x^2 - 7x}$$

$$10. \frac{x^2 + 7x - 8}{x^2 - 7x - 8} \cdot \frac{x^2 + 3x + 2}{x^2 - 4x + 3}$$

Chapter 12

Division of Rational Expressions

Learning target

Divide rational expressions correctly.

12.1 Recognize the shape

A quotient in which the divisor is itself a rational expression.

12.2 Procedure

Require all original denominators and the divisor numerator to be nonzero, multiply by the reciprocal, factor, and simplify.

Common failure

Forgetting that the divisor itself must be nonzero, so its numerator creates additional restrictions.

Worked example

Problem. $\frac{(x-6)(x-2)}{(x-1)(x+2)} \div \frac{(x-6)(x+7)}{(x-7)(x-1)}$

Diagnosis. The shape is quotient of rational expressions. The appropriate tool is: record restrictions, multiply by the reciprocal, factor, and cancel.

Result. $\frac{x^2 - 9x + 14}{x^2 + 9x + 14}$, with $x \neq -7, -2, 1, 6, 7$.

Check. Multiply by the original divisor and compare at an allowed value.

12.3 Mastery exercises

Level I: direct recognition

$$1. \frac{(x+3)(x+5)}{x(x+8)} \div \frac{(x-4)(x+5)}{(x-7)(x+8)}$$

$$2. \frac{(x+1)(x+3)}{(x+6)(x+8)} \div \frac{(x-6)(x+1)}{(x-5)(x+6)}$$

$$3. \frac{(x-6)(x-2)}{(x-1)(x+2)} \div \frac{(x-6)(x+7)}{(x-7)(x-1)}$$

$$4. \frac{x(x+4)}{(x-4)(x+1)} \div \frac{x(x+6)}{(x+1)(x+8)}$$

5. $\frac{(x+3)(x+7)}{(x+6)(x+8)} \div \frac{(x-4)(x+7)}{(x-8)(x+8)}$
6. $\frac{(x-7)(x+2)}{(x+1)(x+3)} \div \frac{(x-6)(x+2)}{(x-5)(x+1)}$
7. $\frac{(x-1)(x+8)}{(x-2)(x+6)} \div \frac{(x-1)(x+5)}{x(x-2)}$
8. $\frac{(x-6)(x+5)}{(x-8)(x-3)} \div \frac{(x-6)(x+3)}{(x-3)(x-2)}$
9. $\frac{(x-1)(x+6)}{(x-7)(x-4)} \div \frac{(x-1)(x+2)}{(x-4)(x-3)}$
10. $\frac{(x-1)(x+1)}{(x-3)(x+2)} \div \frac{(x-2)(x+1)}{(x-3)(x+5)}$

Level II: routine variations

1. $\frac{(x-1)(x+6)}{x(x+2)} \div \frac{(x-2)(x+6)}{(x-5)(x+2)}$
2. $\frac{(x-4)(x+3)}{(x-6)(x-1)} \div \frac{(x+3)(x+6)}{(x-5)(x-1)}$
3. $\frac{(x+1)(x+4)}{(x-1)(x+7)} \div \frac{(x+1)(x+8)}{(x-4)(x+7)}$
4. $\frac{(x-8)(x+1)}{(x-4)(x-3)} \div \frac{(x-8)(x-1)}{(x-3)(x+8)}$
5. $\frac{(x-4)(x+4)}{(x-1)(x+2)} \div \frac{(x-4)(x+3)}{(x-3)(x+2)}$
6. $\frac{(x-4)(x+4)}{(x-8)(x-7)} \div \frac{(x-6)(x+4)}{(x-7)(x+5)}$
7. $\frac{(x-7)(x-6)}{(x-2)(x+6)} \div \frac{(x-7)(x+2)}{(x-2)(x+1)}$
8. $\frac{(x-7)(x+4)}{(x-2)(x+7)} \div \frac{(x-7)(x+1)}{(x-3)(x+7)}$
9. $\frac{(x-6)(x+1)}{(x-7)(x+7)} \div \frac{(x+1)(x+3)}{x(x-7)}$
10. $\frac{(x-5)(x+6)}{(x+3)(x+4)} \div \frac{(x+2)(x+6)}{(x-7)(x+3)}$

Level III: disguised or prepared forms

1. $\frac{x^2+8x+15}{x^2+6x-7} \div \frac{x(x+5)}{(x-1)(x+8)}$

2. $\frac{x^2 + 4x - 12}{x^2 - 8x} \div \frac{(x + 4)(x + 6)}{(x - 8)(x - 3)}$
3. $\frac{x^2 + 7x - 8}{x^2 - 8x} \div \frac{(x + 4)(x + 8)}{x(x - 2)}$
4. $\frac{x^2 + x - 6}{x^2 - x - 30} \div \frac{(x - 2)(x + 4)}{(x + 5)(x + 7)}$
5. $\frac{x^2 - 3x - 4}{x^2 + x - 12} \div \frac{(x - 4)(x + 3)}{(x + 4)(x + 7)}$
6. $\frac{x^2 + 3x - 10}{x^2 - x - 20} \div \frac{x(x - 2)}{(x - 5)(x + 1)}$
7. $\frac{x^2 - 4x - 32}{x^2 + 2x - 24} \div \frac{x(x + 4)}{(x - 5)(x - 4)}$
8. $\frac{x^2 - 9x + 18}{x^2 - 2x - 48} \div \frac{(x - 3)(x + 4)}{(x + 2)(x + 6)}$
9. $\frac{x^2 - 12x + 32}{x^2 + 2x} \div \frac{(x - 4)(x + 5)}{x(x + 7)}$
10. $\frac{x^2 - 16}{x^2 + 10x + 21} \div \frac{(x - 4)(x + 2)}{(x - 5)(x + 7)}$

Level IV: multi-stage execution

1. $\frac{x^2 + 8x}{x^2 - 2x - 3} \div \frac{x^2 + 6x - 16}{x^2 - 7x - 8}$
2. $\frac{x^2 + 6x + 5}{x^2 + 9x + 18} \div \frac{x^2 - 25}{x^2 - 9}$
3. $\frac{x^2 - 4}{x^2 - 10x + 24} \div \frac{x^2 + 6x + 8}{x^2 - 5x - 6}$
4. $\frac{x^2 + 11x + 30}{x^2 - x - 6} \div \frac{x^2 + 5x}{x^2 - 2x - 3}$
5. $\frac{x^2 - 7x + 6}{x^2 - 8x} \div \frac{x^2 + x - 2}{x^2 - 64}$
6. $\frac{x^2 + 7x - 8}{x^2 + 3x - 28} \div \frac{x^2 - 4x + 3}{x^2 + 9x + 14}$
7. $\frac{x^2 - 4x - 32}{x^2 - 4} \div \frac{x^2 + 7x + 12}{x^2 + 3x + 2}$
8. $\frac{x^2 - 2x - 35}{x^2 - 9x + 18} \div \frac{x^2 - 4x - 21}{x^2 + 4x - 21}$
9. $\frac{x^2 - x - 6}{x^2 - 10x + 16} \div \frac{x^2 + 5x + 6}{x^2 - 3x - 40}$

10. $\frac{x^2 + 9x + 18}{x^2 - 2x - 3} \div \frac{x^2 - 36}{x^2 - 1}$

Level V: independent mastery

1. $\frac{x^2 + 4x - 21}{x^2 - 7x} \div \frac{x^2 + 13x + 42}{x^2 - 6x}$

2. $\frac{x^2 - 11x + 24}{x^2 + 2x} \div \frac{x^2 - 7x - 8}{x^2 - 5x}$

3. $\frac{x^2 + 3x - 18}{x^2 + 2x - 8} \div \frac{x^2 + 4x - 21}{x^2 - 16}$

4. $\frac{x^2 - 9x + 8}{x^2 - 7x + 12} \div \frac{x^2 + x - 2}{x^2 + 2x - 24}$

5. $\frac{x^2 + 3x - 4}{x^2 - 12x + 35} \div \frac{x^2 + 6x + 8}{x^2 - 13x + 42}$

6. $\frac{x^2 + 7x + 6}{x^2 + 3x - 28} \div \frac{x^2 - 2x - 48}{x^2 - 2x - 8}$

7. $\frac{x^2 - 3x - 18}{x^2 + 5x - 6} \div \frac{x^2 - 2x - 15}{x^2 + 14x + 48}$

8. $\frac{x^2 - 4x - 21}{x^2 - 11x + 24} \div \frac{x^2 - 3x - 18}{x^2 - 7x - 8}$

9. $\frac{x^2 - 1}{x^2 - 36} \div \frac{x^2 - 3x + 2}{x^2 + x - 30}$

10. $\frac{x^2 + x - 56}{x^2 + 6x - 7} \div \frac{x^2 - 4x - 21}{x^2 + x - 42}$

Chapter 13

Addition and Subtraction of Rational Expressions

Learning target

Build a least common denominator and combine rational expressions.

13.1 Recognize the shape

A sum or difference of fractions, often with related denominators.

13.2 Procedure

Factor denominators, choose the LCD using every factor at its greatest required power, rewrite, combine numerators, and simplify.

Common failure

Using a common denominator that omits a repeated factor or uses too small a power.

Worked example

Problem. $\frac{-2}{x-8} + \frac{-1}{x}$

Diagnosis. The shape is sum or difference with unlike denominators. The appropriate tool is: factor denominators, build the LCD, combine, and simplify.

Result. $\frac{-3x+8}{x^2-8x}$, with $x \neq 0, 8$.

Check. Put the answer over the LCD and compare numerators.

13.3 Mastery exercises

Level I: direct recognition

1. $\frac{-2}{x+5} - \frac{1}{x-4}$

2. $\frac{6}{x-4} + \frac{3}{x+4}$

3. $\frac{-2}{x-8} + \frac{-1}{x}$

4. $\frac{5}{x+3} + \frac{-2}{x-8}$

5. $\frac{-2}{x+6} + \frac{-1}{x-7}$
6. $\frac{6}{x+5} + \frac{-1}{x+1}$
7. $\frac{-6}{x-3} - \frac{4}{x-6}$
8. $\frac{-3}{x-5} + \frac{-2}{x-3}$
9. $\frac{-6}{x+8} + \frac{3}{x-5}$
10. $\frac{4}{x-8} - \frac{4}{x+8}$

Level II: routine variations

1. $\frac{2}{x-3} - \frac{-2}{x-5}$
2. $\frac{5}{x} - \frac{-3}{x+7}$
3. $\frac{1}{x-5} - \frac{-5}{x+8}$
4. $\frac{-1}{x+1} + \frac{-6}{x-1}$
5. $\frac{-6}{x-6} - \frac{1}{x+7}$
6. $\frac{2}{x-3} - \frac{2}{x+1}$
7. $\frac{6}{x-6} + \frac{-4}{x-8}$
8. $\frac{3}{x-4} - \frac{1}{x-2}$
9. $\frac{1}{x-4} - \frac{-4}{x-1}$
10. $\frac{-5}{x+7} + \frac{3}{x-8}$

Level III: disguised or prepared forms

1. $\frac{-3}{(x+7)^2} - \frac{-4}{x(x+7)}$
2. $\frac{5}{(x+8)^2} + \frac{6}{(x+6)(x+8)}$

3. $\frac{-4}{(x-6)^2} - \frac{-5}{(x-6)(x+8)}$
4. $\frac{3}{(x+6)^2} + \frac{-4}{(x-6)(x+6)}$
5. $\frac{2}{(x+2)^2} - \frac{3}{(x+2)(x+3)}$
6. $\frac{-6}{(x-8)^2} + \frac{-3}{(x-8)(x+3)}$
7. $\frac{5}{(x+2)^2} + \frac{3}{(x-4)(x+2)}$
8. $\frac{-6}{(x+7)^2} - \frac{6}{(x+4)(x+7)}$
9. $\frac{-3}{(x+4)^2} + \frac{-6}{(x+1)(x+4)}$
10. $\frac{-3}{(x-1)^2} - \frac{4}{(x-2)(x-1)}$

Level IV: multi-stage execution

1. $\frac{3}{(x-2)^2} + \frac{2}{(x-6)(x-2)}$
2. $\frac{4}{(x+3)^2} + \frac{3}{(x-1)(x+3)}$
3. $\frac{5}{(x+6)^2} - \frac{-6}{(x+1)(x+6)}$
4. $\frac{-1}{(x-3)^2} + \frac{3}{(x-3)(x+7)}$
5. $\frac{3}{(x-8)^2} + \frac{6}{(x-8)(x-7)}$
6. $\frac{-2}{(x+4)^2} - \frac{4}{x(x+4)}$
7. $\frac{3}{(x+5)^2} - \frac{2}{(x-6)(x+5)}$
8. $\frac{-6}{(x-6)^2} + \frac{4}{(x-6)(x-4)}$
9. $\frac{-1}{(x+2)^2} - \frac{-4}{(x+2)(x+5)}$
10. $\frac{1}{(x+8)^2} + \frac{2}{(x-2)(x+8)}$

Level V: independent mastery

1. $\frac{-2}{(x-1)(x^2+1)} - \frac{-1}{(x+2)(x^2+1)}$
2. $\frac{-1}{(x+4)(x^2+1)} - \frac{4}{(x-5)(x^2+1)}$
3. $\frac{-5}{(x+5)(x^2+1)} + \frac{-3}{(x-1)(x^2+1)}$
4. $\frac{6}{(x-8)(x^2+1)} + \frac{4}{(x+4)(x^2+1)}$
5. $\frac{-4}{(x+8)(x^2+1)} - \frac{1}{(x+6)(x^2+1)}$
6. $\frac{4}{(x-6)(x^2+1)} - \frac{-6}{x(x^2+1)}$
7. $\frac{-2}{(x-6)(x^2+1)} - \frac{2}{(x-3)(x^2+1)}$
8. $\frac{1}{(x-4)(x^2+1)} + \frac{-5}{(x-7)(x^2+1)}$
9. $\frac{-5}{(x+1)(x^2+1)} + \frac{-1}{(x+6)(x^2+1)}$
10. $\frac{2}{(x-6)(x^2+1)} + \frac{2}{(x+3)(x^2+1)}$

Chapter 14

Complex Rational Expressions

Learning target

Simplify a large fraction containing smaller fractions.

14.1 Recognize the shape

Fractions appear in the numerator, denominator, or both.

14.2 Procedure

Find the LCD of all small denominators and multiply every term in the large numerator and denominator by it.

Common failure

Multiplying only some terms by the internal LCD. Every term in both the large numerator and large denominator must be multiplied.

Worked example

Problem.
$$\frac{\frac{x}{x-2}}{\frac{x+6}{(x-2)(x+5)}}$$

Diagnosis. The shape is fraction whose numerator or denominator contains fractions. The appropriate tool is: identify the internal LCD and clear all small denominators.

Result.
$$\frac{x^2 + 5x}{x + 6}, \text{ with } x \neq -6, -5, 2.$$

Check. Reverse the clearing step or test an allowed value.

14.3 Mastery exercises

Level I: direct recognition

1.
$$\frac{\frac{1}{x+2} + \frac{1}{x-6}}{\frac{1}{x+2} - \frac{1}{x-6}}$$

2.
$$\frac{\frac{3}{x+6} + \frac{3}{x-5}}{\frac{1}{x+6}}$$

3.
$$\frac{\frac{x}{x-2}}{\frac{x+6}{(x-2)(x+5)}}$$

4. $\frac{\frac{5}{x^2-1} + \frac{1}{x+2}}{\frac{1}{x+2}}$
5. $\frac{\frac{1}{x+1} + \frac{1}{x-5}}{-\frac{1}{x+1} + \frac{1}{x-5}}$
6. $\frac{-\frac{5}{x+3} - \frac{1}{x+1}}{\frac{1}{x+3}}$
7. $\frac{\frac{x}{x+6}}{\frac{x+4}{(x+3)(x+6)}}$
8. $\frac{\frac{5}{x^2-1} + \frac{1}{x+5}}{\frac{1}{x+5}}$
9. $\frac{\frac{1}{x+4} + \frac{1}{x-5}}{-\frac{1}{x+4} + \frac{1}{x-5}}$
10. $\frac{\frac{2}{x-1} + \frac{1}{x-3}}{\frac{1}{x-3}}$

Level II: routine variations

1. $\frac{\frac{x}{x+3}}{\frac{x+4}{(x+3)(x+5)}}$
2. $\frac{-\frac{5}{x^2-1} + \frac{1}{x+3}}{\frac{1}{x+3}}$
3. $\frac{\frac{1}{x+4} + \frac{1}{x-5}}{-\frac{1}{x+4} + \frac{1}{x-5}}$
4. $\frac{\frac{2}{x-3} - \frac{2}{x}}{\frac{1}{x}}$
5. $\frac{\frac{x}{x+3}}{\frac{x-2}{(x-3)(x+3)}}$
6. $\frac{\frac{2}{x^2-1} + \frac{1}{x}}{\frac{1}{x}}$
7. $\frac{\frac{1}{x+3} + \frac{1}{x-5}}{-\frac{1}{x+3} + \frac{1}{x-5}}$
8. $\frac{\frac{2}{x+4} + \frac{4}{x}}{\frac{1}{x+4}}$
9. $\frac{1}{\frac{x-4}{x(x-5)}}$

$$10. \frac{\frac{4}{x^2-1} + \frac{1}{x+1}}{\frac{1}{x+1}}$$

Level III: disguised or prepared forms

$$1. \frac{\frac{1}{x+6} + \frac{1}{x-2}}{\frac{1}{x+6} - \frac{1}{x-2}}$$

$$2. \frac{-\frac{1}{x-2} + \frac{2}{x-4}}{\frac{1}{x-2}}$$

$$3. \frac{\frac{x}{x+2}}{\frac{x+3}{(x+2)(x+4)}}$$

$$4. \frac{-\frac{2}{x^2-1} + \frac{1}{x+1}}{\frac{1}{x+1}}$$

$$5. \frac{\frac{1}{x+5} + \frac{1}{x-3}}{-\frac{1}{x+5} + \frac{1}{x-3}}$$

$$6. \frac{-\frac{5}{x-3} + \frac{5}{x-4}}{\frac{1}{x-3}}$$

$$7. \frac{\frac{x}{x+2}}{\frac{x+5}{(x+2)(x+4)}}$$

$$8. \frac{-\frac{3}{x^2-1} + \frac{1}{x}}{\frac{1}{x}}$$

$$9. \frac{\frac{1}{x+6} + \frac{1}{x-4}}{-\frac{1}{x+6} + \frac{1}{x-4}}$$

$$10. \frac{-\frac{4}{x+6} + \frac{2}{x-3}}{\frac{1}{x-3}}$$

Level IV: multi-stage execution

$$1. \frac{\frac{x}{x+3}}{\frac{x+4}{(x+3)(x+5)}}$$

$$2. \frac{-\frac{3}{x^2-1} + \frac{1}{x-3}}{\frac{1}{x-3}}$$

$$3. \frac{\frac{1}{x+2} + \frac{1}{x+1}}{\frac{1}{x+2} - \frac{1}{x+1}}$$

$$4. \frac{\frac{1}{x-2} + \frac{2}{x}}{\frac{1}{x-2}}$$

$$5. \frac{\frac{x}{x+2}}{\frac{x-6}{(x-7)(x+2)}}$$

$$6. \frac{-\frac{2}{x^2-1} + \frac{1}{x-3}}{\frac{1}{x-3}}$$

$$7. \frac{\frac{1}{x+2} + \frac{1}{x+1}}{-\frac{1}{x+2} + \frac{1}{x+1}}$$

$$8. \frac{-\frac{5}{x+6} - \frac{3}{x}}{\frac{1}{x}}$$

$$9. \frac{\frac{x}{x-5}}{\frac{x-2}{(x-5)(x-3)}}$$

$$10. \frac{-\frac{5}{x^2-1} + \frac{1}{x}}{\frac{1}{x}}$$

Level V: independent mastery

$$1. \frac{\frac{1}{x+2} + \frac{1}{x-6}}{\frac{1}{x+2} - \frac{1}{x-6}}$$

$$2. \frac{-\frac{5}{x-6} + \frac{1}{x}}{\frac{1}{x-6}}$$

$$3. \frac{\frac{x}{x-2}}{\frac{x}{(x-2)(x-1)}}$$

$$4. \frac{\frac{2}{x^2-1} + \frac{1}{x-4}}{\frac{1}{x-4}}$$

$$5. \frac{\frac{1}{x+1} + \frac{1}{x-2}}{-\frac{1}{x+1} + \frac{1}{x-2}}$$

$$6. \frac{\frac{5}{x+5} - \frac{4}{x+1}}{\frac{1}{x+5}}$$

$$7. \frac{\frac{x}{x-2}}{\frac{x-5}{(x-6)(x-2)}}$$

$$8. \frac{\frac{5}{x^2-1} + \frac{1}{x+1}}{\frac{1}{x+1}}$$

$$9. \frac{\frac{1}{x+6} + \frac{1}{x-3}}{\frac{1}{x+6} - \frac{1}{x-3}}$$

$$10. \frac{\frac{1}{x+6} + \frac{2}{x}}{\frac{1}{x}}$$

Part III

Part III: Polynomial Division

Polynomial Division as an Algorithm

Polynomial division is organized bookkeeping. At each stage, choose the next quotient term so that multiplication by the divisor cancels the current leading term. The invariant is

$$P(x) = D(x)Q(x) + R(x), \quad \deg R < \deg D.$$

Synthetic division records the same arithmetic in compressed form when the divisor is $x - c$.

Chapter 15

Polynomial Long Division

Learning target

Find quotient and remainder for arbitrary polynomial divisors.

15.1 Recognize the shape

The divisor is a polynomial and the dividend degree is at least as large.

15.2 Procedure

Arrange descending powers, insert zero placeholders, divide leading terms, multiply, subtract, and continue until $\deg R < \deg D$.

Common failure

Subtracting only the leading term of a product rather than the entire polynomial row.

Worked example

Problem. $(6x^3 - 5x^2 - 8x + 4) \div (-2x + 3)$

Diagnosis. The shape is polynomial dividend and nonzero polynomial divisor. The appropriate tool is: polynomial long division with zero placeholders as needed.

Result. $Q(x) = -3x^2 - 2x + 1$, $R(x) = 1$.

Check. Verify $P = DQ + R$ and $\deg R < \deg D$.

15.3 Mastery exercises

Level I: direct recognition

1. $(-4x^2 - 15x + 4) \div (-x - 4)$
2. $(8x^3 - 14x^2 - 8x) \div (-4x - 1)$
3. $(6x^3 - 5x^2 - 8x + 4) \div (-2x + 3)$
4. $(9x^3 - 4x + 5) \div (-3x + 1)$
5. $(-12x^3 - 4x^2 + 4x - 3) \div (4x)$
6. $(-8x^3 + 18x^2 - 8x + 1) \div (-4x + 1)$
7. $(x^2 + x + 1) \div (-x)$

8. $(-3x^3 + 2x^2 + 2) \div (3x - 2)$
9. $(-6x^3 + 16x^2 - 17x + 10) \div (-3x + 2)$
10. $(-16x^2 + 32x - 16) \div (4x - 4)$

Level II: routine variations

1. $(12x^3 - 5x^2 - 15x + 9) \div (-4x + 3)$
2. $(-3x^4 + 12x^3 - 6x^2 - 10x + 5) \div (-x + 3)$
3. $(12x^3 - 25x^2 + 12) \div (-4x + 3)$
4. $(8x^2 - 16x - 5) \div (-4x)$
5. $(-4x^4 - 7x^3 + x^2 + x - 4) \div (x + 1)$
6. $(4x^4 + 14x^3 - 24x^2 + 4x + 2) \div (4x - 2)$
7. $(-4x^3 + 14x^2 + 4x + 3) \div (-4x - 2)$
8. $(-4x^3 - 18x^2 - 9x + 1) \div (x + 4)$
9. $(x^4 + 2x^3 + 2x^2 + 2) \div (x)$
10. $(9x^2 + 24x + 20) \div (-3x - 4)$

Level III: disguised or prepared forms

1. $(6x^3 + x^2 - 14x - 8) \div (3x + 2)$
2. $(-12x^3 + 6x^2 - 17x + 3) \div (3x^2 + 4)$
3. $(-4x^5 + 15x^4 - 6x^3 - 4x^2 + 4x + 2) \div (-4x^2 + 3x - 1)$
4. $(x^6 - 21x^4 + 24x^2 - 4x) \div (-x^2 - 4x + 2)$
5. $(-8x^4 + 16x^3 + 2x^2 - 12x + 5) \div (-2x + 3)$
6. $(-16x^4 + 25x^2 - 9) \div (4x + 3)$
7. $(-4x^4 + 4x^3 - 20x^2 + 4x - 18) \div (x^2 - x + 4)$
8. $(9x^3 - 3x^2 + 7x - 2) \div (3x^2 + x + 3)$
9. $(-4x^4 - 18x^3 - 26x^2 - 14x - 8) \div (-4x^2 - 2x - 2)$
10. $(3x^2 - 5x - 3) \div (3x - 2)$

Level IV: multi-stage execution

1. $(-6x^6 - 8x^5 - 11x^4 - 15x^3 - 7x^2 - 7x - 2) \div (-3x - 1)$
2. $(-4x^7 - 14x^6 + 9x^5 - 3x^4 - 14x^2 - 9x - 7) \div (4x^2 + 2x + 1)$

3. $(6x^7 - 5x^6 - 5x^5 + 9x^4 + 7x^3 + 3x^2 + x + 1) \div (-2x^2 - x + 1)$
4. $(3x^3 - 7x^2 - 3x + 6) \div (3x + 2)$
5. $(-4x^4 - 5x^3 - 7x^2 + 18x + 7) \div (x^2 + 2x + 4)$
6. $(6x^4 + 3x^3 - 8x^2 + 9x - 2) \div (-3x^2 + 3x - 2)$
7. $(12x^7 + x^6 - 20x^5 + 3x^4 - 8x^3 - 16x^2 + 10x + 15) \div (3x^2 + x - 4)$
8. $(8x^6 - 4x^5 + 2x^4 + 4x^3 + 6x^2 - 8x + 4) \div (-2x)$
9. $(3x^7 - 8x^6 + x^5 + 2x^4 + 8x^3 + 3x^2 - 6x + 5) \div (-x^2 + 2x)$
10. $(9x^4 + 24x^3 + 4x^2 - 14x - 3) \div (-3x^2 - 4x + 4)$

Level V: independent mastery

1. $(-2x^5 + 6x^4 + 14x^3 + 16x^2 + 8x) \div (-x^3 + 4x^2 + 4x)$
2. $(-4x^6 + 12x^5 - x^4 + 6x^3 - 6x^2 - 5x + 3) \div (-x^2 + 3x)$
3. $(-9x^6 - 12x^5 - 3x^4 - 2x^3 + 4x^2 + 8x - 2) \div (3x^3 + 3x^2 + 2x + 3)$
4. $(3x^3 + 12x^2 - 3x + 3) \div (-x^2 - 4x + 1)$
5. $(16x^7 - 8x^6 + 5x^5 - x^4 + 14x^3 + 2x^2 - 5x + 11) \div (4x^3 - x^2 - x + 2)$
6. $(2x^6 - 9x^5 + 8x^4 - 12x^3 - 20x^2 + 13x + 12) \div (x - 4)$
7. $(6x^8 - 6x^7 + x^6 + 3x^5 - 3x^4 - 5x^3 - 5x^2 - 5x + 5) \div (-3x^3 - 2x + 1)$
8. $(6x^7 - 14x^6 + 14x^5 - 11x^4 - 8x^3 + 4x^2 + 7x + 8) \div (3x - 4)$
9. $(2x^7 + 5x^6 - 3x^5 + 12x^4 - 3x^3 - 21x^2 + 2x + 8) \div (-x^3 - 3x^2 + 2x + 2)$
10. $(-3x^8 - 2x^7 + 20x^6 - 9x^5 - 11x^4 + 5x^3 + 5x^2 + 18x + 5) \div (-3x^2 + 4x + 3)$

Chapter 16

Synthetic Division

Learning target

Use the compressed division algorithm for divisors $x - c$.

16.1 Recognize the shape

The divisor is exactly monic linear: $x - c$.

16.2 Procedure

Use c , list every coefficient including zeros, bring down, multiply, add, and interpret the last number as the remainder.

Common failure

Using the visible constant instead of c : the divisor $x + 4$ means $c = -4$.

Worked example

Problem. Divide $-2x^4 + 2x^3 - x^2 + 13x - 2$ by $x - 2$.

Diagnosis. The shape is linear divisor of the form $x - c$. The appropriate tool is: synthetic division using c and every coefficient, including zeros.

Result. $Q(x) = -2x^3 - 2x^2 - 5x + 3$, $R = 4$.

Check. Verify $P = (x - c)Q + R$.

16.3 Mastery exercises

Level I: direct recognition

1. Divide $2x^4 - x^3 - 13x^2 - 5x - 3$ by $x - 3$.
2. Divide $-2x^3 - 12x^2 + 10x - 37$ by $x + 7$.
3. Divide $-2x^4 + 2x^3 - x^2 + 13x - 2$ by $x - 2$.
4. Divide $-4x^4 + 10x^3 - x^2 - x - 1$ by $x - 2$.
5. Divide $-3x^3 - 23x^2 - 35x - 30$ by $x + 6$.
6. Divide $-2x^4 - 3x^3 - 5x^2 - 7x + 4$ by $x + 1$.
7. Divide $-2x^4 + 8x^3 - 6x^2 - 2x - 5$ by $x - 2$.

8. Divide $2x^4 + 4x^3 + 7x^2 + 3x + 3$ by $x + 1$.
9. Divide $-5x^3 - 9x^2 - 5x - 1$ by $x + 1$.
10. Divide $2x^3 + 2x^2 + x - 2$ by $x + 1$.

Level II: routine variations

1. Divide $-x^5 + 4x^3 + 3x^2 - 3x - 5$ by $x - 2$.
2. Divide $4x^4 - 17x^3 + 7x^2 - 16x + 21$ by $x - 4$.
3. Divide $-4x^4 - 19x^3 + 3x^2 - 11x - 5$ by $x + 5$.
4. Divide $-3x^5 - 11x^4 + 16x^3 - 20x^2 - 2x - 11$ by $x + 5$.
5. Divide $-x^3 + 3x^2 + 2x - 15$ by $x - 3$.
6. Divide $x^5 + 6x^4 + 4x^3 - 13x^2 + 13x - 4$ by $x + 4$.
7. Divide $x^4 + 4x^3 - 4x^2 + 9x + 20$ by $x + 5$.
8. Divide $5x^3 - 28x^2 - 10x - 12$ by $x - 6$.
9. Divide $-2x^4 - 10x^3 - 11x^2 + 2x$ by $x + 3$.
10. Divide $-3x^3 + 11x^2 + 9$ by $x - 4$.

Level III: disguised or prepared forms

1. Divide $4x^4 + 26x^3 + 13x^2 + 11x + 30$ by $x + 6$.
2. Divide $4x^5 - 5x^4 - 3x^3 - 2x^2 - 5x - 7$ by $x - 2$.
3. Divide $2x^6 - 11x^5 + 8x^4 + 15x^3 + 5x^2 - 5x + 12$ by $x - 4$.
4. Divide $4x^5 + 13x^4 + 4x^3 + 4x^2 + 2x + 6$ by $x + 3$.
5. Divide $-4x^5 + 18x^4 + 13x^3 - 14x^2 - 6x + 5$ by $x - 5$.
6. Divide $-4x^3 + 3x^2 - 2x - 6$ by $x - 1$.
7. Divide $-x^3 - 4x^2 + 26x + 31$ by $x + 7$.
8. Divide $-2x^4 - 4x^3 + 20x^2 + 15x + 4$ by $x + 4$.
9. Divide $x^5 - 10x^4 + 25x^3 - 2x^2 - 25x + 6$ by $x - 6$.
10. Divide $-3x^6 + 17x^5 + 3x^4 + 19x^3 - x^2 - 29x + 2$ by $x - 6$.

Level IV: multi-stage execution

1. Divide $3x^5 - 24x^4 + 22x^3 - 2x^2 - 40x + 40$ by $x - 7$.
2. Divide $4x^5 - 20x^4 + 20x^3 - 20x^2 + 19x - 11$ by $x - 4$.
3. Divide $-2x^4 + x^3 + 6x^2 - 13x - 10$ by $x + 2$.

4. Divide $4x^4 - 12x^3 + 5x^2 + 10x - 10$ by $x - 2$.
5. Divide $-x^4 - 4x^3 - 2x^2 + 5x + 1$ by $x + 3$.
6. Divide $x^5 + 6x^4 + 4x^3 - 5x^2 + 3x - 9$ by $x + 2$.
7. Divide $-2x^5 - 6x^4 + 2x^3 + 6x^2 + 5x + 15$ by $x + 3$.
8. Divide $-5x^7 - 21x^6 + 21x^5 + 4x^4 - 2x^3 + 19x^2 + 15x - 22$ by $x + 5$.
9. Divide $x^6 + x^5 - 25x^4 + 19x^3 + 9x^2 - 18x - 3$ by $x - 4$.
10. Divide $-4x^7 + x^6 + 10x^5 + 8x^4 + 3x^3 + 4x^2 - 8$ by $x + 1$.

Level V: independent mastery

1. Divide $-4x^3 + 28x^2 - 25x + 6$ by $x - 6$.
2. Divide $-4x^3 + 25x^2 - 23x - 16$ by $x - 5$.
3. Divide $-3x^5 - 19x^4 + 12x^3 - 11x^2 + 23x + 22$ by $x + 7$.
4. Divide $2x^3 - 5x^2 + 6x - 8$ by $x - 1$.
5. Divide $-5x^5 - 22x^4 + 13x^3 - 5x^2 + 25x$ by $x + 5$.
6. Divide $-4x^4 + 5x^3 + 11x^2 - 6x - 10$ by $x - 2$.
7. Divide $5x^5 - 16x^4 + 4x^3 + 2x^2 - 16x + 4$ by $x - 3$.
8. Divide $3x^6 - 11x^5 + 9x^4 + 2x^3 + 5x^2 - 12x + 2$ by $x - 2$.
9. Divide $5x^7 - 38x^6 + 18x^5 + 26x^4 - 37x^3 + 12x^2 + 18x - 28$ by $x - 7$.
10. Divide $-2x^3 - 4x^2 + 2x + 2$ by $x + 2$.

Chapter 17

Choosing Long or Synthetic Division

Learning target

Select the correct polynomial division method.

17.1 Recognize the shape

Inspect the divisor before calculating.

17.2 Procedure

Ordinary synthetic division applies to $x - c$; polynomial long division always applies to a nonzero polynomial divisor.

Common failure

Trying ordinary synthetic division on a divisor that is not exactly of the form $x - c$.

Worked example

Problem. Choose the ordinary method and divide $(3x^4 - 11x^3 + 14x^2 - 7x + 19)$ by $(x^2 - 4x + 5)$.

Diagnosis. The shape is inspect the divisor. The appropriate tool is: use synthetic division only for $x - c$; otherwise use long division.

Result. polynomial long division; $Q(x) = 3x^2 + x + 3$, $R(x) = 4$.

Check. Verify $P = DQ + R$.

17.3 Mastery exercises

Level I: direct recognition

1. Choose the ordinary method and divide $(-3x^2 + 7x - 4)$ by $(x - 2)$.
2. Choose the ordinary method and divide $(-16x^3 + 16x^2 + 5x - 4)$ by $(4x - 1)$.
3. Choose the ordinary method and divide $(3x^4 - 11x^3 + 14x^2 - 7x + 19)$ by $(x^2 - 4x + 5)$.
4. Choose the ordinary method and divide $(-x^3 + 14x - 11)$ by $(x + 4)$.
5. Choose the ordinary method and divide $(6x^2 - 11x - 6)$ by $(2x - 5)$.
6. Choose the ordinary method and divide $(-3x^4 + 16x^3 - 26x^2 + 16x - 2)$ by $(x^2 - 4x + 3)$.
7. Choose the ordinary method and divide $(-x^3 + 2x^2 - 4)$ by $(x + 1)$.

8. Choose the ordinary method and divide $(-4x^2 - x - 1)$ by $(4x + 1)$.
9. Choose the ordinary method and divide $(-3x^3 - x^2 + x - 6)$ by $(x^2 - x + 2)$.
10. Choose the ordinary method and divide $(2x^4 - 2x^3 - 13x^2 + 3x - 1)$ by $(x - 3)$.

Level II: routine variations

1. Choose the ordinary method and divide $(-4x^5 + 4x^4 + 8x^3 - 6x^2 - 8x + 9)$ by $(2x - 2)$.
2. Choose the ordinary method and divide $(4x^3 + 7x^2 + 2x - 5)$ by $(x^2 + 2x + 1)$.
3. Choose the ordinary method and divide $(-2x^4 + 2x^2 - 2x - 3)$ by $(x + 1)$.
4. Choose the ordinary method and divide $(-9x^3 - 21x^2 + 3x + 5)$ by $(3x + 7)$.
5. Choose the ordinary method and divide $(-3x^5 + 10x^4 - 16x^3 + 10x^2 + 12x + 6)$ by $(x^2 - 2x + 4)$.
6. Choose the ordinary method and divide $(4x^3 + 18x^2 + 5x - 12)$ by $(x + 4)$.
7. Choose the ordinary method and divide $(8x^4 + 6x^3 - 12x^2 + 6x - 16)$ by $(4x - 5)$.
8. Choose the ordinary method and divide $(3x^5 + 9x^3 - 8x^2 + 3x + 20)$ by $(x^2 + x + 4)$.
9. Choose the ordinary method and divide $(2x^2 - 8x - 11)$ by $(x - 5)$.
10. Choose the ordinary method and divide $(16x^5 - 8x^4 - 7x^3 + 19x^2 - 28x + 13)$ by $(4x - 3)$.

Level III: disguised or prepared forms

1. Choose the ordinary method and divide $(-x^5 + 2x^4 - 7x^3 + 14x^2 - 24x - 2)$ by $(x^2 + x + 6)$.
2. Choose the ordinary method and divide $(x^6 + 7x^5 + 7x^4 + 5x^3 - 5x^2 + 8x + 14)$ by $(x + 6)$.
3. Choose the ordinary method and divide $(6x^3 - 13x^2 + 8x - 6)$ by $(2x - 3)$.
4. Choose the ordinary method and divide $(-x^4 + 4x^3 - 4x^2 + 5x + 5)$ by $(x^2 - x + 2)$.
5. Choose the ordinary method and divide $(x^5 - 3x^4 - 12x^3 + 6x^2 + 22x - 6)$ by $(x - 5)$.
6. Choose the ordinary method and divide $(8x^5 - 34x^4 + 13x^3 + 28x^2 - 6x + 22)$ by $(2x - 7)$.
7. Choose the ordinary method and divide $(4x^7 + 2x^5 + x^3 - 4x^2 - 3x - 2)$ by $(x^2 + x + 1)$.
8. Choose the ordinary method and divide $(-x^3 + 3x^2 + 8x + 5)$ by $(x + 1)$.
9. Choose the ordinary method and divide $(-4x^2 + 8x + 1)$ by $(2x - 5)$.
10. Choose the ordinary method and divide $(x^3 - 2x^2 + 3x - 2)$ by $(x^2 - x + 2)$.

Level IV: multi-stage execution

1. Choose the ordinary method and divide $(4x^5 - 8x^4 - 15x^3 + 10x^2 - 5)$ by $(x - 3)$.
2. Choose the ordinary method and divide $(-6x^5 - 11x^4 - 4x^3 + 8x^2 + 10x + 3)$ by $(2x + 1)$.
3. Choose the ordinary method and divide $(-x^7 + 6x^6 - 11x^5 + 4x^4 + 8x^3 - 10x^2 - 4x + 4)$ by $(x^2 - 3x + 1)$.

4. Choose the ordinary method and divide $(-x^5 - 4x^4 + 15x^3 + 16x^2 - 15x - 17)$ by $(x + 6)$.
5. Choose the ordinary method and divide $(-16x^2 - 20x + 13)$ by $(4x + 7)$.
6. Choose the ordinary method and divide $(-2x^7 + 4x^6 - 2x^5 + 5x^4 - 2x^3 + 5x^2 - 4x + 1)$ by $(x^2 + 1)$.
7. Choose the ordinary method and divide $(2x^6 + 7x^5 + 5x^4 + 8x^3 + 2x^2 - 15x - 13)$ by $(x + 3)$.
8. Choose the ordinary method and divide $(4x^4 - 20x^3 + 27x^2 - 15x - 25)$ by $(2x - 7)$.
9. Choose the ordinary method and divide $(-4x^6 + 13x^5 - 19x^4 - 9x^3 + 22x^2 + 2x - 20)$ by $(x^2 - 3x + 5)$.
10. Choose the ordinary method and divide $(x^6 - 2x^5 - 24x^4 - 2x^3 + 16x^2 - 20x - 27)$ by $(x - 6)$.

Level V: independent mastery

1. Choose the ordinary method and divide $(-8x^6 - 8x^5 + 22x^4 - 24x^3 + 24x^2 - 12x)$ by $(4x - 2)$.
2. Choose the ordinary method and divide $(3x^3 - 4x^2 + 16x - 5)$ by $(x^2 - x + 5)$.
3. Choose the ordinary method and divide $(-x^7 - 5x^6 - 2x^5 + 6x^4 - 7x^3 - 18x - 6)$ by $(x + 4)$.
4. Choose the ordinary method and divide $(-4x^6 + 13x^5 + 24x^4 + x^3 - 14x^2 - 10x - 4)$ by $(4x + 3)$.
5. Choose the ordinary method and divide $(-x^5 + 6x^4 - 14x^3 + 18x^2 - 15x + 4)$ by $(x^2 - 3x + 2)$.
6. Choose the ordinary method and divide $(3x^5 - 8x^4 - 19x^3 + 14x^2 - 4x - 17)$ by $(x - 4)$.
7. Choose the ordinary method and divide $(-6x^5 - 11x^4 + x^3 - 11x^2 + x - 14)$ by $(3x + 7)$.
8. Choose the ordinary method and divide $(2x^7 + 4x^6 - 2x^5 + 15x^3 + 19x^2 + 11x + 3)$ by $(x^2 + 3x + 2)$.
9. Choose the ordinary method and divide $(-2x^8 + 6x^7 + 23x^6 - 19x^5 + 24x^4 - 16x^3 - 19x^2 - 4x - 8)$ by $(x - 5)$.
10. Choose the ordinary method and divide $(2x^6 + 9x^5 + x^4 - 12x^3 + 6x^2 + 5x - 10)$ by $(2x + 3)$.

Part IV

Part IV: Partial-Fraction Decomposition

Partial Fractions: The Master Setup

Partial-fraction decomposition reverses the addition of rational expressions. The denominator shape determines the template; the numerator determines the coefficients.

Prerequisite checklist

Before solving for any coefficient: (1) divide if the fraction is improper; (2) factor the denominator completely; (3) classify every factor and multiplicity; (4) write the complete template.

Distinct linear factors and cover-up

For

$$\frac{5x + 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2},$$

multiply by the denominator:

$$5x + 1 = A(x + 2) + B(x - 1).$$

Set $x = 1$ to obtain $6 = 3A$, so $A = 2$. Set $x = -2$ to obtain $-9 = -3B$, so $B = 3$. Therefore

$$\frac{5x + 1}{(x - 1)(x + 2)} = \frac{2}{x - 1} + \frac{3}{x + 2}.$$

Recombining the right side verifies the result.

Repeated linear factor

A denominator $(x - 1)^2(x + 2)$ requires

$$\frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{C}{x + 2}.$$

Both powers of the repeated factor must appear. Root substitution isolates B and C ; a convenient extra value or coefficient comparison determines A .

Irreducible quadratic factor

A denominator $(x - 1)(x^2 + 4)$ requires

$$\frac{A}{x - 1} + \frac{Bx + C}{x^2 + 4}.$$

The numerator over the quadratic is linear because its degree must be strictly smaller than the quadratic degree.

Chapter 18

Distinct Linear Factors

Learning target

Write and compute a partial-fraction decomposition with distinct linear factors.

18.1 Recognize the shape

A proper rational expression whose denominator factors as distinct $x - r$.

18.2 Procedure

Place one constant numerator over each linear factor, clear denominators, and determine the constants.

Common failure

Writing a template before making the fraction proper and factoring the denominator completely.

Worked example

Problem. $\frac{-2x - 46}{(x + 2)(x + 8)}$

Diagnosis. The shape is proper rational expression with distinct linear denominator factors. The appropriate tool is: write the distinct-linear template and determine its constants.

Result. $\frac{5}{x + 8} - \frac{7}{x + 2}$.

Check. Recombine the simple fractions.

18.3 Mastery exercises

Level I: direct recognition

1. $\frac{6x + 54}{(x - 5)(x + 7)}$

2. $\frac{-3x + 57}{(x - 8)(x + 3)}$

3. $\frac{-2x - 46}{(x + 2)(x + 8)}$

4. $\frac{7x + 10}{(x - 5)(x + 4)}$

5. $\frac{-5x}{(x - 7)(x - 2)}$

6. $\frac{-4x + 42}{(x - 8)(x + 2)}$
7. $\frac{-x - 62}{(x - 1)(x + 8)}$
8. $\frac{-2x + 18}{(x - 6)(x - 5)}$
9. $\frac{8x + 14}{x(x + 7)}$
10. $\frac{-84}{(x - 6)(x + 8)}$

Level II: routine variations

1. $\frac{3x + 28}{(x + 4)(x + 6)}$
2. $\frac{-7x - 9}{(x - 8)(x + 5)}$
3. $\frac{9x + 18}{(x - 6)(x + 3)}$
4. $\frac{-3x - 21}{(x - 7)(x - 1)}$
5. $\frac{-10x + 14}{(x - 2)(x - 1)}$
6. $\frac{-6x - 30}{x(x + 6)}$
7. $\frac{2x + 19}{(x + 2)(x + 5)}$
8. $\frac{15x + 98}{(x + 6)(x + 7)}$
9. $\frac{x + 20}{(x - 1)(x + 2)}$
10. $\frac{-2x + 18}{(x - 5)(x - 3)}$

Level III: disguised or prepared forms

1. $\frac{-6x^2 - 15x + 89}{(x + 1)(x + 3)(x + 8)}$
2. $\frac{-5x^2 - 53x - 134}{(x + 4)(x + 5)(x + 6)}$
3. $\frac{4x^2 - 78x - 236}{(x - 6)(x + 2)(x + 4)}$

4. $\frac{7x^2 + 23x - 134}{(x - 4)(x - 2)(x + 3)}$
5. $\frac{8x^2 + 20x + 24}{x(x + 1)(x + 4)}$
6. $\frac{-12x^2 + 30x - 10}{x(x - 5)(x - 1)}$
7. $\frac{-5x^2 - 36x - 87}{(x - 3)(x + 2)(x + 3)}$
8. $\frac{4x^2 - 3x + 5}{x(x + 1)(x + 5)}$
9. $\frac{x^2 - 16x - 9}{(x - 5)(x - 3)(x + 3)}$
10. $\frac{-9x^2 - 31x - 76}{(x - 3)(x + 2)(x + 7)}$

Level IV: multi-stage execution

1. $\frac{-10x^2 - 81x - 113}{(x - 2)(x + 5)(x + 7)}$
2. $\frac{-10x + 150}{x(x - 5)(x + 5)}$
3. $\frac{15x^2 - 74x - 200}{(x - 8)(x - 2)(x + 6)}$
4. $\frac{-13x - 35}{x(x - 1)(x + 5)}$
5. $\frac{-10x^2 + 57x - 84}{x(x - 4)(x - 3)}$
6. $\frac{-8x^2 + 7x + 44}{(x - 4)(x - 3)(x + 4)}$
7. $\frac{15x^2 + 53x - 16}{(x - 2)(x + 3)(x + 4)}$
8. $\frac{-2x^2 - 38x - 240}{x(x + 6)(x + 8)}$
9. $\frac{-5x^2 - 61x - 174}{(x + 5)(x + 7)(x + 8)}$
10. $\frac{19x^2 - 117x - 60}{(x - 8)(x - 3)(x + 3)}$

Level V: independent mastery

1. $\frac{x^3 + 18x^2 - 53x - 470}{(x-4)(x-1)(x+6)(x+7)}$
2. $\frac{-13x^3 + 138x^2 - 208x - 1024}{(x-8)(x-6)(x-4)(x+8)}$
3. $\frac{-x^3 - 2x^2 + 229x - 646}{(x-8)(x-6)(x-1)(x+5)}$
4. $\frac{7x^3 + 103x^2 - 692x - 1824}{(x-8)(x-6)(x+3)(x+8)}$
5. $\frac{-3x^3 + 37x^2 - 128x + 120}{x(x-6)(x-5)(x-4)}$
6. $\frac{4x^3 - 78x^2 - 570x - 56}{x(x-8)(x+1)(x+7)}$
7. $\frac{x^3 - 96x^2 + 615x - 868}{(x-7)(x-4)(x-3)(x+4)}$
8. $\frac{5x^3 - 112x^2 + 527x - 280}{x(x-8)(x-5)(x-1)}$
9. $\frac{25x^3 + 246x^2 + 387x - 338}{(x-2)(x+1)(x+7)(x+8)}$
10. $\frac{-8x^3 - 138x^2 - 222x + 260}{(x-7)(x+2)(x+3)(x+8)}$

Chapter 19

Heaviside Cover-Up Method

Learning target

Rapidly isolate coefficients attached to distinct linear factors.

19.1 Recognize the shape

A proper rational expression with distinct linear factors and no repeated power for the coefficient being covered.

19.2 Procedure

For the coefficient over $x - r$, multiply by $x - r$ and substitute $x = r$; then verify the complete decomposition.

Limit of ordinary cover-up

Cover-up immediately isolates coefficients attached to distinct linear factors. Repeated factors and irreducible quadratics require additional substitution, coefficient comparison, or an advanced derivative formula. This workbook uses the elementary methods first.

Common failure

Using ordinary cover-up as though it determined every coefficient of a repeated factor.

Worked example

Problem. $\frac{8x + 37}{(x - 1)(x + 8)}$

Diagnosis. The shape is proper rational expression with distinct linear denominator factors. The appropriate tool is: use the Heaviside cover-up method, then verify.

Result. $\frac{3}{x + 8} + \frac{5}{x - 1}$.

Check. Recombine after using cover-up.

19.3 Mastery exercises

Level I: direct recognition

1. $\frac{-2x - 24}{(x - 3)(x + 2)}$

2. $\frac{4x - 22}{(x - 1)(x + 8)}$

3. $\frac{8x + 37}{(x - 1)(x + 8)}$

4. $\frac{-5x - 49}{(x - 1)(x + 8)}$

5. $\frac{5x + 35}{x(x + 5)}$

6. $\frac{-7x - 4}{(x - 3)(x + 2)}$

7. $\frac{-2x + 74}{(x - 7)(x + 8)}$

8. $\frac{7x - 3}{x(x - 3)}$

9. $\frac{-6x + 42}{(x - 4)(x + 5)}$

10. $\frac{-9x + 10}{x(x - 2)}$

Level II: routine variations

1. $\frac{5x + 20}{(x - 4)(x + 1)}$

2. $\frac{-5x + 42}{x(x - 7)}$

3. $\frac{-2x + 15}{x(x + 5)}$

4. $\frac{5}{x(x + 5)}$

5. $\frac{-2x + 21}{(x - 6)(x + 3)}$

6. $\frac{12}{(x - 8)(x - 4)}$

7. $\frac{6x - 57}{(x - 8)(x + 1)}$

8. $\frac{-x + 19}{(x + 1)(x + 6)}$

9. $\frac{-7x + 3}{(x - 3)(x + 6)}$

10. $\frac{-10x - 28}{(x + 1)(x + 7)}$

Level III: disguised or prepared forms

1. $\frac{10x^2 + 106x + 256}{(x+1)(x+5)(x+6)}$
2. $\frac{5x^2 - 8x - 312}{(x-1)(x+6)(x+8)}$
3. $\frac{3x^2 - 63x - 192}{(x-5)(x+1)(x+4)}$
4. $\frac{-3x^2 + 14x + 56}{(x-8)(x-4)(x-2)}$
5. $\frac{-9x^2 - 71x + 500}{(x-5)(x-4)(x+5)}$
6. $\frac{-19x^2 - 182x - 384}{(x+2)(x+4)(x+8)}$
7. $\frac{3x^2 + 31x - 40}{x(x+5)(x+8)}$
8. $\frac{-3x^2 - 15x + 12}{x(x-2)(x+3)}$
9. $\frac{-8x^2 + 54x + 74}{(x-7)(x-1)(x+3)}$
10. $\frac{13x^2 + 109x + 66}{(x-5)(x+7)(x+8)}$

Level IV: multi-stage execution

1. $\frac{3x^2 + 2x + 107}{(x-5)(x-1)(x+3)}$
2. $\frac{24x + 52}{(x+1)(x+3)(x+8)}$
3. $\frac{5x^2 + 45x - 140}{x(x-7)(x+5)}$
4. $\frac{-4x^2 + 53x - 85}{(x-7)(x-2)(x-1)}$
5. $\frac{6x^2 - 5x - 64}{x(x-1)(x+8)}$
6. $\frac{3x^2 - 82x + 202}{(x-7)(x-2)(x+8)}$
7. $\frac{5x^2 - 21x - 80}{x(x-8)(x-5)}$

8. $\frac{-14x + 40}{x(x-8)(x-5)}$
9. $\frac{14x^2 - 53x + 8}{(x-4)(x-2)(x+1)}$
10. $\frac{-2x^2 + 11x + 40}{(x-2)(x+4)(x+7)}$

Level V: independent mastery

1. $\frac{9x^3 - 138x^2 + 416x - 256}{x(x-8)(x-4)(x-2)}$
2. $\frac{6x^3 - 7x^2 - 119x - 1168}{(x-7)(x-5)(x+4)(x+6)}$
3. $\frac{-2x^3 - 64x^2 - 544x - 1448}{(x-8)(x+5)(x+6)(x+7)}$
4. $\frac{19x^3 - 55x^2 - 508x - 296}{(x-8)(x-1)(x+3)(x+4)}$
5. $\frac{14x^3 - 126x^2 + 218x - 26}{(x-5)(x-3)(x-1)(x+1)}$
6. $\frac{-19x^3 + 56x^2 + 824x - 3360}{(x-7)(x-6)(x+6)(x+8)}$
7. $\frac{3x^3 + 50x^2 - 209x - 456}{(x-7)(x-4)(x+1)(x+6)}$
8. $\frac{-2x^3 + 58x^2 + 34x - 602}{(x-5)(x-1)(x+3)(x+7)}$
9. $\frac{-19x^3 + 107x^2 - 52x - 336}{(x-6)(x-3)(x-1)(x+4)}$
10. $\frac{-x^3 + 5x^2 + 203x - 147}{x(x-7)(x-3)(x+7)}$

Chapter 20

Strategic Substitution

Learning target

Choose values that eliminate many unknown terms.

20.1 Recognize the shape

After clearing denominators, special values make factors zero or simplify the identity.

20.2 Procedure

Use roots first, then convenient values such as 0, 1, -1 when they reduce the remaining system.

Common failure

Substituting into the original fractional equation before first clearing denominators and forming a polynomial identity.

Worked example

Problem. Determine the constants by strategic substitution: $\frac{11x + 20}{(x - 2)(x + 5)} = \frac{A}{x + 5} + \frac{B}{x - 2}$.

Diagnosis. The shape is coefficient identity with values that make terms vanish. The appropriate tool is: clear denominators, then choose roots and other convenient values.

Result. $\frac{5}{x + 5} + \frac{6}{x - 2}$.

Check. Substitute the coefficients into the cleared identity.

20.3 Mastery exercises

Level I: direct recognition

1. Determine the constants by strategic substitution: $\frac{-7x + 24}{(x - 5)(x + 6)} = \frac{A}{x + 6} + \frac{B}{x - 5}$.
2. Determine the constants by strategic substitution: $\frac{-3x + 26}{(x - 7)(x - 2)} = \frac{A}{x - 2} + \frac{B}{x - 7}$.
3. Determine the constants by strategic substitution: $\frac{11x + 20}{(x - 2)(x + 5)} = \frac{A}{x + 5} + \frac{B}{x - 2}$.
4. Determine the constants by strategic substitution: $\frac{-4x - 10}{(x + 4)(x + 7)} = \frac{A}{x + 7} + \frac{B}{x + 4}$.

5. Determine the constants by strategic substitution: $\frac{3x-21}{x(x+7)} = \frac{A}{x} + \frac{B}{x+7}$.
6. Determine the constants by strategic substitution: $\frac{-5x+27}{(x-3)(x+3)} = \frac{A}{x+3} + \frac{B}{x-3}$.
7. Determine the constants by strategic substitution: $\frac{-3x+6}{(x-6)(x-4)} = \frac{A}{x-4} + \frac{B}{x-6}$.
8. Determine the constants by strategic substitution: $\frac{-10}{(x-7)(x+3)} = \frac{A}{x-7} + \frac{B}{x+3}$.
9. Determine the constants by strategic substitution: $\frac{8x-6}{(x-6)(x+1)} = \frac{A}{x-6} + \frac{B}{x+1}$.
10. Determine the constants by strategic substitution: $\frac{-6x-6}{(x-1)(x+2)} = \frac{A}{x+2} + \frac{B}{x-1}$.

Level II: routine variations

1. Determine the constants by strategic substitution: $\frac{11x-38}{(x-4)(x-3)} = \frac{A}{x-4} + \frac{B}{x-3}$.
2. Determine the constants by strategic substitution: $\frac{-4x-22}{(x-5)(x+2)} = \frac{A}{x+2} + \frac{B}{x-5}$.
3. Determine the constants by strategic substitution: $\frac{3x-27}{(x-6)(x+3)} = \frac{A}{x+3} + \frac{B}{x-6}$.
4. Determine the constants by strategic substitution: $\frac{7x-4}{(x-1)(x+2)} = \frac{A}{x+2} + \frac{B}{x-1}$.
5. Determine the constants by strategic substitution: $\frac{-2x-44}{(x-2)(x+6)} = \frac{A}{x-2} + \frac{B}{x+6}$.
6. Determine the constants by strategic substitution: $\frac{-7x-20}{(x-5)(x+6)} = \frac{A}{x-5} + \frac{B}{x+6}$.
7. Determine the constants by strategic substitution: $\frac{x-2}{x(x+2)} = \frac{A}{x+2} + \frac{B}{x}$.
8. Determine the constants by strategic substitution: $\frac{2x+16}{(x-7)(x-1)} = \frac{A}{x-1} + \frac{B}{x-7}$.
9. Determine the constants by strategic substitution: $\frac{4x+22}{(x-2)(x+3)} = \frac{A}{x-2} + \frac{B}{x+3}$.
10. Determine the constants by strategic substitution: $\frac{-10x+45}{(x-6)(x-3)} = \frac{A}{x-6} + \frac{B}{x-3}$.

Level III: disguised or prepared forms

1. Determine the constants by strategic substitution: $\frac{-7x+12}{x(x-6)(x-1)} = \frac{A}{x-1} + \frac{B}{x} + \frac{C}{x-6}$.

2. Determine the constants by strategic substitution: $\frac{x^2 + 19x - 160}{(x-7)(x-6)(x+4)} = \frac{A}{x+4} + \frac{B}{x-6} + \frac{C}{x-7}$.
3. Determine the constants by strategic substitution: $\frac{13x^2 - 63x - 144}{(x-7)(x-3)(x+6)} = \frac{A}{x+6} + \frac{B}{x-3} + \frac{C}{x-7}$.
4. Determine the constants by strategic substitution: $\frac{4x^2 - 40x + 4}{(x-5)(x-1)(x+3)} = \frac{A}{x-5} + \frac{B}{x+3} + \frac{C}{x-1}$.
5. Determine the constants by strategic substitution: $\frac{-12x^2 - 96x - 190}{(x+3)(x+4)(x+5)} = \frac{A}{x+3} + \frac{B}{x+5} + \frac{C}{x+4}$.
6. Determine the constants by strategic substitution: $\frac{-2x^2 - 14x + 172}{(x-6)(x-4)(x+2)} = \frac{A}{x-4} + \frac{B}{x-6} + \frac{C}{x+2}$.
7. Determine the constants by strategic substitution: $\frac{11x^2 + 22x - 15}{x(x-1)(x+5)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+5}$.
8. Determine the constants by strategic substitution: $\frac{-11x^2 - 61x - 48}{(x-1)(x+4)(x+5)} = \frac{A}{x-1} + \frac{B}{x+5} + \frac{C}{x+4}$.
9. Determine the constants by strategic substitution: $\frac{5x^2 - 2x + 9}{(x-7)(x+1)(x+3)} = \frac{A}{x+3} + \frac{B}{x+1} + \frac{C}{x-7}$.
10. Determine the constants by strategic substitution: $\frac{-5x^2 + 47x + 48}{(x-7)(x-1)(x+4)} = \frac{A}{x-7} + \frac{B}{x+4} + \frac{C}{x-1}$.

Level IV: multi-stage execution

1. Determine the constants by strategic substitution: $\frac{-11x^2 - 7x + 72}{(x-3)(x-2)(x+5)} = \frac{A}{x+5} + \frac{B}{x-2} + \frac{C}{x-3}$.
2. Determine the constants by strategic substitution: $\frac{x^2 - 34x - 15}{x(x-5)(x+3)} = \frac{A}{x-5} + \frac{B}{x+3} + \frac{C}{x}$.
3. Determine the constants by strategic substitution: $\frac{-5x^2 - 29x + 18}{(x-5)(x+2)(x+4)} = \frac{A}{x-5} + \frac{B}{x+4} + \frac{C}{x+2}$.
4. Determine the constants by strategic substitution: $\frac{-10x^2 + 81x - 108}{x(x-6)(x-3)} = \frac{A}{x-3} + \frac{B}{x-6} + \frac{C}{x}$.
5. Determine the constants by strategic substitution: $\frac{-3x^2 - 22x - 21}{(x+3)(x+5)(x+6)} = \frac{A}{x+3} + \frac{B}{x+6} + \frac{C}{x+5}$.
6. Determine the constants by strategic substitution: $\frac{2x^2 - 4x + 6}{x(x+1)(x+3)} = \frac{A}{x+3} + \frac{B}{x+1} + \frac{C}{x}$.
7. Determine the constants by strategic substitution: $\frac{20x + 50}{(x-1)(x+4)(x+6)} = \frac{A}{x-1} + \frac{B}{x+4} + \frac{C}{x+6}$.
8. Determine the constants by strategic substitution: $\frac{3x^2 - 4x - 36}{x(x-4)(x-3)} = \frac{A}{x-4} + \frac{B}{x} + \frac{C}{x-3}$.
9. Determine the constants by strategic substitution: $\frac{-6x^2 - 4x + 106}{(x-1)(x+2)(x+7)} = \frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{x+7}$.

10. Determine the constants by strategic substitution: $\frac{-6x^2 - 7x + 3}{x(x-3)(x+1)} = \frac{A}{x} + \frac{B}{x-3} + \frac{C}{x+1}$.

Level V: independent mastery

1. Determine the constants by strategic substitution: $\frac{7x^2 - 77x - 84}{x(x-7)(x+3)} = \frac{A}{x-7} + \frac{B}{x+3} + \frac{C}{x}$.

2. Determine the constants by strategic substitution: $\frac{-4x^2 + 51x + 45}{x(x-5)(x+3)} = \frac{A}{x-5} + \frac{B}{x+3} + \frac{C}{x}$.

3. Determine the constants by strategic substitution: $\frac{-9x^2 + 40x - 12}{x(x-6)(x-2)} = \frac{A}{x-2} + \frac{B}{x} + \frac{C}{x-6}$.

4. Determine the constants by strategic substitution: $\frac{-7x^2 - 3x + 76}{(x-2)(x+1)(x+5)} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{x+5}$.

5. Determine the constants by strategic substitution: $\frac{-4x^2 - 5x + 129}{(x-6)(x-1)(x+3)} = \frac{A}{x-6} + \frac{B}{x-1} + \frac{C}{x+3}$.

6. Determine the constants by strategic substitution: $\frac{10x^2 - 111x + 293}{(x-7)(x-5)(x-4)} = \frac{A}{x-4} + \frac{B}{x-7} + \frac{C}{x-5}$.

7. Determine the constants by strategic substitution: $\frac{-x^2 + 26x - 97}{(x-7)(x-4)(x-3)} = \frac{A}{x-3} + \frac{B}{x-4} + \frac{C}{x-7}$.

8. Determine the constants by strategic substitution: $\frac{11x^2 - 65x - 4}{(x-7)(x-2)(x+1)} = \frac{A}{x-7} + \frac{B}{x-2} + \frac{C}{x+1}$.

9. Determine the constants by strategic substitution: $\frac{-69x + 366}{(x-7)(x-4)(x+6)} = \frac{A}{x+6} + \frac{B}{x-7} + \frac{C}{x-4}$.

10. Determine the constants by strategic substitution: $\frac{-24x - 156}{(x-4)(x+3)(x+5)} = \frac{A}{x+5} + \frac{B}{x+3} + \frac{C}{x-4}$.

Chapter 21

Equating Coefficients

Learning target

Determine constants from a polynomial identity.

21.1 Recognize the shape

Clearing denominators produces polynomials equal for every allowed x .

21.2 Procedure

Expand, collect like powers, equate matching coefficients, and solve the resulting linear system.

Common failure

Equating unlike powers or overlooking a missing power whose coefficient is zero.

Worked example

Problem. Use coefficient comparison to decompose $\frac{-6x + 12}{x(x - 3)}$.

Diagnosis. The shape is polynomial identity after clearing denominators. The appropriate tool is: expand both sides and equate coefficients of equal powers.

Result. $-\frac{2}{x - 3} - \frac{4}{x}$.

Check. Compare the expanded coefficients on both sides.

21.3 Mastery exercises

Level I: direct recognition

1. Use coefficient comparison to decompose $\frac{7x + 4}{(x - 4)(x + 4)}$.
2. Use coefficient comparison to decompose $\frac{-6}{(x - 5)(x + 1)}$.
3. Use coefficient comparison to decompose $\frac{-6x + 12}{x(x - 3)}$.
4. Use coefficient comparison to decompose $\frac{-2x + 18}{x(x - 6)}$.
5. Use coefficient comparison to decompose $\frac{-8x + 32}{(x - 5)(x - 1)}$.

6. Use coefficient comparison to decompose $\frac{-x-22}{(x-5)(x+4)}$.
7. Use coefficient comparison to decompose $\frac{-9x}{(x-4)(x+5)}$.
8. Use coefficient comparison to decompose $\frac{-4x-13}{(x+2)(x+3)}$.
9. Use coefficient comparison to decompose $\frac{-4x-36}{x(x+6)}$.
10. Use coefficient comparison to decompose $\frac{-8x+27}{(x-6)(x+1)}$.

Level II: routine variations

1. Use coefficient comparison to decompose $\frac{5x-15}{(x-6)(x-1)}$.
2. Use coefficient comparison to decompose $\frac{-x-3}{x(x+1)}$.
3. Use coefficient comparison to decompose $\frac{2x-20}{(x-4)(x-1)}$.
4. Use coefficient comparison to decompose $\frac{-10x+48}{(x-6)(x-3)}$.
5. Use coefficient comparison to decompose $\frac{7x-17}{(x-3)(x-1)}$.
6. Use coefficient comparison to decompose $\frac{5x-15}{(x-6)(x-1)}$.
7. Use coefficient comparison to decompose $\frac{8x+30}{x(x+5)}$.
8. Use coefficient comparison to decompose $\frac{-11x-25}{x(x+5)}$.
9. Use coefficient comparison to decompose $\frac{-4x-14}{(x-4)(x+6)}$.
10. Use coefficient comparison to decompose $\frac{5x+8}{(x+2)(x+4)}$.

Level III: disguised or prepared forms

1. Use coefficient comparison to decompose $\frac{9x^2+39x+37}{(x+4)(x^2+4x+5)}$.
2. Use coefficient comparison to decompose $\frac{6x^2+12x+4}{(x-6)(x^2+5x+7)}$.
3. Use coefficient comparison to decompose $\frac{-2x^2+4x-7}{(x-1)(x^2+4)}$.

4. Use coefficient comparison to decompose $\frac{-4x^2 + 17x - 35}{(x - 5)(x^2 - 4x + 5)}$.
5. Use coefficient comparison to decompose $\frac{-9x^2 + 11x - 14}{(x - 1)(x^2 - x + 3)}$.
6. Use coefficient comparison to decompose $\frac{x^2 + 7x - 3}{(x + 5)(x^2 + 4x + 8)}$.
7. Use coefficient comparison to decompose $\frac{4x^2 + 5x - 2}{(x + 4)(x^2 + 2x + 6)}$.
8. Use coefficient comparison to decompose $\frac{-x^2 - 27x - 18}{(x - 6)(x^2 + 2x + 6)}$.
9. Use coefficient comparison to decompose $\frac{-3x^2 + 23x - 38}{(x + 5)(x^2 - 5x + 7)}$.
10. Use coefficient comparison to decompose $\frac{6x^2 - 2x + 41}{(x - 4)(x^2 + 5x + 7)}$.

Level IV: multi-stage execution

1. Use coefficient comparison to decompose $\frac{3x^2 - 12x + 9}{(x + 5)(x^2 - x + 6)}$.
2. Use coefficient comparison to decompose $\frac{-4x^2 + 9x - 4}{(x - 3)(x^2 + x + 1)}$.
3. Use coefficient comparison to decompose $\frac{8x^2 + 25x + 54}{(x + 6)(x^2 - x + 6)}$.
4. Use coefficient comparison to decompose $\frac{2x^2 + 20x + 13}{(x + 6)(x^2 + x + 5)}$.
5. Use coefficient comparison to decompose $\frac{-2x^2 - 6x - 34}{(x + 5)(x^2 + 3x + 8)}$.
6. Use coefficient comparison to decompose $\frac{-8x^2 + 5x + 20}{(x - 5)(x^2 + x + 1)}$.
7. Use coefficient comparison to decompose $\frac{4x^2 + 22x + 17}{(x + 4)(x^2 + 3x + 3)}$.
8. Use coefficient comparison to decompose $\frac{-x^2 + 3x}{(x - 2)(x^2 - 4x + 5)}$.
9. Use coefficient comparison to decompose $\frac{-3x^2 - 6x - 13}{(x + 1)(x^2 + x + 5)}$.
10. Use coefficient comparison to decompose $\frac{2x^2 + 2x + 10}{(x + 1)(x^2 - 2x + 7)}$.

Level V: independent mastery

1. Use coefficient comparison to decompose $\frac{2x^3 + 20x^2 + 4x}{(x^2 + 7)(x^2 + 5x + 7)}$.
2. Use coefficient comparison to decompose $\frac{-17x^2 + 16x + 4}{(x^2 - 2x + 8)(x^2 + 4x + 6)}$.
3. Use coefficient comparison to decompose $\frac{-2x^3 - x^2 + 12x + 9}{(x^2 - 3x + 3)(x^2 + 4x + 6)}$.
4. Use coefficient comparison to decompose $\frac{6x^3 + 8x^2 + 36x + 26}{(x^2 - 3x + 4)(x^2 + 4x + 6)}$.
5. Use coefficient comparison to decompose $\frac{-2x^3 - 13x^2 - 24x + 24}{(x^2 - x + 4)(x^2 + 4x + 8)}$.
6. Use coefficient comparison to decompose $\frac{4x^3 - 14x^2 + 18x + 19}{(x^2 - 4x + 5)(x^2 - 4x + 8)}$.
7. Use coefficient comparison to decompose $\frac{-x^3 - 7x + 2}{(x^2 + x + 4)(x^2 + x + 6)}$.
8. Use coefficient comparison to decompose $\frac{-4x^3 + x^2 - 24x - 8}{(x^2 + 6)(x^2 - 2x + 4)}$.
9. Use coefficient comparison to decompose $\frac{x^3 + 12x^2 + 15x - 5}{(x^2 + 2x + 4)(x^2 + 5x + 7)}$.
10. Use coefficient comparison to decompose $\frac{-5x^3 - 5x^2 - 26x}{(x^2 + 4)(x^2 + x + 6)}$.

Chapter 22

Repeated Linear Factors

Learning target

Build the complete template for repeated linear factors.

22.1 Recognize the shape

The denominator contains $(x - r)^m$ with $m > 1$.

22.2 Procedure

Include $A_1/(x - r) + A_2/(x - r)^2 + \cdots + A_m/(x - r)^m$.

Template completeness

Do not skip intermediate powers. A repeated factor contributes a separate numerator over every power from 1 through its multiplicity.

Common failure

Omitting one of the powers $1, 2, \dots, m$ of a repeated linear factor.

Worked example

Problem. $\frac{-5x - 10}{(x + 1)^2}$

Diagnosis. The shape is one or more repeated linear factors. The appropriate tool is: include a term for every power from 1 through the multiplicity.

Result. $-\frac{5}{x + 1} - \frac{5}{(x + 1)^2}$.

Check. Recombine over the repeated denominator.

22.3 Mastery exercises

Level I: direct recognition

1. $\frac{3x - 18}{(x - 4)^2}$

2. $\frac{6x + 16}{(x + 3)^2}$

3. $\frac{-5x - 10}{(x + 1)^2}$
4. $\frac{-x - 4}{(x + 3)^2}$
5. $\frac{-6x - 18}{(x + 2)^2}$
6. $\frac{-6x - 12}{(x + 1)^2}$
7. $\frac{-4x - 17}{(x + 4)^2}$
8. $\frac{-2x - 5}{(x + 1)^2}$
9. $\frac{-5x + 10}{(x - 1)^2}$
10. $\frac{-x - 6}{(x + 2)^2}$

Level II: routine variations

1. $\frac{-2x}{(x + 1)^2}$
2. $\frac{-6x - 21}{(x + 3)^2}$
3. $\frac{-2x + 1}{(x - 2)^2}$
4. $\frac{-4x + 15}{(x - 4)^2}$
5. $\frac{3x - 10}{(x - 5)^2}$
6. $\frac{-5x + 21}{(x - 3)^2}$
7. $\frac{5x + 31}{(x + 6)^2}$
8. $\frac{6x + 22}{(x + 3)^2}$
9. $\frac{3x - 16}{(x - 6)^2}$
10. $\frac{-2x - 1}{(x - 2)^2}$

Level III: disguised or prepared forms

1. $\frac{-3x^2 + 5}{(x + 1)^3}$
2. $\frac{6x^2 + 50x + 107}{(x + 4)^3}$
3. $\frac{-6x^2 - 8x - 6}{(x + 1)^3}$
4. $\frac{3x^2 + 36x + 99}{(x + 5)^3}$
5. $\frac{-5x^2 + 15x - 13}{(x - 1)^3}$
6. $\frac{-5x^2 + 34x - 61}{(x - 4)^3}$
7. $\frac{-4x^2 + 6x - 7}{(x - 1)^3}$
8. $\frac{-6x^2 + 14x - 2}{(x - 1)^3}$
9. $\frac{-4x^2 + 53x - 173}{(x - 6)^3}$
10. $\frac{2x^2 - 27x + 92}{(x - 6)^3}$

Level IV: multi-stage execution

1. $\frac{4x^3 + 18x^2 + 34x + 26}{(x + 1)^3 (x + 2)}$
2. $\frac{8x^3 - 31x^2 + 35x - 8}{(x - 2)(x - 1)^3}$
3. $\frac{8x^3 - 39x^2 + 46x - 15}{(x - 2)^3 (x + 1)}$
4. $\frac{-2x^3 + 6x^2 + 133x + 287}{(x - 3)(x + 4)^3}$
5. $\frac{10x^3 - 5x^2 - 14x - 4}{(x - 4)(x + 1)^3}$
6. $\frac{2x^3 - 40x^2 + 280x - 578}{(x - 5)^3 (x + 7)}$

7. $\frac{-x^3 - 15x^2 + 146x - 269}{(x-3)^3(x+4)}$
8. $\frac{3x^3 + 40x^2 + 169x + 234}{(x+3)(x+4)^3}$
9. $\frac{x^3 - 56x^2 - 644x - 1731}{(x-7)(x+5)^3}$
10. $\frac{2x^3 - 42x^2 - 373x - 682}{(x-6)(x+4)^3}$

Level V: independent mastery

1. $\frac{-4x^4 + 67x^3 - 405x^2 + 1034x - 935}{(x-4)^4(x-1)}$
2. $\frac{3x^4 + 60x^3 + 422x^2 + 1150x + 788}{(x+2)(x+6)^4}$
3. $\frac{-7x^4 - 152x^3 - 1200x^2 - 4048x - 4868}{(x+1)(x+6)^4}$
4. $\frac{14x^3 + 59x^2 + 96x + 53}{(x-3)(x+2)^4}$
5. $\frac{6x^4 - 40x^3 + 42x^2 + 139x - 231}{(x-3)^4(x+3)}$
6. $\frac{2x^4 - 25x^3 + 105x^2 - 150x + 5}{(x-5)(x-4)^4}$
7. $\frac{-11x^4 - 100x^3 - 343x^2 - 529x - 308}{(x+2)^4(x+4)}$
8. $\frac{-6x^4 + 56x^3 - 214x^2 + 404x - 320}{(x-6)(x-2)^4}$
9. $\frac{-8x^4 + 56x^3 - 145x^2 + 169x - 74}{(x-2)^4(x-1)}$
10. $\frac{-8x^4 + 46x^3 - 60x^2 - 23x + 51}{(x-7)(x-1)^4}$

Chapter 23

Distinct Irreducible Quadratic Factors

Learning target

Decompose over irreducible quadratic denominator factors.

23.1 Recognize the shape

A quadratic factor has negative discriminant and does not factor over the reals.

23.2 Procedure

Use a general linear numerator $Ax + B$ over each irreducible quadratic.

Common failure

Putting only a constant over an irreducible quadratic; the numerator must be a general linear polynomial.

Worked example

Problem. $\frac{-4x^3 - 9x^2 - 48x + 14}{(x^2 - 2x + 6)(x^2 + 2x + 8)}$

Diagnosis. The shape is distinct irreducible quadratic factors. The appropriate tool is: place a general linear numerator over each quadratic factor.

Result. $\frac{-5x - 2}{x^2 - 2x + 6} + \frac{x + 5}{x^2 + 2x + 8}$.

Check. Recombine over the product of quadratics.

23.3 Mastery exercises

Level I: direct recognition

1. $\frac{-6x^3 + 6x^2 - 7x - 7}{(x^2 - 2x + 3)(x^2 - x + 1)}$

2. $\frac{-3x^3 - 4x^2 - 6x + 16}{(x^2 + x + 2)(x^2 + 2x + 6)}$

3. $\frac{-4x^3 - 9x^2 - 48x + 14}{(x^2 - 2x + 6)(x^2 + 2x + 8)}$

4. $\frac{7x^3 - 4x^2 + 43x - 54}{(x^2 + 8)(x^2 + x + 6)}$

5. $\frac{2x^3 - 7x^2 + 17x + 41}{(x^2 - x + 5)(x^2 + 5x + 7)}$
6. $\frac{9x^3 + 20x^2 + 27x - 22}{(x^2 - x + 1)(x^2 + 5x + 8)}$
7. $\frac{3x^3 + 15x^2 + 49x + 38}{(x^2 + 2x + 2)(x^2 + 2x + 7)}$
8. $\frac{-7x^3 - 14x^2 - 30x - 20}{(x^2 + 2)(x^2 + 2x + 4)}$
9. $\frac{-7x^3 - 9x^2 - 11x}{(x^2 + x + 1)(x^2 + 3x + 3)}$
10. $\frac{13x - 6}{(x^2 + 4)(x^2 - x + 2)}$

Level II: routine variations

1. $\frac{6x^3 - x^2 + 33x + 20}{(x^2 + 4)(x^2 - x + 8)}$
2. $\frac{-x^3 + 2x^2 + x + 6}{(x^2 - 4x + 7)(x^2 - 3x + 4)}$
3. $\frac{-7x^3 + 7x^2 - 30x + 18}{(x^2 + 5)(x^2 - x + 3)}$
4. $\frac{x^2 + 6x - 22}{(x^2 - 3x + 5)(x^2 - x + 4)}$
5. $\frac{6x^3 + 12x^2 + 44x + 38}{(x^2 + 6)(x^2 + 2x + 8)}$
6. $\frac{11x^2 - 29x + 34}{(x^2 - x + 8)(x^2 + 2x + 2)}$
7. $\frac{10x^3 + 16x^2 + 37x - 18}{(x^2 + 3)(x^2 + 4x + 6)}$
8. $\frac{4x^3 + 18x^2 + 27x + 55}{(x^2 - 3x + 3)(x^2 + x + 8)}$
9. $\frac{2x^3 + 14x^2 + 11x - 25}{(x^2 - x + 1)(x^2 + 4x + 5)}$
10. $\frac{x^3 - 8x^2 - 12x - 9}{(x^2 - 2x + 3)(x^2 + x + 6)}$

Level III: disguised or prepared forms

1. $\frac{x^3 - 3x^2 - x - 1}{(x^2 - 3x + 3)(x^2 - 3x + 4)}$

2. $\frac{-x^3 + 3x^2 - x - 7}{(x^2 - x + 2)(x^2 + 3x + 5)}$
3. $\frac{-5x^3 + 15x^2 - 24x + 6}{(x^2 - 5x + 8)(x^2 - x + 6)}$
4. $\frac{-7x^3 + 6x^2 - 46x - 4}{(x^2 + 8)(x^2 - 2x + 4)}$
5. $\frac{-2x^3 + 6x^2 - 10x - 11}{(x^2 - 4x + 7)(x^2 - 4x + 8)}$
6. $\frac{-2x^3 + 2x^2 + 4x + 24}{(x^2 + x + 2)(x^2 + 3x + 6)}$
7. $\frac{-x^3 + 23x^2 - 25x + 33}{(x^2 - 5x + 7)(x^2 + x + 4)}$
8. $\frac{-x^3 + 16x^2 + 5x + 19}{(x^2 + 6)(x^2 + 3x + 5)}$
9. $\frac{-9x^3 - 68x - 23}{(x^2 - 4x + 8)(x^2 + 4x + 5)}$
10. $\frac{5x^3 + 22x - 22}{(x^2 + 6)(x^2 + 2x + 8)}$

Level IV: multi-stage execution

1. $\frac{-7x^3 - 13x^2 - 53x - 29}{(x^2 - x + 5)(x^2 + 4x + 6)}$
2. $\frac{-4x^3 + 9x^2 - 39x + 12}{(x^2 - 2x + 8)(x^2 - x + 5)}$
3. $\frac{11x^2 + 14x - 3}{(x^2 - x + 6)(x^2 + 2x + 7)}$
4. $\frac{x^3 + 13x^2 + 27x + 29}{(x^2 + x + 5)(x^2 + 3x + 8)}$
5. $\frac{8x^3 - 27x^2 + 47x - 8}{(x^2 - 3x + 4)(x^2 - 3x + 6)}$
6. $\frac{-8x^3 - 10x^2 - 54x - 34}{(x^2 - 3x + 5)(x^2 + 4x + 6)}$
7. $\frac{5x^3 + 11x^2 + 59x + 3}{(x^2 - 2x + 8)(x^2 + 3x + 7)}$
8. $\frac{4x^3 - 7x^2 + 16x + 8}{(x^2 - 2x + 2)(x^2 - 2x + 4)}$
9. $\frac{10x^3 + 38x^2 + 60x + 13}{(x^2 + 3x + 3)(x^2 + 4x + 7)}$

$$10. \frac{-2x^3 - 11x^2 - 27x - 32}{(x^2 + 2x + 6)(x^2 + 4x + 7)}$$

Level V: independent mastery

$$1. \frac{-23x^2 + 20x + 1}{(x^2 - 5x + 7)(x^2 + 3x + 4)}$$

$$2. \frac{-6x^3 - 10x^2 - 51x - 51}{(x^2 + 7)(x^2 + 2x + 6)}$$

$$3. \frac{x^3 + 10x^2 + 16x - 25}{(x^2 - 5x + 7)(x^2 + x + 3)}$$

$$4. \frac{4x^3 + x^2 + 9x - 26}{(x^2 + x + 2)(x^2 + 3x + 8)}$$

$$5. \frac{-4x^3 + 14x^2 - 8x - 2}{(x^2 - 2x + 6)(x^2 + 3x + 4)}$$

$$6. \frac{5x^3 + 13x^2 + 26x + 33}{(x^2 + x + 3)(x^2 + x + 6)}$$

$$7. \frac{5x^2 + x - 13}{(x^2 + x + 3)(x^2 + 5x + 7)}$$

$$8. \frac{-2x^3 - 3x^2 - 9x + 17}{(x^2 + 3x + 3)(x^2 + 3x + 7)}$$

$$9. \frac{2x^3 - 16x^2 + 30x - 17}{(x^2 - 5x + 8)(x^2 - 3x + 7)}$$

$$10. \frac{3x^3 + x^2 - 46}{(x^2 + 8)(x^2 + 2x + 3)}$$

Chapter 24

Repeated Irreducible Quadratic Factors

Learning target

Build the complete template for repeated quadratic factors.

24.1 Recognize the shape

The denominator contains $q(x)^m$ where q is irreducible quadratic.

24.2 Procedure

Include a separate linear numerator over every power $q, q^2, \dots, q^m$.

Template completeness

Do not skip intermediate powers. A repeated factor contributes a separate numerator over every power from 1 through its multiplicity.

Common failure

Using one numerator for the highest power only and omitting the lower powers.

Worked example

Problem.
$$\frac{-2x^3 + 8x^2 - 18x + 7}{(x^2 - 3x + 5)^2}$$

Diagnosis. The shape is repeated irreducible quadratic factor. The appropriate tool is: include a linear numerator over every power of the quadratic.

Result.
$$\frac{2 - 2x}{x^2 - 3x + 5} + \frac{-2x - 3}{(x^2 - 3x + 5)^2}.$$

Check. Recombine over the highest quadratic power.

24.3 Mastery exercises

Level I: direct recognition

1.
$$\frac{-x^3 - 6x^2 - 14x - 16}{(x^2 + 4x + 7)^2}$$

2.
$$\frac{-2x^3 - 6x^2 - 5x}{(x^2 + x + 1)^2}$$

3. $\frac{-2x^3 + 8x^2 - 18x + 7}{(x^2 - 3x + 5)^2}$
4. $\frac{-4x^3 + 7x^2 - 6x - 29}{(x^2 - 3x + 5)^2}$
5. $\frac{-3x^3 + 11x^2 - 14x - 9}{(x^2 - 4x + 7)^2}$
6. $\frac{3x^3 + 12x^2 + 25x + 15}{(x^2 + 3x + 4)^2}$
7. $\frac{4x^3 - 3x^2 + 9x + 29}{(x^2 - 2x + 5)^2}$
8. $\frac{-2x^3 + 6x^2 - 13x + 17}{(x^2 - x + 5)^2}$
9. $\frac{4x^3 - 5x^2 + 10x + 10}{(x^2 - 2x + 5)^2}$
10. $\frac{-3x^3 - x^2 + 17}{(x^2 + 2x + 4)^2}$

Level II: routine variations

1. $\frac{-2x^3 - 5x^2 - 4x - 4}{(x^2 + 2x + 3)^2}$
2. $\frac{-2x^3 + 4x^2 + 3x - 15}{(x^2 - 4x + 5)^2}$
3. $\frac{-x^3 - 2x^2 + 14x - 16}{(x^2 - 3x + 4)^2}$
4. $\frac{2x^3 + 7x^2 + 6x + 4}{(x^2 + 3x + 3)^2}$
5. $\frac{4x^3 - 7x^2 + 18x + 41}{(x^2 - 3x + 8)^2}$
6. $\frac{-2x^3 - 5x^2 - 9x + 1}{(x^2 + 2x + 2)^2}$
7. $\frac{-x^3 + x^2 + 4x + 16}{(x^2 + 2x + 4)^2}$
8. $\frac{-2x^3 - 5x^2 - 9x - 3}{(x^2 + 2x + 4)^2}$
9. $\frac{3x^3 - 2x^2 - 3x + 5}{(x^2 - 2x + 2)^2}$

$$10. \frac{-4x^3 + 16x^2 - 31x + 12}{(x^2 - 3x + 4)^2}$$

Level III: disguised or prepared forms

$$1. \frac{2x^3 - 6x^2 + 17x - 27}{(x^2 - x + 7)^2}$$

$$2. \frac{4x^3 - 2x^2 + 36x - 19}{(x^2 + 8)^2}$$

$$3. \frac{3x^3 + 4x^2 - 7x - 16}{(x^2 + 3x + 4)^2}$$

$$4. \frac{-4x^3 - 10x^2 - 12x - 5}{(x^2 + 2x + 3)^2}$$

$$5. \frac{-2x^3 - 2x^2 - 5x - 23}{(x^2 - x + 5)^2}$$

$$6. \frac{-3x^3 - 16x^2 - 33x - 25}{(x^2 + 4x + 7)^2}$$

$$7. \frac{3x^3 - 9x^2 + 29x - 16}{(x^2 - 2x + 7)^2}$$

$$8. \frac{-2x^3 - 12x^2 - 27x - 24}{(x^2 + 4x + 5)^2}$$

$$9. \frac{-2x^3 + 2x^2 + 10x - 15}{(x^2 - 3x + 3)^2}$$

$$10. \frac{2x^3 + 11x^2 + 23x + 16}{(x^2 + 4x + 7)^2}$$

Level IV: multi-stage execution

$$1. \frac{-4x^5 + 20x^4 - 73x^3 + 135x^2 - 175x + 90}{(x^2 - 2x + 5)^3}$$

$$2. \frac{3x^5 - 23x^4 + 81x^3 - 130x^2 + 70x + 75}{(x^2 - 4x + 7)^3}$$

$$3. \frac{3x^5 - 4x^4 + 7x^3 + 2x}{(x^2 - x + 2)^3}$$

$$4. \frac{-2x^5 - 16x^4 - 72x^3 - 192x^2 - 306x - 270}{(x^2 + 3x + 8)^3}$$

$$5. \frac{-2x^5 - 21x^4 - 95x^3 - 221x^2 - 260x - 117}{(x^2 + 4x + 5)^3}$$

6. $\frac{3x^5 - 15x^4 + 31x^3 - 23x^2 - 12x + 24}{(x^2 - 3x + 3)^3}$
7. $\frac{4x^5 - 21x^4 + 53x^3 - 59x^2 + 16x + 24}{(x^2 - 3x + 4)^3}$
8. $\frac{-3x^5 - 13x^4 - 48x^3 - 79x^2 - 109x - 33}{(x^2 + 2x + 5)^3}$
9. $\frac{-3x^5 + 7x^4 - 20x^3 - 9x^2 + 28x - 104}{(x^2 - 2x + 5)^3}$
10. $\frac{3x^5 + 3x^4 + 50x^3 + 51x^2 + 204x + 220}{(x^2 + 8)^3}$

Level V: independent mastery

1. $\frac{7x^6 - 4x^5 + 89x^4 - 33x^3 + 329x^2 - 55x + 255}{(x - 3)(x^2 + 6)^3}$
2. $\frac{-2x^6 - 52x^5 - 406x^4 - 1505x^3 - 2856x^2 - 2520x - 556}{(x - 4)(x^2 + 5x + 7)^3}$
3. $\frac{9x^6 - 97x^5 + 485x^4 - 1391x^3 + 2414x^2 - 2409x + 1124}{(x - 1)(x^2 - 4x + 6)^3}$
4. $\frac{-x^6 - 2x^5 - 21x^4 - 13x^3 - 97x^2 - 43x - 101}{(x - 5)(x^2 + 2)^3}$
5. $\frac{-2x^6 + 11x^5 - 2x^4 - 135x^3 + 436x^2 - 573x + 281}{(x - 1)(x^2 - 4x + 5)^3}$
6. $\frac{7x^6 + x^5 + 25x^4 + 5x^3 + 67x^2 + 24x + 51}{(x - 3)(x^2 + x + 2)^3}$
7. $\frac{5x^6 - 25x^5 + 92x^4 - 179x^3 + 385x^2 - 488x + 714}{(x + 3)(x^2 - 3x + 6)^3}$
8. $\frac{7x^6 + 60x^5 + 281x^4 + 783x^3 + 1432x^2 + 1541x + 896}{(x - 1)(x^2 + 3x + 6)^3}$
9. $\frac{-x^6 - 3x^5 - 42x^4 - 45x^3 - 407x^2 - 181x - 1116}{(x + 2)(x^2 + 7)^3}$
10. $\frac{5x^6 - 62x^5 + 355x^4 - 1250x^3 + 2820x^2 - 3645x + 2047}{(x + 5)(x^2 - 5x + 7)^3}$

Chapter 25

Mixed Denominator Types

Learning target

Combine all partial-fraction template rules.

25.1 Recognize the shape

The factored denominator contains several factor types or multiplicities.

25.2 Procedure

Classify every factor first; write the entire template before solving any coefficients.

Template completeness

Do not skip intermediate powers. A repeated factor contributes a separate numerator over every power from 1 through its multiplicity.

Common failure

Solving for coefficients before checking that the full mixed template is complete.

Worked example

Problem. $\frac{-3x^3 - 4x^2 + 6x - 67}{(x-2)(x+3)(x^2+4x+7)}$

Diagnosis. The shape is mixed linear, repeated, and quadratic factors. The appropriate tool is: classify each factor and write every required numerator and power.

Result. $\frac{3-4x}{x^2+4x+7} + \frac{2}{x+3} - \frac{1}{x-2}$.

Check. Recombine the full mixed decomposition.

25.3 Mastery exercises

Level I: direct recognition

1. $\frac{5x^3 + 22x^2 + 55x + 8}{x(x+1)(x^2+3x+8)}$

2. $\frac{-10x^3 + 21x^2 + 15x + 76}{(x-5)(x+2)(x^2+2x+6)}$

3. $\frac{-3x^3 - 4x^2 + 6x - 67}{(x-2)(x+3)(x^2+4x+7)}$
4. $\frac{x^3 - 19x^2 + 9x + 44}{(x-6)(x-1)(x^2-x+7)}$
5. $\frac{-9x^3 - 25x^2 + 34x + 195}{(x-4)(x-1)(x^2+5x+7)}$
6. $\frac{x^3 + 7x^2 + 3x + 36}{x(x-3)(x^2+x+3)}$
7. $\frac{7x^3 - 48x^2 + 120x - 112}{(x-4)(x-2)(x^2-4x+6)}$
8. $\frac{-5x^3 - 16x^2 - 80x + 50}{(x-5)(x+6)(x^2-x+5)}$
9. $\frac{-7x^3 - 6x^2 + 94x - 132}{(x-4)(x+6)(x^2-3x+6)}$
10. $\frac{-14x + 7}{(x-3)(x+2)(x^2-x+1)}$

Level II: routine variations

1. $\frac{x^4 - 7x^3 - 9x^2 + 64x - 68}{(x-3)(x-2)^2(x^2+4)}$
2. $\frac{-5x^4 + 54x^3 - 114x^2 + 211x - 1020}{(x-5)^2(x+5)(x^2-x+7)}$
3. $\frac{-4x^4 - 13x^3 + 22x^2 + 27x - 127}{(x+3)^2(x+4)(x^2-x+5)}$
4. $\frac{4x^4 + 24x^3 + 13x^2 - 123x + 307}{(x-1)(x+4)^2(x^2-3x+5)}$
5. $\frac{-12x^4 - 90x^3 - 239x^2 - 466x - 480}{(x+1)^2(x+6)(x^2+2x+8)}$
6. $\frac{-9x^4 + 15x^3 - 41x^2 + 52x - 48}{x^2(x-3)(x^2+4)}$
7. $\frac{-7x^4 + 24x^3 + 78x^2 - 326x + 636}{(x-4)^2(x+5)(x^2-2x+4)}$
8. $\frac{-6x^4 - 19x^3 - 28x^2 - 28x - 48}{x^2(x+3)(x^2+2x+4)}$
9. $\frac{16x^3 + 14x^2 - 15x + 31}{(x+1)(x+3)^2(x^2-2x+8)}$

$$10. \frac{2x^4 - 44x^3 + 208x^2 - 272x + 544}{(x-6)^2(x-2)(x^2-2x+8)}$$

Level III: disguised or prepared forms

$$1. \frac{6x^4 - 58x^3 + 101x^2 - 97x + 1564}{(x-6)^2(x-5)(x^2+4x+7)}$$

$$2. \frac{x^4 + 25x^3 + 134x^2 + 40x + 512}{(x-2)(x+6)^2(x^2+x+1)}$$

$$3. \frac{-9x^3 + 21x^2 - 22x + 13}{(x-4)(x-1)^2(x^2-3x+3)}$$

$$4. \frac{-5x^4 + 52x^3 - 152x^2 + 37x + 310}{(x-6)(x-5)^2(x^2+x+5)}$$

$$5. \frac{2x^4 + 46x^3 + 78x^2 - 544x + 1756}{(x-5)(x+6)^2(x^2+8)}$$

$$6. \frac{-2x^4 + 41x^3 + 6x^2 + 162x + 108}{x^2(x-6)(x^2+6)}$$

$$7. \frac{8x^4 + 2x^3 - 36x^2 - 56x - 368}{(x-6)(x+4)^2(x^2+8)}$$

$$8. \frac{-5x^4 - 34x^3 + 20x^2 + 86x - 555}{(x-3)(x+3)^2(x^2-2x+5)}$$

$$9. \frac{9x^4 - 82x^3 + 249x^2 - 525x + 1079}{(x-5)^2(x-3)(x^2-x+7)}$$

$$10. \frac{33x^3 - 20x^2 - 65x - 204}{(x-3)^2(x+1)(x^2+3x+8)}$$

Level IV: multi-stage execution

$$1. \frac{-4x^6 + 26x^5 - 115x^4 + 180x^3 - 187x^2 - 284x + 999}{(x-5)(x+2)^2(x^2-x+7)^2}$$

$$2. \frac{-2x^6 + 17x^5 - 16x^4 + 78x^3 - 30x^2 + 82x - 79}{(x-3)(x-1)^2(x^2+4)^2}$$

$$3. \frac{4x^6 - 50x^5 + 261x^4 - 785x^3 + 1573x^2 - 2105x + 1386}{(x-3)^2(x+2)(x^2-5x+8)^2}$$

$$4. \frac{-5x^6 + 4x^5 + 90x^4 + 199x^3 + 936x^2 + 663x + 2180}{(x-4)(x+3)^2(x^2+4)^2}$$

$$5. \frac{-13x^5 + 187x^4 - 1081x^3 + 3098x^2 - 4256x + 1888}{(x-4)^2(x+2)(x^2-5x+8)^2}$$

6. $\frac{-5x^6 + 11x^5 - 65x^4 + 84x^3 - 269x^2 + 101x - 185}{(x-3)(x+1)^2(x^2+5)^2}$
7. $\frac{-x^6 - 13x^5 + 75x^4 + 425x^3 - 481x^2 + 2248x + 716}{(x-5)(x+6)^2(x^2-2x+4)^2}$
8. $\frac{6x^6 - 27x^5 + 75x^4 - 201x^3 + 47x^2 - 152x - 3036}{(x-3)(x+3)^2(x^2-3x+8)^2}$
9. $\frac{-8x^6 + 31x^5 - 131x^4 + 405x^3 - 874x^2 + 1358x - 2073}{(x-3)^2(x+2)(x^2+7)^2}$
10. $\frac{-3x^6 - 72x^5 - 675x^4 - 3164x^3 - 7833x^2 - 9750x - 4871}{(x+5)(x+6)^2(x^2+5x+7)^2}$

Level V: independent mastery

1. $\frac{-7x^6 + 51x^5 - 136x^4 + 63x^3 + 428x^2 - 894x + 535}{(x-3)(x+1)^2(x^2-4x+5)^2}$
2. $\frac{4x^6 - 5x^5 - 25x^4 - 151x^3 - 7x^2 + 84x + 1252}{(x-5)^2(x-1)(x^2+x+4)^2}$
3. $\frac{-6x^6 + 19x^5 - 14x^4 - 135x^3 + 564x^2 + 854x - 572}{(x-4)^2(x-2)(x^2+4x+5)^2}$
4. $\frac{9x^6 + 2x^5 - 17x^4 + 176x^3 + 176x^2 - 64x + 848}{(x+2)^2(x+4)(x^2-3x+6)^2}$
5. $\frac{-4x^6 - 19x^5 + 47x^4 + 8x^3 - 252x^2 - 501x + 177}{(x+3)^2(x+5)(x^2-4x+7)^2}$
6. $\frac{7x^6 - 36x^5 - 146x^4 + 409x^3 - 462x^2 + 460x + 488}{(x-6)(x+4)^2(x^2-2x+2)^2}$
7. $\frac{-x^6 - 7x^5 - 47x^4 - 407x^3 - 1574x^2 - 2935x - 3475}{(x+4)^2(x+5)(x^2+2x+5)^2}$
8. $\frac{10x^6 - 63x^5 + 202x^4 - 433x^3 + 621x^2 - 600x + 276}{(x-2)^2(x+1)(x^2-2x+4)^2}$
9. $\frac{10x^6 + 11x^5 + 36x^4 - 25x^3 - 38x^2 - 192x - 144}{x^2(x-4)(x^2+2x+3)^2}$
10. $\frac{3x^6 - 16x^5 + 59x^4 - 130x^3 + 124x^2 - 164x - 96}{(x-4)^2(x+1)(x^2-x+2)^2}$

Chapter 26

Improper Rational Expressions and Partial Fractions

Learning target

Decompose rational expressions whose numerator degree is too large.

26.1 Recognize the shape

The numerator degree is at least the denominator degree.

26.2 Procedure

Use polynomial division first; decompose only the proper remainder fraction.

Common failure

Attempting partial fractions while the rational expression is still improper.

Worked example

Problem. $\frac{-3x^3 - 6x^2 + 4x - 9}{x(x+3)}$

Diagnosis. The shape is improper rational expression. The appropriate tool is: perform polynomial division first, then decompose the proper remainder.

Result. $(3 - 3x) - \frac{2}{x+3} - \frac{3}{x}$.

Check. Combine the polynomial quotient and decomposed remainder.

26.3 Mastery exercises

Level I: direct recognition

1. $\frac{-3x^3 - 31x^2 - 69x + 58}{(x+5)(x+6)}$

2. $\frac{-x^3 + 4x^2 + 13x - 32}{(x-5)(x+3)}$

3. $\frac{-3x^3 - 6x^2 + 4x - 9}{x(x+3)}$

4. $\frac{-2x^3 + 3x^2 + 26x + 10}{(x-5)(x+2)}$
5. $\frac{-2x^3 + 8x^2 + 11x + 13}{(x-5)(x+1)}$
6. $\frac{-3x^3 - 12x^2 + 32x - 10}{(x-1)(x+6)}$
7. $\frac{x^3 - 4x^2 - 15x + 28}{(x-6)(x+4)}$
8. $\frac{-x^3 + 2x^2 + 35x - 18}{(x-6)(x+6)}$
9. $\frac{-3x^3 + 4x^2 + 8x + 9}{(x-3)(x+1)}$
10. $\frac{x^3 + 4x^2 - 2x - 16}{(x+2)(x+4)}$

Level II: routine variations

1. $\frac{-x^3 - x^2 + 6x - 10}{x(x-2)}$
2. $\frac{3x^3 + 9x^2 - 47x + 10}{(x-2)(x+6)}$
3. $\frac{-x^3 + 5x^2 + 12x - 8}{(x-6)(x+1)}$
4. $\frac{2x^3 - 19x^2 + 41x}{(x-6)(x-3)}$
5. $\frac{-x^3 + x^2 + 16x + 38}{(x-5)(x+1)}$
6. $\frac{-x^3 - 10x^2 - 28x - 19}{(x+3)(x+4)}$
7. $\frac{x^3 - x^2 - 22x + 22}{(x-4)(x+5)}$
8. $\frac{2x^3 - 20x^2 + 50x - 14}{(x-6)(x-4)}$
9. $\frac{x^3 + 7x^2 + 11x + 21}{(x+1)(x+5)}$
10. $\frac{3x^3 + 3x^2 + 7x + 5}{x(x+1)}$

Level III: disguised or prepared forms

1. $\frac{x^4 - x^3 - 25x^2 + 23x}{(x - 5)(x + 5)}$
2. $\frac{x^4 + x^3 - 9x^2 + 6x + 25}{(x - 2)(x + 3)}$
3. $\frac{-3x^4 + 5x^3 + 59x^2 - 35x - 23}{(x - 5)(x + 4)}$
4. $\frac{2x^4 + 2x^3 - 17x^2 - 17x - 24}{(x - 3)(x + 3)}$
5. $\frac{2x^4 - 15x^3 + 18x^2 + 4x - 30}{x(x - 6)}$
6. $\frac{x^4 + 2x^3 - 6x^2 - 7x - 14}{(x - 2)(x + 3)}$
7. $\frac{-x^4 - 3x^3 + 5x^2 - 5x - 11}{(x - 1)(x + 4)}$
8. $\frac{-2x^4 + 10x^3 + 14x^2 - 21x + 46}{(x - 6)(x + 2)}$
9. $\frac{-3x^4 - 9x^3 + 10x^2 - 9x - 8}{x(x + 4)}$
10. $\frac{2x^4 - 8x^3 - 24x^2 + 6x - 12}{(x - 6)(x + 2)}$

Level IV: multi-stage execution

1. $\frac{-2x^5 + 5x^4 + 38x^3 + 12x^2 + 4x + 93}{(x - 6)(x + 1)(x + 3)}$
2. $\frac{-3x^5 + 14x^4 - 14x^3 + 3x^2 - 11x + 3}{x(x - 3)(x - 1)}$
3. $\frac{-3x^5 - 13x^4 + 23x^3 + 127x^2 + 47x - 81}{(x - 3)(x + 3)(x + 4)}$
4. $\frac{2x^5 + 3x^4 - 39x^3 - 49x^2 + 99x - 44}{(x - 4)(x + 3)(x + 4)}$
5. $\frac{3x^5 - 22x^4 - 5x^3 + 173x^2 + 24x - 44}{(x - 6)(x - 4)(x + 2)}$
6. $\frac{-2x^5 + 16x^4 - 34x^3 + 27x^2 - 25x + 10}{(x - 5)(x - 2)(x - 1)}$
7. $\frac{-2x^5 - 2x^4 + 74x^3 + 68x^2 - 80x + 288}{(x - 6)(x + 2)(x + 6)}$

8. $\frac{-2x^5 - 10x^4 - 13x^3 - 2x^2 - 5x - 12}{x(x+1)(x+3)}$
9. $\frac{-2x^5 + 5x^4 + 85x^3 - 185x^2 - 474x + 264}{(x-6)(x-4)(x+6)}$
10. $\frac{x^5 - 9x^4 - 5x^3 + 218x^2 - 494x + 95}{(x-6)(x-5)(x+5)}$

Level V: independent mastery

1. $\frac{2x^6 - 8x^5 - 19x^4 + 49x^3 - 37x^2 + 99x + 50}{(x-5)(x-2)(x+3)}$
2. $\frac{2x^6 - 8x^5 - 2x^4 + 5x^3 + 12x^2 - x - 20}{x(x-4)(x-1)}$
3. $\frac{-2x^6 + 16x^5 - 16x^4 - 51x^3 + 11x^2 + 18x + 72}{x(x-6)(x-3)}$
4. $\frac{-x^6 + 29x^4 + 46x^3 - 24x^2 + 6x - 96}{x(x-6)(x+4)}$
5. $\frac{x^6 - 17x^4 + 10x^3 + 58x^2 + x + 28}{(x-3)(x+2)(x+4)}$
6. $\frac{-x^6 + 9x^5 - 22x^4 + 4x^3 + 42x^2 - 69x + 15}{x(x-5)(x-3)}$
7. $\frac{3x^6 - 4x^5 - 25x^4 + 32x^3 - 10x^2 + 26x + 18}{(x-3)(x-1)(x+3)}$
8. $\frac{2x^6 - 17x^5 + 29x^4 + 40x^3 - 53x^2 - 6x + 99}{(x-5)(x-4)(x+1)}$
9. $\frac{-2x^6 + 22x^5 - 68x^4 + 5x^3 + 186x^2 - 38x - 77}{(x-5)(x-4)(x-3)}$
10. $\frac{x^6 - 8x^5 + 10x^4 + 15x^3 - 39x^2 + 121x - 42}{(x-6)(x-3)(x+2)}$

Chapter 27

Hybrid Coefficient Methods

Learning target

Use the most efficient mixture of coefficient-finding methods.

27.1 Recognize the shape

Some coefficients isolate at roots while others remain coupled.

27.2 Procedure

Use cover-up or substitution for easy coefficients, then coefficient comparison for the rest; verify at the end.

Common failure

Using one method mechanically when a short substitution can isolate some coefficients immediately.

Worked example

Problem. Use an efficient mixture of methods to decompose $\frac{8x^3 - 16x^2 - 39x - 10}{(x - 4)(x - 3)(x^2 + 2)}$.

Diagnosis. The shape is some coefficients isolate immediately while others do not. The appropriate tool is: use cover-up/substitution for easy constants and coefficient comparison for the rest.

Result. $\frac{5 - 2x}{x^2 + 2} + \frac{5}{x - 3} + \frac{5}{x - 4}$.

Check. Recombine all terms.

27.3 Mastery exercises

Level I: direct recognition

1. Use an efficient mixture of methods to decompose $\frac{-9x^3 - 21x^2 + 17x - 20}{(x - 4)(x + 4)(x^2 + x + 7)}$.
2. Use an efficient mixture of methods to decompose $\frac{-5x^3 - 23x^2 - 15x - 75}{x(x + 3)(x^2 - x + 5)}$.
3. Use an efficient mixture of methods to decompose $\frac{8x^3 - 16x^2 - 39x - 10}{(x - 4)(x - 3)(x^2 + 2)}$.
4. Use an efficient mixture of methods to decompose $\frac{-3x^3 + 12x^2 - 63x + 42}{x(x - 3)(x^2 - 2x + 7)}$.

5. Use an efficient mixture of methods to decompose $\frac{4x^3 + 16x^2 + 21x + 17}{(x+2)(x+3)(x^2+2x+7)}$.
6. Use an efficient mixture of methods to decompose $\frac{-4x^3 - 3x^2 - 29x - 12}{x(x+1)(x^2+x+6)}$.
7. Use an efficient mixture of methods to decompose $\frac{-2x^3 - 17x^2 + 80x - 112}{(x-4)(x+4)(x^2-3x+8)}$.
8. Use an efficient mixture of methods to decompose $\frac{-2x^3 + 8x^2 - 89x - 54}{(x-6)(x+6)(x^2+3x+7)}$.
9. Use an efficient mixture of methods to decompose $\frac{12x^3 + 63x^2 + 26x + 215}{(x+5)(x+6)(x^2-2x+5)}$.
10. Use an efficient mixture of methods to decompose $\frac{-6x^3 + 43x^2 - 139x + 126}{x(x-6)(x^2-4x+7)}$.

Level II: routine variations

1. Use an efficient mixture of methods to decompose $\frac{-5x^3 - 25x^2 - 60x - 42}{x(x+3)(x^2+4x+7)}$.
2. Use an efficient mixture of methods to decompose $\frac{-2x^3 - 15x^2 + 27x + 30}{x(x-6)(x^2+4x+5)}$.
3. Use an efficient mixture of methods to decompose $\frac{3x^2 - 6x - 42}{(x+3)(x+4)(x^2+4x+6)}$.
4. Use an efficient mixture of methods to decompose $\frac{-4x^3 - 5x^2 + 5x - 48}{(x-2)(x+1)(x^2-x+4)}$.
5. Use an efficient mixture of methods to decompose $\frac{7x^3 - 25x^2 + 67x - 205}{(x-5)(x-3)(x^2+x+8)}$.
6. Use an efficient mixture of methods to decompose $\frac{3x^3 + 34x^2 - 253x - 94}{(x-6)(x-4)(x^2+4x+5)}$.
7. Use an efficient mixture of methods to decompose $\frac{3x^3 - 35x^2 + 125x - 39}{(x-5)(x-4)(x^2+3x+3)}$.
8. Use an efficient mixture of methods to decompose $\frac{-x^3 - 13x^2 + 69x - 81}{(x-3)(x+3)(x^2-3x+3)}$.
9. Use an efficient mixture of methods to decompose $\frac{-10x^3 - 26x^2 + 24x - 44}{(x-1)(x+6)(x^2+1)}$.
10. Use an efficient mixture of methods to decompose $\frac{10x^3 - 41x^2 + 83x - 117}{(x-3)(x-2)(x^2-x+5)}$.

Level III: disguised or prepared forms

1. Use an efficient mixture of methods to decompose $\frac{-9x^4 + 52x^3 + 111x^2 - 358x - 668}{(x-6)^2(x+5)(x^2-x+4)}$.

2. Use an efficient mixture of methods to decompose $\frac{-x^4 - 9x^3 - 33x^2 - 52x - 39}{(x-3)(x+1)^2(x^2+2x+2)}$.
3. Use an efficient mixture of methods to decompose $\frac{-4x^4 + 23x^3 + 170x^2 + 226x + 280}{(x-4)(x+4)^2(x^2-x+5)}$.
4. Use an efficient mixture of methods to decompose $\frac{-5x^4 + 39x^3 - 97x^2 + 82x + 401}{(x-5)^2(x+3)(x^2-2x+2)}$.
5. Use an efficient mixture of methods to decompose $\frac{8x^4 + 22x^3 - 161x^2 + 378x + 1108}{(x-4)(x+6)^2(x^2-4x+7)}$.
6. Use an efficient mixture of methods to decompose $\frac{-5x^4 + 49x^3 - 158x^2 + 216x - 64}{x^2(x-4)(x^2-5x+8)}$.
7. Use an efficient mixture of methods to decompose $\frac{-4x^4 - 26x^3 - 34x^2 + 14x - 114}{(x-3)(x+3)^2(x^2+x+1)}$.
8. Use an efficient mixture of methods to decompose $\frac{-3x^4 - 19x^3 - 79x^2 - 97x + 72}{(x+2)^2(x+3)(x^2-3x+8)}$.
9. Use an efficient mixture of methods to decompose $\frac{-x^4 + 16x^3 - 60x^2 + 227x - 228}{(x-2)^2(x+5)(x^2+x+8)}$.
10. Use an efficient mixture of methods to decompose $\frac{-x^4 - 4x^3 + 57x^2 + 39x - 88}{(x-4)(x+2)^2(x^2-2x+5)}$.

Level IV: multi-stage execution

1. Use an efficient mixture of methods to decompose $\frac{-2x^4 + 48x^3 - 125x^2 - 261x - 504}{x(x-6)^2(x^2+4x+7)}$.
2. Use an efficient mixture of methods to decompose $\frac{7x^4 + 12x^3 + 6x^2 - 253x + 204}{(x-3)^2(x+2)(x^2+3x+8)}$.
3. Use an efficient mixture of methods to decompose $\frac{2x^4 - 7x^3 - 119x^2 + 384x + 455}{(x-5)^2(x+4)(x^2-x+5)}$.
4. Use an efficient mixture of methods to decompose $\frac{-11x^4 - 10x^3 - 15x^2 - 92x - 103}{(x+1)^2(x+2)(x^2-3x+5)}$.
5. Use an efficient mixture of methods to decompose $\frac{x^4 - 6x^3 + 22x^2 - 18x + 37}{(x-1)^2(x+1)(x^2+x+7)}$.
6. Use an efficient mixture of methods to decompose $\frac{-x^4 + 32x^3 - 36x^2 - 429x - 1233}{(x-6)^2(x+3)(x^2+3x+3)}$.
7. Use an efficient mixture of methods to decompose $\frac{-2x^4 - 24x^3 + 190x^2 - 430x + 1756}{(x-6)^2(x+5)(x^2-2x+8)}$.
8. Use an efficient mixture of methods to decompose $\frac{-7x^4 + 58x^3 - 67x^2 - 74x - 284}{(x-5)^2(x+2)(x^2+1)}$.

9. Use an efficient mixture of methods to decompose $\frac{4x^4 + 27x^3 - 4x^2 - 130x + 184}{(x-4)(x+3)^2(x^2-2x+4)}$.
10. Use an efficient mixture of methods to decompose $\frac{-x^4 + 41x^3 - 120x^2 + 124x - 1845}{(x-5)^2(x+6)(x^2-x+5)}$.

Level V: independent mastery

1. Use an efficient mixture of methods to decompose $\frac{3x^4 + 35x^3 + 76x^2 + 14x + 262}{(x-3)(x+5)^2(x^2+2x+2)}$.
2. Use an efficient mixture of methods to decompose $\frac{-2x^4 - 32x^3 - 144x^2 - 266x - 928}{(x-1)(x+6)^2(x^2+5x+8)}$.
3. Use an efficient mixture of methods to decompose $\frac{x^4 - 5x^3 - 34x^2 + 271x - 533}{(x-5)^2(x-3)(x^2-3x+4)}$.
4. Use an efficient mixture of methods to decompose $\frac{3x^4 - 19x^3 - 58x^2 - 92x - 20}{(x-2)(x+1)^2(x^2+3x+5)}$.
5. Use an efficient mixture of methods to decompose $\frac{2x^4 + 5x^3 + 23x^2 - 5x + 23}{(x+1)^2(x+5)(x^2+x+6)}$.
6. Use an efficient mixture of methods to decompose $\frac{-5x^4 + 49x^3 - 136x^2 + 296x - 1149}{(x-6)(x-5)^2(x^2-x+3)}$.
7. Use an efficient mixture of methods to decompose $\frac{2x^4 - 50x^3 + 371x^2 - 971x + 508}{(x-6)^2(x-5)(x^2-x+4)}$.
8. Use an efficient mixture of methods to decompose $\frac{4x^4 + 47x^3 + 222x^2 + 537x + 438}{(x+5)^2(x+6)(x^2-x+6)}$.
9. Use an efficient mixture of methods to decompose $\frac{-3x^4 + 18x^3 - 10x^2 + 48x - 225}{x(x-5)^2(x^2+3)}$.
10. Use an efficient mixture of methods to decompose $\frac{-9x^4 + 9x^3 + 28x^2 - 83x + 75}{(x-1)^2(x+4)(x^2-2x+3)}$.

Chapter 28

Verifying Decompositions

Learning target

Check a proposed partial-fraction identity and diagnose errors.

28.1 Recognize the shape

A decomposition is supplied rather than requested.

28.2 Procedure

Recombine the right side or clear denominators; compare the resulting numerator with the original numerator.

Common failure

Checking a proposed identity at only one numerical value. Agreement at one value does not prove an identity.

Worked example

Problem. Determine whether $\frac{-2x - 56}{(x - 5)(x + 6)} = \frac{4}{x + 6} - \frac{6}{x - 5}$ is valid.

Diagnosis. The shape is proposed rational identity. The appropriate tool is: recombine over a common denominator or clear denominators and compare.

Result. Valid.

Check. Clear denominators and compare the resulting polynomial identity.

28.3 Mastery exercises

Level I: direct recognition

1. Determine whether $\frac{-x - 26}{(x - 6)(x + 2)} = \frac{3}{x + 2} - \frac{4}{x - 6}$ is valid.
2. Determine whether $\frac{33}{(x - 6)(x + 5)} = -\frac{5}{x + 5} + \frac{3}{x - 6}$ is valid.
3. Determine whether $\frac{-2x - 56}{(x - 5)(x + 6)} = \frac{4}{x + 6} - \frac{6}{x - 5}$ is valid.
4. Determine whether $\frac{-8x + 18}{x(x - 3)} = -\frac{2}{x - 3} - \frac{4}{x}$ is valid.

5. Determine whether $\frac{6x+33}{(x+5)(x+6)} = \frac{3}{x+6} + \frac{3}{x+5}$ is valid.
6. Determine whether $\frac{10}{(x+1)(x+6)} = -\frac{4}{x+6} + \frac{2}{x+1}$ is valid.
7. Determine whether $\frac{-7x-16}{(x+2)(x+3)} = -\frac{5}{x+3} - \frac{2}{x+2}$ is valid.
8. Determine whether $\frac{-8x-32}{(x+3)(x+7)} = -\frac{8}{x+7} - \frac{2}{x+3}$ is valid.
9. Determine whether $\frac{5x+29}{(x+1)(x+5)} = -\frac{1}{x+5} + \frac{6}{x+1}$ is valid.
10. Determine whether $\frac{6x-36}{(x-7)(x-4)} = \frac{6}{x-4} + \frac{2}{x-7}$ is valid.

Level II: routine variations

1. Determine whether $\frac{-5x+13}{(x-1)(x+7)} = -\frac{6}{x+7} + \frac{1}{x-1}$ is valid.
2. Determine whether $\frac{-2x+10}{(x-1)(x+1)} = -\frac{6}{x+1} + \frac{3}{x-1}$ is valid.
3. Determine whether $\frac{2x-4}{(x-5)(x-4)} = -\frac{4}{x-4} + \frac{6}{x-5}$ is valid.
4. Determine whether $\frac{7x+2}{x(x+1)} = \frac{4}{x+1} + \frac{2}{x}$ is valid.
5. Determine whether $\frac{11x-19}{(x-4)(x+1)} = \frac{6}{x+1} + \frac{5}{x-4}$ is valid.
6. Determine whether $\frac{7x+36}{(x+5)(x+6)} = \frac{8}{x+6} + \frac{1}{x+5}$ is valid.
7. Determine whether $\frac{-12x+18}{(x-5)(x+2)} = -\frac{6}{x+2} - \frac{6}{x-5}$ is valid.
8. Determine whether $\frac{x+7}{(x+3)(x+4)} = -\frac{2}{x+4} + \frac{4}{x+3}$ is valid.
9. Determine whether $\frac{x+5}{(x+3)(x+4)} = -\frac{1}{x+4} + \frac{2}{x+3}$ is valid.
10. Determine whether $\frac{2x+7}{(x+1)(x+2)} = -\frac{1}{x+2} + \frac{5}{x+1}$ is valid.

Level III: disguised or prepared forms

1. Determine whether $\frac{8x+22}{(x+2)(x+5)} = \frac{6}{x+5} + \frac{2}{x+2}$ is valid.
2. Determine whether $\frac{5x+6}{(x-6)(x+3)} = \frac{1}{x+3} + \frac{3}{x-6}$ is valid.

3. Determine whether $\frac{-5x+2}{(x-1)(x+2)} = -\frac{4}{x+2} - \frac{1}{x-1}$ is valid.
4. Determine whether $\frac{-x-11}{(x-4)(x+1)} = \frac{2}{x+1} - \frac{4}{x-4}$ is valid.
5. Determine whether $\frac{-7x+4}{(x-4)(x+2)} = -\frac{3}{x+2} - \frac{4}{x-4}$ is valid.
6. Determine whether $\frac{-2x^2+58x+240}{(x-5)(x+3)(x+5)} = -\frac{4}{x+5} - \frac{3}{x+3} + \frac{6}{x-5}$ is valid.
7. Determine whether $\frac{-12x^2-41x+160}{(x-7)(x+4)(x+6)} = -\frac{1}{x+6} - \frac{6}{x+4} - \frac{5}{x-7}$ is valid.
8. Determine whether $\frac{-2x^2+35x+99}{(x-5)(x+2)(x+3)} = -\frac{3}{x+3} - \frac{1}{x+2} + \frac{4}{x-5}$ is valid.
9. Determine whether $\frac{-x^2+19x+122}{(x-1)(x+4)(x+6)} = -\frac{2}{x+6} - \frac{3}{x+4} + \frac{4}{x-1}$ is valid.
10. Determine whether $\frac{7x^2+43x+20}{(x-1)(x+4)(x+6)} = \frac{1}{x+6} + \frac{3}{x+4} + \frac{2}{x-1}$ is valid.

Level IV: multi-stage execution

1. Determine whether $\frac{-7x^2+30x+13}{(x-4)(x-1)(x+3)} = -\frac{5}{x+3} - \frac{3}{x-1} + \frac{1}{x-4}$ is valid.
2. Determine whether $\frac{-3x^2-25x+112}{(x-5)(x-1)(x+6)} = \frac{1}{x+6} - \frac{3}{x-1} - \frac{2}{x-5}$ is valid.
3. Determine whether $\frac{3x^2+x-126}{x(x+6)(x+7)} = \frac{2}{x+7} + \frac{4}{x+6} - \frac{3}{x}$ is valid.
4. Determine whether $\frac{3x^2-18x+6}{(x-3)(x-2)(x+4)} = \frac{2}{x+4} + \frac{3}{x-2} - \frac{3}{x-3}$ is valid.
5. Determine whether $\frac{-2x^2-77x-402}{(x-6)(x+6)(x+7)} = \frac{3}{x+7} + \frac{1}{x+6} - \frac{6}{x-6}$ is valid.
6. Determine whether $\frac{-x^2-47x-228}{(x-3)(x+4)(x+6)} = \frac{1}{x+6} + \frac{6}{x+4} - \frac{6}{x-3}$ is valid.
7. Determine whether $\frac{-6x^2-15x+18}{x(x-3)(x+6)} = -\frac{2}{x+6} - \frac{3}{x-3} - \frac{1}{x}$ is valid.
8. Determine whether $\frac{-5x^2+34x+64}{(x-6)(x-2)(x+5)} = -\frac{3}{x+5} - \frac{6}{x-2} + \frac{2}{x-6}$ is valid.
9. Determine whether $\frac{20x+168}{x(x-7)(x+4)} = \frac{2}{x+4} + \frac{4}{x-7} - \frac{6}{x}$ is valid.
10. Determine whether $\frac{-4x^2-46x-40}{(x-1)(x+2)(x+5)} = \frac{5}{x+5} - \frac{5}{x+2} - \frac{5}{x-1}$ is valid.

Level V: independent mastery

1. Determine whether $\frac{2x^2 - 72x - 360}{(x - 6)(x + 4)(x + 6)} = \frac{6}{x + 6} + \frac{2}{x + 4} - \frac{6}{x - 6}$ is valid.
2. Determine whether $\frac{x^2 + 18x - 16}{x(x - 4)(x + 2)} = -\frac{4}{x + 2} + \frac{3}{x - 4}$ is valid.
3. Determine whether $\frac{x^2 - 16x + 27}{(x - 5)(x - 1)(x + 2)} = \frac{3}{x + 2} - \frac{1}{x - 1} - \frac{1}{x - 5}$ is valid.
4. Determine whether $\frac{-5x^2 + 63x - 220}{(x - 6)(x - 5)(x + 5)} = -\frac{6}{x + 5} + \frac{2}{x - 5} - \frac{2}{x - 6}$ is valid.
5. Determine whether $\frac{5x^2 - 17}{(x - 4)(x + 3)(x + 5)} = \frac{6}{x + 5} - \frac{2}{x + 3} + \frac{1}{x - 4}$ is valid.
6. Determine whether $\frac{-8x^2 - 34x + 46}{(x - 5)(x + 1)(x + 4)} = -\frac{4}{x + 1} - \frac{6}{x - 5}$ is valid.
7. Determine whether $\frac{-x^2 - 28x - 48}{(x - 4)(x + 4)(x + 7)} = \frac{3}{x + 7} - \frac{2}{x + 4} - \frac{2}{x - 4}$ is valid.
8. Determine whether $\frac{6x^2 + 72x - 216}{(x - 3)(x - 2)(x + 6)} = -\frac{6}{x + 6} + \frac{6}{x - 2} + \frac{7}{x - 3}$ is valid.
9. Determine whether $\frac{-5x^2 - 24x + 9}{(x - 3)(x + 3)(x + 6)} = -\frac{1}{x + 6} - \frac{2}{x + 3} - \frac{2}{x - 3}$ is valid.
10. Determine whether $\frac{11x^2 + 17x + 36}{(x - 5)(x + 1)(x + 6)} = \frac{6}{x + 6} - \frac{2}{x + 1} + \frac{6}{x - 5}$ is valid.

Part V

Part V: Tool Selection Capstone

Chapter 29

Systematic Decision Process

Classification before calculation

1. Is the task factorization, a rational-expression operation, polynomial division, or partial-fraction decomposition?
2. Is there a greatest common factor or leading negative sign?
3. Which factoring shape is present: squares, cubes, trinomial, grouping, or substitution pattern?
4. For a rational operation, what values are excluded by the original expression?
5. For polynomial division, is the divisor exactly $x - c$? If yes, synthetic division is available; otherwise use long division.
6. For partial fractions, is the expression proper? If not, divide first.
7. Is the denominator completely factored? Classify every factor as distinct/repeated linear or distinct/repeated irreducible quadratic.
8. Write the complete template before determining coefficients.
9. Choose coefficient tools: cover-up, strategic substitution, equating coefficients, or a hybrid.
10. Verify by multiplication, recombination, or the identity $P = DQ + R$.

Do not let the operation sign hide the structure

The visible symbol $+$, $-$, $\times$, or $\div$ tells you the requested operation, but the internal polynomial shapes determine the tools required to complete it.

Chapter 30

Capstone: Identify, Select, Execute, Verify

For every problem, record four items before doing the algebra:

Shape Tool(s) Order of use Verification method.

No problem is labeled by technique. Some require one tool; others require several in sequence.

Level I: direct recognition

1. Factor completely: $24x^3 + 60x^2$
2. Factor completely: $36x^2 - 81$
3. Factor completely: $16x^2 - 8x + 1$
4. Factor completely: $8x^3 + 27$
5. Factor completely: $x^2 - 6x + 8$
6. Factor completely: $3x^2 + 26x + 48$
7. Factor completely: $2x^3 - 8x^2 - 11x + 5$
8. Factor completely: $x^4 + 8x^2 + 7$
9. Factor completely: $2x^6 - 8x^4 - 16x^3 + 64x$
10. Perform the indicated operation or simplification; state all restrictions: $\frac{x^2 + 7x + 12}{x^2 + 8x + 15}$
11. Perform the indicated operation or simplification; state all restrictions: $\frac{(x-7)(x-2)}{(x-8)(x+2)} \cdot \frac{(x-8)(x-4)}{x(x-2)}$
12. Perform the indicated operation or simplification; state all restrictions: $\frac{x(x+4)}{(x-4)(x+1)} \div \frac{x(x+6)}{(x+1)(x+8)}$
13. Perform the indicated operation or simplification; state all restrictions: $\frac{-6}{x-3} - \frac{4}{x-6}$
14. Perform the indicated operation or simplification; state all restrictions: $\frac{\frac{2}{x-1} + \frac{1}{x-3}}{\frac{1}{x-3}}$
15. $(6x^3 - 5x^2 - 8x + 4) \div (-2x + 3)$
16. Divide $-2x^4 - 3x^3 - 5x^2 - 7x + 4$ by $x + 1$.
17. Choose the ordinary method and divide $(-3x^3 - x^2 + x - 6)$ by $(x^2 - x + 2)$.
18. Decompose into partial fractions: $\frac{-3x + 57}{(x-8)(x+3)}$

19. Decompose into partial fractions: $\frac{-6x - 18}{(x + 2)^2}$
20. Decompose into partial fractions: $\frac{-7x^3 - 14x^2 - 30x - 20}{(x^2 + 2)(x^2 + 2x + 4)}$

Level II: routine variations

1. Decompose into partial fractions: $\frac{-2x^3 + 4x^2 + 3x - 15}{(x^2 - 4x + 5)^2}$
2. Decompose into partial fractions: $\frac{-12x^4 - 90x^3 - 239x^2 - 466x - 480}{(x + 1)^2(x + 6)(x^2 + 2x + 8)}$
3. Decompose into partial fractions: $\frac{2x^3 - 20x^2 + 50x - 14}{(x - 6)(x - 4)}$
4. Factor completely: $12x^5 - 96x^3$
5. Factor completely: $4x^6 - 25$
6. Factor completely: $x^2 - 10x + 25$
7. Factor completely: $x^3 - 216$
8. Factor completely: $x^2 + 10x + 21$
9. Factor completely: $9x^2 + 18x - 7$
10. Factor completely: $-8x^3 - 2x^2 + 28x + 7$
11. Factor completely: $x^4 - 3x^2 - 28$
12. Factor completely: $5x^6 - 5x^4 - 5x^3 + 5x$
13. Perform the indicated operation or simplification; state all restrictions: $\frac{x^2 + 4x - 21}{x^2 + 9x + 14}$
14. Perform the indicated operation or simplification; state all restrictions: $\frac{(x - 8)(x + 7)}{(x - 7)(x - 2)} \cdot \frac{(x - 2)(x + 6)}{(x - 1)(x + 7)}$
15. Perform the indicated operation or simplification; state all restrictions: $\frac{(x - 8)(x + 1)}{(x - 4)(x - 3)} \div \frac{(x - 8)(x - 1)}{(x - 3)(x + 8)}$
16. Perform the indicated operation or simplification; state all restrictions: $\frac{6}{x - 6} + \frac{-4}{x - 8}$
17. Perform the indicated operation or simplification; state all restrictions: $\frac{\frac{4}{x^2 - 1} + \frac{1}{x + 1}}{\frac{1}{x + 1}}$
18. $(12x^3 - 25x^2 + 12) \div (-4x + 3)$
19. Divide $x^5 + 6x^4 + 4x^3 - 13x^2 + 13x - 4$ by $x + 4$.
20. Choose the ordinary method and divide $(2x^2 - 8x - 11)$ by $(x - 5)$.

Level III: disguised or prepared forms

1. Decompose into partial fractions: $\frac{4x^2 - 78x - 236}{(x - 6)(x + 2)(x + 4)}$
2. Decompose into partial fractions: $\frac{-5x^2 + 34x - 61}{(x - 4)^3}$
3. Decompose into partial fractions: $\frac{-9x^3 - 68x - 23}{(x^2 - 4x + 8)(x^2 + 4x + 5)}$
4. Decompose into partial fractions: $\frac{4x^3 - 2x^2 + 36x - 19}{(x^2 + 8)^2}$
5. Decompose into partial fractions: $\frac{2x^4 + 46x^3 + 78x^2 - 544x + 1756}{(x - 5)(x + 6)^2(x^2 + 8)}$
6. Decompose into partial fractions: $\frac{-2x^4 + 10x^3 + 14x^2 - 21x + 46}{(x - 6)(x + 2)}$
7. Factor completely: $-35x^8 - 40x^6 + 35x^4$
8. Factor completely: $x^2 + 6x + 8$
9. Factor completely: $16x^4 + 24x^2 + 9$
10. Factor completely: $64x^3 - 27$
11. Factor completely: $x^2 - 10x + 24$
12. Factor completely: $6x^2 + 11x + 3$
13. Factor completely: $8x^3 - 12x^2 - 12x - 2$
14. Factor completely: $x^6 + 6x^3 + 5$
15. Factor completely: $-6x^8 + 24x^6 + 48x^5 - 192x^3$
16. Perform the indicated operation or simplification; state all restrictions: $\frac{x^3 - 8x^2 + 9x + 18}{x^3 - 10x^2 + 19x + 30}$
17. Perform the indicated operation or simplification; state all restrictions: $\frac{x^2 + 4x - 5}{x^2 - 2x - 8} \cdot \frac{(x - 4)(x + 8)}{(x + 3)(x + 5)}$
18. Perform the indicated operation or simplification; state all restrictions: $\frac{x^2 + x - 6}{x^2 - x - 30} \div \frac{(x - 2)(x + 4)}{(x + 5)(x + 7)}$
19. Perform the indicated operation or simplification; state all restrictions: $\frac{5}{(x + 2)^2} + \frac{3}{(x - 4)(x + 2)}$
20. Perform the indicated operation or simplification; state all restrictions: $\frac{-\frac{4}{x+6} + \frac{2}{x-3}}{\frac{1}{x-3}}$

Level IV: multi-stage execution

1. $(3x^3 - 7x^2 - 3x + 6) \div (3x + 2)$

2. Divide $-2x^5 - 6x^4 + 2x^3 + 6x^2 + 5x + 15$ by $x + 3$.
3. Choose the ordinary method and divide $(x^6 - 2x^5 - 24x^4 - 2x^3 + 16x^2 - 20x - 27)$ by $(x - 6)$.
4. Decompose into partial fractions: $\frac{15x^2 - 74x - 200}{(x - 8)(x - 2)(x + 6)}$
5. Decompose into partial fractions: $\frac{2x^3 - 40x^2 + 280x - 578}{(x - 5)^3(x + 7)}$
6. Decompose into partial fractions: $\frac{10x^3 + 38x^2 + 60x + 13}{(x^2 + 3x + 3)(x^2 + 4x + 7)}$
7. Decompose into partial fractions: $\frac{3x^5 - 23x^4 + 81x^3 - 130x^2 + 70x + 75}{(x^2 - 4x + 7)^3}$
8. Decompose into partial fractions: $\frac{-13x^5 + 187x^4 - 1081x^3 + 3098x^2 - 4256x + 1888}{(x - 4)^2(x + 2)(x^2 - 5x + 8)^2}$
9. Decompose into partial fractions: $\frac{-2x^5 - 10x^4 - 13x^3 - 2x^2 - 5x - 12}{x(x + 1)(x + 3)}$
10. Factor completely: $-24x^{10} + 12x^9 + 72x^8$
11. Factor completely: $6x^5 - 24x^3$
12. Factor completely: $196x^3 - 392x^2 + 196x$
13. Factor completely: $2x^5 + 16x^2$
14. Factor completely: $x^2 + 10x + 21$
15. Factor completely: $9x^2 - 3x - 20$
16. Factor completely: $6x^3 - 7x + 1$
17. Factor completely: $3x^6 + 9x^4 - 12x^2$
18. Factor completely: $-3x^6 + 3x^4 + 3x^3 - 3x$
19. Perform the indicated operation or simplification; state all restrictions: $\frac{x^5 + 2x^4 - 4x^3 - 4x^2 - 5x - 6}{x^3 + 4x^2 - 7x - 10}$
20. Perform the indicated operation or simplification; state all restrictions: $\frac{x^2 + 7x + 10}{x^2 + 3x - 40} \cdot \frac{x^2 + x - 30}{x^2 - x - 30}$

Level V: independent mastery

1. Perform the indicated operation or simplification; state all restrictions: $\frac{x^2 + 3x - 4}{x^2 - 12x + 35} \div \frac{x^2 + 6x + 8}{x^2 - 13x + 42}$
2. Perform the indicated operation or simplification; state all restrictions: $\frac{1}{(x - 4)(x^2 + 1)} + \frac{-5}{(x - 7)(x^2 + 1)}$
3. Perform the indicated operation or simplification; state all restrictions: $\frac{\frac{1}{x+2} + \frac{1}{x-6}}{\frac{1}{x+2} - \frac{1}{x-6}}$

4. $(3x^3 + 12x^2 - 3x + 3) \div (-x^2 - 4x + 1)$
5. Divide $5x^5 - 16x^4 + 4x^3 + 2x^2 - 16x + 4$ by $x - 3$.
6. Choose the ordinary method and divide $(2x^6 + 9x^5 + x^4 - 12x^3 + 6x^2 + 5x - 10)$ by $(2x + 3)$.
7. Decompose into partial fractions: $\frac{-x^3 - 2x^2 + 229x - 646}{(x - 8)(x - 6)(x - 1)(x + 5)}$
8. Decompose into partial fractions: $\frac{2x^4 - 25x^3 + 105x^2 - 150x + 5}{(x - 5)(x - 4)^4}$
9. Decompose into partial fractions: $\frac{2x^3 - 16x^2 + 30x - 17}{(x^2 - 5x + 8)(x^2 - 3x + 7)}$
10. Decompose into partial fractions: $\frac{-2x^6 - 52x^5 - 406x^4 - 1505x^3 - 2856x^2 - 2520x - 556}{(x - 4)(x^2 + 5x + 7)^3}$
11. Decompose into partial fractions: $\frac{-4x^6 - 19x^5 + 47x^4 + 8x^3 - 252x^2 - 501x + 177}{(x + 3)^2(x + 5)(x^2 - 4x + 7)^2}$
12. Decompose into partial fractions: $\frac{2x^6 - 17x^5 + 29x^4 + 40x^3 - 53x^2 - 6x + 99}{(x - 5)(x - 4)(x + 1)}$
13. Factor completely: $56x^3 - 576x^2 + 648x$
14. Factor completely: $x^4 - 2401$
15. Factor completely: $243x^6 + 54x^3 + 3$
16. Factor completely: $2x^7 - 2x$
17. Factor completely: $x^2 - 3x - 10$
18. Factor completely: $35x^2 - 9x - 56$
19. Factor completely: $12x^3 + x^2 - 3x - 2$
20. Factor completely: $2x^8 - 18x^5 + 40x^2$

Part VI

Part VI: Reference and Answer Key

Chapter 31

Formula and Procedure Reference

Core factoring identities

$$\begin{aligned}
 A^2 - B^2 &= (A - B)(A + B), \\
 A^2 \pm 2AB + B^2 &= (A \pm B)^2, \\
 A^3 - B^3 &= (A - B)(A^2 + AB + B^2), \quad A^3 + B^3 = (A + B)(A^2 - AB + B^2).
 \end{aligned}$$

Polynomial division

For nonzero $D(x)$ there are unique polynomials $Q(x), R(x)$ such that

$$P(x) = D(x)Q(x) + R(x), \quad \deg R < \deg D.$$

Partial-fraction templates

$$\begin{aligned}
 (x - r)^m : & \quad \frac{A_1}{x - r} + \frac{A_2}{(x - r)^2} + \cdots + \frac{A_m}{(x - r)^m}, \\
 q(x)^m : & \quad \frac{A_1x + B_1}{q(x)} + \frac{A_2x + B_2}{q(x)^2} + \cdots + \frac{A_mx + B_m}{q(x)^m},
 \end{aligned}$$

where q is irreducible quadratic. Add the contributions from every distinct denominator factor.

Chapter 32

Answer Key

Answers are intentionally concise, as in the companion division workbook. Rework any missed problem from its recognition cue rather than memorizing the final form.

A. Greatest Common Factor and Sign Extraction

- | | |
|---------------------------------------|--------------------------------------|
| A1. $12x^2(2x + 5)$ | A26. $8x^3(6x^5 - x^3 + 6)$ |
| A2. $3x(2x + 5)$ | A27. $6x^3(2x^4 - 6x^3 - 1)$ |
| A3. $6x^2(7x + 11)$ | A28. $-10x^3(5x^2 - 5x + 2)$ |
| A4. $-5x(7x^2 - 10)$ | A29. $-9x^6(x + 1)(8x - 3)$ |
| A5. $8x^2(5x - 11)$ | A30. $-8x^5(8x^3 + 5x^2 - 7)$ |
| A6. $4x^2(9x + 4)$ | A31. $-12x^8(x - 2)(2x + 3)$ |
| A7. $-6x^2(2x^2 + 9)$ | A32. $6x^4(5x^3 - 6x^2 + 6)$ |
| A8. $-12x^2(7x^2 - 4)$ | A33. $-8x^4(x^6 - 8x^2 + 5)$ |
| A9. $-8x(x + 11)$ | A34. $6x^2(4x^3 + 8x^2 + 7)$ |
| A10. $-12x^2(x - 7)$ | A35. $9x^7(8x^2 - 7x - 4)$ |
| A11. $12x^3(x^2 - 8)$ | A36. $-10x^4(4x^3 - x + 1)$ |
| A12. $-12x(2x^2 - 3)$ | A37. $6x^5(2x^5 + 8x + 3)$ |
| A13. $4x(3x + 8)$ | A38. $-9x^4(6x^5 - 7x^4 + 2)$ |
| A14. $-9x^3(5x - 1)$ | A39. $4x^7(5x^2 + 3x - 6)$ |
| A15. $12x^2(9x^2 + 4)$ | A40. $-3x^6(7x^4 + 5x - 6)$ |
| A16. $-12x(5x^2 + 7)$ | A41. $8x(x - 9)(7x - 9)$ |
| A17. $-3x^3(6x + 7)$ | A42. $4x^2(x + 2)(x + 9)$ |
| A18. $4x^3(4x + 11)$ | A43. $-5x(x - 3)(x + 2)$ |
| A19. $-5x^2(6x^2 + 1)$ | A44. $6x^4(x - 6)(3x - 1)$ |
| A20. $10x^3(7x - 4)$ | A45. $-6x^2(x - 6)(8x + 1)$ |
| A21. $-5x^4(7x^4 + 8x^2 - 7)$ | A46. $8x^3(x + 5)(5x + 2)$ |
| A22. $-12x(8x^3 - x^2 + 6)$ | A47. $3x^2(x + 2)(x + 3)$ |
| A23. $-5x^5(5x^3 - 2x + 2)$ | A48. $12x^5(x + 9)(7x - 8)$ |
| A24. $5x^2(4x^5 - 6x^4 + 5)$ | A49. $4x^2(x - 3)(9x + 8)$ |
| A25. $-10x^4(2x^4 - 3x^2 + 6)$ | A50. $4x^3(x - 9)(x + 3)$ |

B. Difference of Squares

- B1.** $(2x - 1)(2x + 1)$
B2. $(5x - 2)(5x + 2)$
B3. $(x - 3)(x + 3)$
B4. $9(2x - 3)(2x + 3)$
B5. $(5x - 9)(5x + 9)$
B6. $(5x - 2)(5x + 2)$
B7. $(6x - 1)(6x + 1)$
B8. $4(2x - 1)(2x + 1)$
B9. $49(x - 1)(x + 1)$
B10. $(2x - 9)(2x + 9)$
B11. $9(x^2 - 2)(x^2 + 2)$
B12. $(2x^2 - 3)(2x^2 + 3)$
B13. $(5x^2 - 1)(5x^2 + 1)$
B14. $(2x^3 - 5)(2x^3 + 5)$
B15. $(5x^2 - 7)(5x^2 + 7)$
B16. $(x^2 - 6)(x^2 + 6)$
B17. $(x - 1)(x + 1)(x^2 + 1)$
B18. $(4x^4 - 7)(4x^4 + 7)$
B19. $(x^2 - 3)(x^2 + 3)$
B20. $(2x^3 - 5)(2x^3 + 5)$
B21. $(4x + 1)(4x + 5)$
B22. $(3x + 1)(3x + 7)$
B23. $9x(x + 4)$
B24. $(x + 2)(x + 4)$
B25. $(x + 2)(x + 10)$
B26. $4(x + 1)(x + 3)$
B27. $5(x + 2)(5x + 2)$
B28. $4(x + 2)(x + 3)$
B29. $(2x + 1)(2x + 9)$
B30. $(x + 1)(x + 3)$
B31. $6x^2(x - 4)(x + 4)$
B32. $5x(2x - 1)(2x + 1)$
B33. $2x^2(3x - 1)(3x + 1)$
B34. $6x^3(x - 2)(x + 2)$
B35. $5x^2(x - 1)(x + 1)$
B36. $4x^2(x - 7)(x + 7)$
B37. $2x^2(x - 4)(x + 4)$
B38. $16x^2(2x - 1)(2x + 1)$
B39. $8x(x - 1)(x + 1)$
B40. $32x(x - 2)(x + 2)$
B41. $16(x - 3)(x + 3)(x^2 + 9)$
B42. $(4x - 3)(4x + 3)(16x^2 + 9)$
B43. $16(x - 2)(x + 2)(x^2 + 4)$
B44. $(x - 7)(x + 7)(x^2 + 49)$
B45. $(2x - 5)(2x + 5)(4x^2 + 25)$
B46. $16(x - 2)(x + 2)(x^2 + 4)$
B47. $(3x - 5)(3x + 5)(9x^2 + 25)$
B48. $(3x - 4)(3x + 4)(9x^2 + 16)$
B49. $81(x - 1)(x + 1)(x^2 + 1)$
B50. $(4x - 1)(4x + 1)(16x^2 + 1)$

C. Perfect-Square Trinomials

- C1.** $(5x - 8)^2$
C2. $(x + 1)^2$
C3. $4(2x - 1)^2$
C4. $9(x + 1)^2$
C5. $(5x + 6)^2$
C6. $(x + 5)^2$
C7. $(4x - 1)^2$
C8. $(3x + 4)^2$
C9. $(3x + 4)^2$
C10. $(x + 4)^2$

- C11.** $(x - 5)^2$
C12. $16(x - 2)^2$
C13. $(x + 7)^2$
C14. $(x + 6)^2$
C15. $(3x - 5)^2$
C16. $(6x + 1)^2$
C17. $(x - 5)^2$
C18. $4(3x + 2)^2$
C19. $(3x + 4)^2$
C20. $4(3x + 1)^2$
C21. $16(x + 1)^2$
C22. $4(2x + 3)^2$
C23. $(3x^2 - 7)^2$
C24. $(7x^2 - 5)^2$
C25. $(4x - 5)^2$
C26. $(x + 1)^2$
C27. $(4x^2 + 3)^2$
C28. $(x - 7)^2$
C29. $(x^2 - 6)^2$
C30. $(2x + 5)^2$
C31. $5x^2(7x - 5)^2$
C32. $8x(4x + 3)^2$
C33. $18x(2x - 3)^2$
C34. $8x(x^2 + 2)^2$
C35. $45x(x + 1)^2(x^2 - x + 1)^2$
C36. $2x^2(x^2 - 7)^2$
C37. $196x(x - 1)^2$
C38. $24x(4x + 1)^2$
C39. $54x^2(2x^2 - 1)^2$
C40. $45x(x + 3)^2$
C41. $6x(3x + 4)^2$
C42. $3x(5x - 9)^2$
C43. $20x^2(3x^2 + 2)^2$
C44. $4x^2(3x - 5)^2$
C45. $4(3x + 8)^2$
C46. $5x^2(8x^4 - 3)^2$
C47. $3(9x^3 + 1)^2$
C48. $125x^2(x + 1)^2$
C49. $16x(2x - 1)^2(2x + 1)^2$
C50. $2x(2x^4 - 1)^2$

D. Sum and Difference of Cubes

- D1.** $(x - 6)(x^2 + 6x + 36)$
D2. $8(x - 1)(x^2 + x + 1)$
D3. $8(2x + 1)(4x^2 - 2x + 1)$
D4. $(2x + 3)(4x^2 - 6x + 9)$
D5. $(x + 6)(x^2 - 6x + 36)$
D6. $(x + 3)(x^2 - 3x + 9)$
D7. $(4x + 5)(16x^2 - 20x + 25)$
D8. $27(x + 2)(x^2 - 2x + 4)$
D9. $8(x + 1)(x^2 - x + 1)$
D10. $(2x + 3)(4x^2 - 6x + 9)$
D11. $27(x + 1)(x^2 - x + 1)$
D12. $(x + 2)(x^2 - 2x + 4)$
D13. $(2x + 1)(4x^2 - 2x + 1)$
D14. $(x + 6)(x^2 - 6x + 36)$
D15. $(4x + 5)(16x^2 - 20x + 25)$
D16. $(2x + 1)(4x^2 - 2x + 1)$
D17. $(x + 4)(x^2 - 4x + 16)$
D18. $(x - 1)(x^2 + x + 1)$
D19. $64(x + 1)(x^2 - x + 1)$
D20. $(x - 6)(x^2 + 6x + 36)$
D21. $(4x^2 + 3)(4x^2 - 6x + 3)(4x^2 + 6x + 3)$
D22. $(2x^2 + 1)(4x^4 - 2x^2 + 1)$

- D23.** $(x^2 + 6)(x^4 - 6x^2 + 36)$
D24. $(3x^2 - 5)(9x^4 + 15x^2 + 25)$
D25. $(3x^2 - 5)(9x^4 + 15x^2 + 25)$
D26. $(3x - 5)(9x^2 + 15x + 25)$
D27. $(x^2 + 5)(x^4 - 5x^2 + 25)$
D28. $(2x - 1)(2x + 1)(4x^2 - 2x + 1)(4x^2 + 2x + 1)$
D29. $(3x - 4)(9x^2 + 12x + 16)$
D30. $(4x - 3)(16x^2 + 12x + 9)$
D31. $2x(3x + 4)(9x^2 - 12x + 16)$
D32. $40x^2(x - 1)(x + 1)(x^2 - x + 1)(x^2 + x + 1)$
D33. $3(2x - 5)(4x^2 + 10x + 25)$
D34. $2(x - 5)(x^2 + 5x + 25)$
D35. $5x(3x^2 - 5)(9x^4 + 15x^2 + 25)$
D36. $40(x^2 + 1)(x^4 - x^2 + 1)$
D37. $135x^2(x + 1)(x^2 - x + 1)$
D38. $5(3x + 4)(9x^2 - 12x + 16)$
D39. $5x^2(x^2 - 2)(x^4 + 2x^2 + 4)$
D40. $2x^2(x + 2)(x^2 - 2x + 4)$
D41. $16x^2(x^2 + 1)(x^4 - x^2 + 1)$
D42. $3(x + 1)(x^2 - x + 1)(x^6 - x^3 + 1)$
D43. $24x(x - 1)(x^2 + x + 1)$
D44. $3x^2(x - 1)(x^2 + x + 1)$
D45. $24x(x + 3)(x^2 - 3x + 9)$
D46. $2x(3x + 5)(9x^2 - 15x + 25)$
D47. $5x^2(x^3 + 6)(x^6 - 6x^3 + 36)$
D48. $5x^2(x^2 + 6)(x^4 - 6x^2 + 36)$
D49. $2x^2(x + 3)(x^2 - 3x + 9)$
D50. $2x(x - 1)(x + 1)(x^2 - x + 1)(x^2 + x + 1)$

E. Monic Quadratic Trinomials

- E1.** $(x - 8)(x - 1)$
E2. $(x - 5)(x - 4)$
E3. $(x - 4)(x - 2)$
E4. $(x - 6)(x - 2)$
E5. $(x - 7)(x - 2)$
E6. $(x - 2)(x - 1)$
E7. $(x - 7)(x - 4)$
E8. $(x - 9)(x - 7)$
E9. $(x - 8)(x - 3)$
E10. $(x - 7)(x - 4)$
E11. $(x + 6)(x + 8)$
E12. $(x + 7)(x + 9)$
E13. $(x + 3)(x + 7)$
E14. $(x + 2)(x + 6)$
E15. $(x + 4)(x + 8)$
E16. $(x + 1)^2$
E17. $(x + 2)(x + 3)$
E18. $(x + 7)(x + 9)$
E19. $(x + 6)(x + 8)$
E20. $(x + 2)(x + 10)$
E21. $(x - 11)(x + 10)$
E22. $(x + 5)(x + 9)$
E23. $(x - 6)(x - 4)$
E24. $(x - 6)(x + 2)$
E25. $(x - 8)(x - 3)$
E26. $(x - 11)(x + 1)$
E27. $(x - 10)(x - 4)$
E28. $(x - 7)(x + 10)$
E29. $(x - 10)(x + 5)$
E30. $(x - 5)(x + 5)$
E31. $(x - 5)(x + 3)$
E32. $(x - 8)(x + 9)$
E33. $(x + 3)(x + 7)$
E34. $(x - 10)(x - 9)$
E35. $(x + 5)(x + 9)$
E36. $(x + 3)(x + 9)$

E37. $x(x + 7)$

E38. $(x - 12)(x + 12)$

E39. $(x + 4)(x + 8)$

E40. $(x + 5)(x + 7)$

E41. $(x - 4)(x - 2)$

E42. $(x - 1)(x + 11)$

E43. $(x - 5)(x + 2)$

E44. $(x - 3)(x + 4)$

E45. $(x + 3)(x + 5)$

E46. $(x - 5)(x - 3)$

E47. $(x - 10)(x + 3)$

E48. $(x - 8)(x - 5)$

E49. $(x + 1)(x + 11)$

E50. $(x - 9)(x + 4)$

F. General Quadratic Trinomials

F1. $(x + 6)(2x + 1)$

F2. $(x + 8)(2x - 5)$

F3. $(2x + 3)(3x + 7)$

F4. $(2x - 5)(2x + 1)$

F5. $(x - 4)(2x + 5)$

F6. $(x + 6)(3x + 8)$

F7. $(x - 8)(3x - 2)$

F8. $(2x - 1)(3x + 8)$

F9. $(x + 8)(2x - 3)$

F10. $(x + 8)(2x - 3)$

F11. $(4x - 5)(4x + 1)$

F12. $(2x + 7)(3x + 5)$

F13. $(3x + 5)(4x + 5)$

F14. $(3x - 2)(4x - 1)$

F15. $(x - 3)(2x - 7)$

F16. $(3x - 1)(3x + 7)$

F17. $(x + 2)(2x + 5)$

F18. $(3x - 7)(3x - 5)$

F19. $(x - 7)(2x + 3)$

F20. $(3x + 5)(4x + 1)$

F21. $(3x - 5)(5x - 7)$

F22. $(4x - 3)(5x + 8)$

F23. $(x - 4)(2x - 5)$

F24. $(2x - 3)(3x + 2)$

F25. $(x - 3)(3x + 5)$

F26. $(2x + 3)(3x + 1)$

F27. $(2x + 3)(2x + 7)$

F28. $(x - 1)(4x + 1)$

F29. $(3x + 4)(5x + 2)$

F30. $(3x - 5)(3x + 7)$

F31. $(5x - 6)(6x + 7)$

F32. $(6x + 1)(6x + 5)$

F33. $(2x + 7)(5x - 7)$

F34. $(x - 6)(3x - 2)$

F35. $(2x + 3)(5x + 8)$

F36. $(3x - 5)(3x + 4)$

F37. $(2x + 5)(5x - 3)$

F38. $(2x + 3)(3x + 2)$

F39. $(x + 3)(2x + 5)$

F40. $(3x - 2)(4x + 5)$

F41. $(x + 6)(5x + 2)$

F42. $(2x - 5)(7x + 2)$

F43. $(2x + 5)(3x - 5)$

F44. $(x + 7)(3x - 5)$

F45. $(3x - 4)(3x + 7)$

F46. $(5x - 7)(7x + 8)$

F47. $(6x - 7)(7x - 5)$

F48. $(5x - 8)(5x - 6)$

F49. $(2x + 3)(5x - 4)$

F50. $(x - 2)(7x - 1)$

G. Factoring by Grouping

- G1.** $2(x+1)(2x+1)(5x-1)$
G2. $(5x+4)(x^2-3x-1)$
G3. $-(x-4)(2x^2-4x-1)$
G4. $(6x+1)(x^2-x-1)$
G5. $4(3x-1)(2x^2+1)$
G6. $2(3x-2)(5x^2-5x+1)$
G7. $5(x+1)(4x^2+1)$
G8. $-2(x+2)(5x^2-3x-1)$
G9. $(x-5)(2x^2+2x-1)$
G10. $(5x-1)(3x^2+6x-1)$
G11. $-(4x-7)(2x^2-5x-1)$
G12. $-(6x+7)(5x^2+4)$
G13. $-(x+3)(5x^2+7)$
G14. $-6(x-1)(x+1)(x+7)$
G15. $(x-4)(2x^2+7x-1)$
G16. $(2x-1)(x^2-6x-1)$
G17. $-2(4x-7)(3x^2-2)$
G18. $-(3x+5)(x^2+1)$
G19. $-(4x+1)(2x^2-7)$
G20. $(5x-1)(x^2-7)$
G21. $(2x-5)(4x^3+3)$
G22. $-(5x+2)(x^2+7x+1)$
G23. $-(3x-4)(5x^2+2x+1)$
G24. $(2x-7)(x^2+6x-1)$
G25. $-2(3x-1)(6x^3-5)$
G26. $-(x+1)(2x-1)(5x-1)$
G27. $-(x-6)(2x^2+5)$
G28. $3(x+1)(6x^3-1)$
G29. $2(2x+1)(2x^2-4x-1)$
G30. $-(5x-6)(5x^2-6)$
G31. $-3(3x-5)(2x^3+1)$
G32. $9(x-1)(2x^3+1)$
G33. $(3x-4)(6x^2+6x+1)$
G34. $5(x+1)(5x^2-2)$
G35. $-2(4x-3)(3x^3+2)$
G36. $2(x+1)(3x-1)(5x-1)$
G37. $2(x-2)(x-1)(x+2)$
G38. $2(x-1)(x+1)(x+4)$
G39. $(x-1)(6x^2+6x-1)$
G40. $(3x+5)(4x^3+3)$
G41. $-(6x+1)(6x^2-1)$
G42. $(4x-5)(3x^2+6x+1)$
G43. $(5x-3)(5x^2+4x+1)$
G44. $-3(2x+1)(5x^3+4)$
G45. $(4x-5)(4x^2+1)$
G46. $(4x+5)(6x^2-3x-1)$
G47. $-4(3x+2)(3x^2-2)$
G48. $-2(6x-5)(3x^4+2)$
G49. $(3x-2)(4x^2+3x+1)$
G50. $(3x+5)(3x^3-1)$

H. Substitution and Higher-Power Patterns

- H1.** $(x^2-3)(x^2-2)$
H2. $(x^2+1)(x^2+7)$
H3. $(x-1)(x+1)(x^2-2)$
H4. $(x^2-7)(x^2-6)$
H5. $(x^2+2)(x^2+3)$
H6. $(x^2-6)(x^2-5)$
H7. $(x^2-3)(x^2+4)$
H8. $(x-2)(x+2)(x^2-5)$
H9. $(x^2-3)(x^2+2)$
H10. $(x^2+5)(x^2+6)$

H11. $(x-2)(x+2)(x^2+3)$

H12. $(x^2-7)(x^2+4)$

H13. $(x^2+4)(x^2+5)$

H14. $(x^2+2)(x^2+7)$

H15. $(x^2+6)(x^2+7)$

H16. $(x-1)(x+1)(x^2-x+1)(x^2+x+1)$

H17. $(x^3-4)(x^3-2)$

H18. $(x^2+2)(x^2+3)$

H19. $(x^2-5)(x^2+2)$

H20. $(x^2-7)(x^2+2)$

H21. $(x-1)(x^3-3)(x^2+x+1)$

H22. $(x+1)(x^3+5)(x^2-x+1)$

H23. $(x^2+3)(x^2+4)$

H24. $(x^2-5)(x^2+7)$

H25. $(x^3-7)(x^3+4)$

H26. $(x^2+3)(x^2+5)$

H27. $(x^2+1)(x^2+2)$

H28. $(x^2-3)(x^2+1)$

H29. $(x^3-6)(x^3+4)$

H30. $(x^2-5)(x^2+1)$

H31. $2x^2(x-1)(x^3+2)(x^2+x+1)$

H32. $3x^2(x-1)(x+1)(x^2+4)$

H33. $2x(x-1)(x+1)(x^2-7)$

H34. $2x(x^2-7)(x^2-5)$

H35. $3x(x^2+1)(x^2+2)$

H36. $3x(x^3-7)(x^3+5)$

H37. $(x^2+4)(x^2+7)$

H38. $(x-2)(x+2)(x^2-2)$

H39. $(x^2+1)(x^2+5)$

H40. $(x^2+1)(x^2+5)$

H41. $3x^2(x-2)(x+2)(x^2-5)$

H42. $2x^2(x^3-5)(x^3-4)$

H43. $3x(x^2-5)(x^2+1)$

H44. $(x^3-4)(x^3+6)$

H45. $(x-2)(x+2)(x^2-6)$

H46. $(x-1)(x^3-5)(x^2+x+1)$

H47. $(x^3-6)(x^3-5)$

H48. $3x^2(x^4+2)(x^4+3)$

H49. $(x^2-7)(x^2+4)$

H50. $5x(x^2+2)(x^2+4)$

I. Multi-Stage Complete Factorization

I1. $5(x-2)(x-1)(x+1)(x+2)$

I2. $4x^2(x-1)^2(x+1)(x^2+x+1)$

I3. $5x(x-1)(x+4)(x^2+1)$

I4. $5x^2(x+1)(x+2)(2x-5)(2x+5)$

I5. $2x(x-2)^2(x+2)(x^2+2x+4)$

I6. $4x^2(x-2)(x-1)(x+1)(x+2)$

I7. $-3(x-2)^2(x+2)(x^2+2x+4)$

I8. $3x(x-5)(x+3)(x^2+1)$

I9. $3(x-5)(x+1)(x+2)(x+5)$

I10. $5x^2(x-3)^2(x+3)(x^2+3x+9)$

I11. $-6(x-2)(x-1)(x+1)(x+2)$

I12. $2x(x-2)^2(x+2)(x^2+2x+4)$

I13. $2x(x-6)(x+1)(x^2+1)$

I14. $3x(x+1)(x+2)(3x-2)(3x+2)$

I15. $5x(x-1)^2(x+1)(x^2+x+1)$

I16. $5x(x-2)(x-1)(x+1)(x+2)$

I17. $4x^2(x-5)^2(x+5)(x^2+5x+25)$

I18. $-4x(x-2)(x+3)(x^2+1)$

I19. $6x(x+1)(x+2)(3x-1)(3x+1)$

I20. $4(x-2)^2(x+2)(x^2+2x+4)$

I21. $5(x-2)(x-1)(x+1)(x+2)$

I22. $-3x(x-1)^2(x+1)(x^2+x+1)$

I23. $-4x^3(x+2)(x+6)(x^2+1)$

I24. $2x^2(x+1)(x+2)(2x-3)(2x+3)$

I25. $-6x^3(x-2)^2(x+2)(x^2+2x+4)$

I26. $-6x^3(x-2)(x-1)(x+1)(x+2)$

I27. $5x(x-1)^2(x+1)(x^2+x+1)$

I28. $6x^3(x-2)(x+5)(x^2+1)$

I29. $20(x-1)(x+1)^2(x+2)$

I30. $4(x-2)^2(x+2)(x^2+2x+4)$

I31. $2(x-2)(x-1)(x+1)(x+2)$

I32. $-6x(x-1)^2(x+1)(x^2+x+1)$

I33. $-3x^2(x-3)(x+6)(x^2+1)$

I34. $5x^3(x+1)(x+2)(3x-2)(3x+2)$

I35. $-3x(x-1)^2(x+1)(x^2+x+1)$

I36. $3x^3(x-2)(x-1)(x+1)(x+2)$

I37. $2x(x-3)^2(x+3)(x^2+3x+9)$

I38. $-4x^2(x-4)(x-2)(x^2+1)$

I39. $4(x+1)(x+2)(3x-1)(3x+1)$

I40. $2x(x-4)^2(x+4)(x^2+4x+16)$

I41. $2x^2(x-2)(x-1)(x+1)(x+2)$

I42. $-4(x-5)^2(x+5)(x^2+5x+25)$

I43. $-3x^2(x-5)(x+3)(x^2+1)$

I44. $-6(x-4)(x+1)(x+2)(x+4)$

I45. $-4x^3(x-2)^2(x+2)(x^2+2x+4)$

I46. $4x^2(x-2)(x-1)(x+1)(x+2)$

I47. $4x^4(x-2)^2(x+2)(x^2+2x+4)$

I48. $-6x^4(x-1)(x+4)(x^2+1)$

I49. $-4x^2(x+1)(x+2)(2x-3)(2x+3)$

I50. $-4(x-2)^2(x+2)(x^2+2x+4)$

J. Simplification by Factoring and Cancellation

J1. $\frac{x+6}{x-5}$, with $x \neq 5, 6$.

J2. $\frac{x+4}{x+3}$, with $x \neq -3, 7$.

J3. $\frac{x-3}{x+1}$, with $x \neq -1, 4$.

J4. $\frac{x+2}{x}$, with $x \neq -5, 0$.

J5. $\frac{x-2}{x+5}$, with $x \neq -6, -5$.

J6. $\frac{x-4}{x-6}$, with $x \neq -7, 6$.

J7. $\frac{x-5}{x-3}$, with $x \neq 1, 3$.

J8. $\frac{x+4}{x+5}$, with $x \neq -5, -3$.

J9. $\frac{x-2}{x+7}$, with $x \neq -7, 5$.

J10. $\frac{x+5}{x+2}$, with $x \neq -7, -2$.

J11. $\frac{x+5}{x-6}$, with $x \neq -2, 6$.

J12. $\frac{x+3}{x+2}$, with $x \neq -2, 3$.

J13. $\frac{x-4}{x+4}$, with $x \neq -4, -3$.

J14. $\frac{x}{x-3}$, with $x \neq 3, 6$.

J15. $\frac{x+1}{x+7}$, with $x \neq -7, -5$.

J16. $\frac{x-4}{x-3}$, with $x \neq 3, 7$.

J17. $\frac{x+1}{x-6}$, with $x \neq -7, 6$.

J18. $\frac{x-3}{x+2}$, with $x \neq -7, -2$.

J19. $\frac{x+3}{x+4}$, with $x \neq -4, 0$.

J20. $\frac{x-5}{x-1}$, with $x \neq -7, 1$.

J21. $\frac{x+6}{x+2}$, with $x \neq -2, 1, 5$.

J22. $\frac{x-1}{x-3}$, with $x \neq -1, 2, 3$.

J23. $\frac{x+4}{x+5}$, with $x \neq -5, -1, 5$.

J24. $\frac{x+5}{x-3}$, with $x \neq -3, 3, 6$.

J25. $\frac{x-7}{x-5}$, with $x \neq -2, 3, 5$.

J26. $\frac{x}{x-5}$, with $x \neq -5, 3, 5$.

$$\mathbf{J27.} \quad \frac{x-1}{x-4}, \text{ with } x \neq -7, 4, 6.$$

$$\mathbf{J28.} \quad \frac{x-3}{x-5}, \text{ with } x \neq -1, 5, 6.$$

$$\mathbf{J29.} \quad \frac{x-4}{x}, \text{ with } x \neq 0, 3, 5.$$

$$\mathbf{J30.} \quad \frac{x-4}{x-5}, \text{ with } x \neq -5, 3, 5.$$

$$\mathbf{J31.} \quad \frac{x^3 - 3x^2 + x - 3}{x+3}, \text{ with } x \neq -6, -3, -2.$$

$$\mathbf{J32.} \quad \frac{x^3 - 7x^2 + x - 7}{x-6}, \text{ with } x \neq -1, 1, 6.$$

$$\mathbf{J33.} \quad \frac{x^3 - 4x^2 + x - 4}{x+4}, \text{ with } x \neq -4, -1, 2.$$

$$\mathbf{J34.} \quad \frac{x^3 + 3x^2 + x + 3}{x+4}, \text{ with } x \neq -4, -2, 4.$$

$$\mathbf{J35.} \quad \frac{x^3 + 4x^2 + x + 4}{x}, \text{ with } x \neq -1, 0, 2.$$

$$\mathbf{J36.} \quad \frac{x^3 + 2x^2 + x + 2}{x+1}, \text{ with } x \neq -6, -1, 3.$$

$$\mathbf{J37.} \quad \frac{x^3 - 4x^2 + x - 4}{x+1}, \text{ with } x \neq -5, -1, 2.$$

$$\mathbf{J38.} \quad \frac{x^3 + 3x^2 + x + 3}{x+5}, \text{ with } x \neq -5, -1, 2.$$

$$\mathbf{J39.} \quad \frac{x^3 + 5x^2 + x + 5}{x+1}, \text{ with } x \neq -4, -1, 5.$$

$$\mathbf{J40.} \quad \frac{x^3 - 6x^2 + x - 6}{x+4}, \text{ with } x \neq -4, 3, 7.$$

$$\mathbf{J41.} \quad \frac{x^3 + 5x^2 + x + 5}{x-3}, \text{ with } x \neq -3, -1, 3.$$

$$\mathbf{J42.} \quad \frac{x^3 + 3x^2 + x + 3}{x+1}, \text{ with } x \neq -1, 3, 7.$$

$$\mathbf{J43.} \quad \frac{x^3 - 4x^2 + x - 4}{x-3}, \text{ with } x \neq -6, -4, 3.$$

$$\mathbf{J44.} \quad \frac{x^3 - 6x^2 + x - 6}{x-7}, \text{ with } x \neq -7, 5, 7.$$

$$\mathbf{J45.} \quad \frac{x^3 + 6x^2 + x + 6}{x-1}, \text{ with } x \neq -5, 1, 4.$$

$$\mathbf{J46.} \quad \frac{x^3 + 7x^2 + x + 7}{x+4}, \text{ with } x \neq -4, -3, 4.$$

$$\mathbf{J47.} \quad \frac{x^3 + 5x^2 + x + 5}{x}, \text{ with } x \neq -3, -1, 0.$$

$$\mathbf{J48.} \quad \frac{x^3 + 5x^2 + x + 5}{x-1}, \text{ with } x \neq -6, 1, 7.$$

$$\mathbf{J49.} \quad \frac{x^3 - 6x^2 + x - 6}{x-5}, \text{ with } x \neq -7, -4, 5.$$

$$\mathbf{J50.} \quad \frac{x^3 + 6x^2 + x + 6}{x}, \text{ with } x \neq -3, 0, 5.$$

K. Multiplication of Rational Expressions

$$\mathbf{K1.} \quad \frac{x^2 - 11x + 28}{x^2 + 2x}, \text{ with } x \neq -2, 0, 2, 8.$$

$$\mathbf{K2.} \quad \frac{x^2 - 7x + 10}{x^2 - x - 56}, \text{ with } x \neq -7, -5, -4, 8.$$

$$\mathbf{K3.} \quad \frac{x^2 - x}{x^2 - x - 20}, \text{ with } x \neq -4, -3, -1, 5.$$

$$\mathbf{K4.} \quad \frac{x^2 + 8x + 7}{x^2 - 2x - 8}, \text{ with } x \neq -2, 1, 3, 4.$$

$$\mathbf{K5.} \quad \frac{x^2 - 8x + 12}{x^2 + 7x + 6}, \text{ with } x \neq -6, -5, -2, -1.$$

$$\mathbf{K6.} \quad \frac{x^2 - x}{x^2 + 6x + 5}, \text{ with } x \neq -6, -5, -1, 3.$$

$$\mathbf{K7.} \quad \frac{x^2 + 4x - 32}{x^2 + 13x + 42}, \text{ with } x \neq -7, -6, -3, 1.$$

$$\mathbf{K8.} \quad \frac{x^2 - 4x}{x^2 - 14x + 48}, \text{ with } x \neq -1, 2, 6, 8.$$

$$\mathbf{K9.} \quad \frac{x^2 + 13x + 42}{x^2 + 5x + 4}, \text{ with } x \neq -4, -1, 0, 7.$$

$$\mathbf{K10.} \quad \frac{x^2 - 3x - 28}{x^2 - 13x + 40}, \text{ with } x \neq -8, 5, 6, 8.$$

$$\mathbf{K11.} \quad \frac{x^2 - 2x - 48}{x^2 - 8x + 7}, \text{ with } x \neq -7, 1, 2, 7.$$

$$\mathbf{K12.} \quad \frac{x^2 + 3x - 10}{x^2 + 4x - 32}, \text{ with } x \neq -8, -6, -3, 4.$$

$$\mathbf{K13.} \quad \frac{x^2 - 12x + 32}{x^2 + x - 42}, \text{ with } x \neq -7, -2, 1, 6.$$

$$\mathbf{K14.} \quad \frac{x^2 + 7x + 10}{x^2 - x - 12}, \text{ with } x \neq -8, -3, 0, 4.$$

$$\mathbf{K15.} \quad \frac{x^2 - 2x - 8}{x^2 - 7x + 6}, \text{ with } x \neq -1, 1, 6, 8.$$

$$\mathbf{K16.} \quad \frac{x^2 - 4x}{x^2 - 9}, \text{ with } x \neq -5, -3, 1, 3.$$

- K17.** $\frac{x^2 + 5x}{x^2 + 12x + 32}$, with $x \neq -8, -4, 6, 7$.
- K18.** $\frac{x^2 + 10x + 16}{x^2 - 9x + 8}$, with $x \neq -6, 1, 7, 8$.
- K19.** $\frac{x^2 - x - 2}{x^2 - 4x}$, with $x \neq -8, -5, 0, 4$.
- K20.** $\frac{x^2 + 8x}{x^2 - 11x + 30}$, with $x \neq 1, 5, 6, 7$.
- K21.** $\frac{x^2 + 7x - 8}{x^2 + 5x + 6}$, with $x \neq -5, -3, -2, 4$.
- K22.** $\frac{x^2 + 13x + 40}{x^2 - 6x}$, with $x \neq -6, 0, 1, 6$.
- K23.** $\frac{x^2 - 4x - 21}{x^2 - x}$, with $x \neq -5, -4, 0, 1$.
- K24.** $\frac{x^2 + 3x - 18}{x^2 + 7x - 8}$, with $x \neq -8, -3, -2, 1$.
- K25.** $\frac{x^2 - 7x + 6}{x^2 - 7x - 8}$, with $x \neq -1, 0, 3, 8$.
- K26.** $\frac{x^2 - 9x + 14}{x^2 - 2x - 24}$, with $x \neq -4, 3, 6, 8$.
- K27.** $\frac{x^2 - x - 2}{x^2 + 7x + 12}$, with $x \neq -8, -5, -4, -3$.
- K28.** $\frac{x^2 - 9x + 14}{x^2 + 5x}$, with $x \neq -5, -2, 0, 4$.
- K29.** $\frac{x^2 + 3x - 10}{x^2 + 4x}$, with $x \neq -7, -4, -1, 0$.
- K30.** $\frac{x^2 + 14x + 48}{x^2 - 4x + 3}$, with $x \neq -7, 1, 3, 4$.
- K31.** $\frac{x^2 + 8x + 12}{x^2 + 2x - 48}$, with $x \neq -8, -5, 5, 6$.
- K32.** $\frac{x^2 + 15x + 56}{x^2 - 3x - 10}$, with $x \neq -3, -2, 3, 5$.
- K33.** $\frac{x^2 - x - 12}{x^2 - 15x + 56}$, with $x \neq -4, 6, 7, 8$.
- K34.** $\frac{x^2 - 9x + 20}{x^2 - 64}$, with $x \neq -8, 3, 7, 8$.
- K35.** $\frac{x^2 - 9x + 20}{x^2 + 15x + 56}$, with $x \neq -8, -7, -6, -3$.
- K36.** $\frac{x^2 - 10x + 21}{x^2 + 14x + 48}$, with $x \neq -8, -6, -4, -1$.
- K37.** $\frac{x^2 + 7x + 12}{x^2 + 8x + 7}$, with $x \neq -8, -7, -2, -1$.
- K38.** $\frac{x^2 + 2x}{x^2 - 9x + 18}$, with $x \neq -8, -7, 3, 6$.
- K39.** $\frac{x^2 + 2x}{x^2 + 15x + 56}$, with $x \neq -8, -7, 1, 8$.
- K40.** $\frac{x^2 - 2x - 15}{x^2 + 3x - 28}$, with $x \neq -8, -7, -5, 4$.
- K41.** $\frac{x^2 - 3x - 18}{x^2 - x - 2}$, with $x \neq -2, -1, 1, 2$.
- K42.** $\frac{x^2 - 5x - 14}{x^2 - 14x + 48}$, with $x \neq -8, -4, 6, 8$.
- K43.** $\frac{x^2 + 5x - 6}{x^2 - 5x - 6}$, with $x \neq -1, 5, 6, 7$.
- K44.** $\frac{x^2 + 4x + 3}{x^2 - 10x + 16}$, with $x \neq -8, 0, 2, 8$.
- K45.** $\frac{x^2 + 2x}{x^2 + 12x + 32}$, with $x \neq -8, -4, 1, 6$.
- K46.** $\frac{x^2 + 6x + 5}{x^2 + 6x}$, with $x \neq -6, -4, -2, 0$.
- K47.** $\frac{x^2 + 10x + 16}{x^2 + 5x}$, with $x \neq -5, -3, 0, 5$.
- K48.** $\frac{x^2 + 9x + 14}{x^2 + 5x - 24}$, with $x \neq -8, 1, 3, 4$.
- K49.** $\frac{x^2 - 7x + 12}{x^2 + 8x}$, with $x \neq -8, 0, 7, 8$.
- K50.** $\frac{x^2 + 10x + 16}{x^2 - 11x + 24}$, with $x \neq -1, 1, 3, 8$.

L. Division of Rational Expressions

- L1.** $\frac{x^2 - 4x - 21}{x^2 - 4x}$, with $x \neq -8, -5, 0, 4, 7$.
- L2.** $\frac{x^2 - 2x - 15}{x^2 + 2x - 48}$, with $x \neq -8, -6, -1, 5, 6$.
- L3.** $\frac{x^2 - 9x + 14}{x^2 + 9x + 14}$, with $x \neq -7, -2, 1, 6, 7$.

- L4.** $\frac{x^2 + 12x + 32}{x^2 + 2x - 24}$, with $x \neq -8, -6, -1, 0, 4$.
- L5.** $\frac{x^2 - 5x - 24}{x^2 + 2x - 24}$, with $x \neq -8, -7, -6, 4, 8$.
- L6.** $\frac{x^2 - 12x + 35}{x^2 - 3x - 18}$, with $x \neq -3, -2, -1, 5, 6$.
- L7.** $\frac{x^2 + 8x}{x^2 + 11x + 30}$, with $x \neq -6, -5, 0, 1, 2$.
- L8.** $\frac{x^2 + 3x - 10}{x^2 - 5x - 24}$, with $x \neq -3, 2, 3, 6, 8$.
- L9.** $\frac{x^2 + 3x - 18}{x^2 - 5x - 14}$, with $x \neq -2, 1, 3, 4, 7$.
- L10.** $\frac{x^2 + 4x - 5}{x^2 - 4}$, with $x \neq -5, -2, -1, 2, 3$.
- L11.** $\frac{x^2 - 6x + 5}{x^2 - 2x}$, with $x \neq -6, -2, 0, 2, 5$.
- L12.** $\frac{x^2 - 9x + 20}{x^2 - 36}$, with $x \neq -6, -3, 1, 5, 6$.
- L13.** $\frac{x^2 - 16}{x^2 + 7x - 8}$, with $x \neq -8, -7, -1, 1, 4$.
- L14.** $\frac{x^2 + 9x + 8}{x^2 - 5x + 4}$, with $x \neq -8, 1, 3, 4, 8$.
- L15.** $\frac{x^2 + x - 12}{x^2 + 2x - 3}$, with $x \neq -3, -2, 1, 3, 4$.
- L16.** $\frac{x^2 + x - 20}{x^2 - 14x + 48}$, with $x \neq -5, -4, 6, 7, 8$.
- L17.** $\frac{x^2 - 5x - 6}{x^2 + 8x + 12}$, with $x \neq -6, -2, -1, 2, 7$.
- L18.** $\frac{x^2 + x - 12}{x^2 - x - 2}$, with $x \neq -7, -1, 2, 3, 7$.
- L19.** $\frac{x^2 - 6x}{x^2 + 10x + 21}$, with $x \neq -7, -3, -1, 0, 7$.
- L20.** $\frac{x^2 - 12x + 35}{x^2 + 6x + 8}$, with $x \neq -6, -4, -3, -2, 7$.
- L21.** $\frac{x^2 + 11x + 24}{x^2 + 7x}$, with $x \neq -8, -7, -5, 0, 1$.
- L22.** $\frac{x^2 - 5x + 6}{x^2 + 4x}$, with $x \neq -6, -4, 0, 3, 8$.
- L23.** $\frac{x^2 - 3x + 2}{x^2 - 4x - 32}$, with $x \neq -8, -4, 0, 2, 8$.
- L24.** $\frac{x^2 + 10x + 21}{x^2 - 2x - 24}$, with $x \neq -7, -5, -4, 2, 6$.

- L25.** $\frac{x^2 + 8x + 7}{x^2 - 9}$, with $x \neq -7, -4, -3, 3, 4$.
- L26.** $\frac{x^2 + 6x + 5}{x^2 + 4x}$, with $x \neq -4, -1, 0, 2, 5$.
- L27.** $\frac{x^2 - 13x + 40}{x^2 + 6x}$, with $x \neq -6, -4, 0, 4, 5$.
- L28.** $\frac{x^2 - 4x - 12}{x^2 - 4x - 32}$, with $x \neq -6, -4, -2, 3, 8$.
- L29.** $\frac{x^2 - x - 56}{x^2 + 7x + 10}$, with $x \neq -7, -5, -2, 0, 4$.
- L30.** $\frac{x^2 - x - 20}{x^2 + 5x + 6}$, with $x \neq -7, -3, -2, 4, 5$.
- L31.** $\frac{x^2 - 8x}{x^2 - 5x + 6}$, with $x \neq -8, -1, 2, 3, 8$.
- L32.** $\frac{x^2 - 2x - 3}{x^2 + x - 30}$, with $x \neq -6, -5, -3, 3, 5$.
- L33.** $\frac{x^2 - x - 2}{x^2 - 16}$, with $x \neq -4, -2, -1, 4, 6$.
- L34.** $\frac{x^2 + 7x + 6}{x^2 + 2x}$, with $x \neq -5, -2, -1, 0, 3$.
- L35.** $\frac{x^2 + 2x - 48}{x^2 + 2x}$, with $x \neq -8, -2, 0, 1, 8$.
- L36.** $\frac{x^2 + 10x + 16}{x^2 - 7x + 12}$, with $x \neq -7, -2, 1, 3, 4$.
- L37.** $\frac{x^2 - 7x - 8}{x^2 + x - 6}$, with $x \neq -4, -3, -2, -1, 2$.
- L38.** $\frac{x^2 + 12x + 35}{x^2 - 3x - 18}$, with $x \neq -7, -3, 3, 6, 7$.
- L39.** $\frac{x^2 + 2x - 15}{x^2 + x - 6}$, with $x \neq -5, -3, -2, 2, 8$.
- L40.** $\frac{x^2 + 2x - 3}{x^2 - 9x + 18}$, with $x \neq -6, -1, 1, 3, 6$.
- L41.** $\frac{x^2 - 9x + 18}{x^2 - x - 42}$, with $x \neq -7, -6, 0, 6, 7$.
- L42.** $\frac{x^2 - 8x + 15}{x^2 + 3x + 2}$, with $x \neq -2, -1, 0, 5, 8$.
- L43.** $\frac{x^2 + 2x - 24}{x^2 + 5x - 14}$, with $x \neq -7, -4, 2, 3, 4$.
- L44.** $\frac{x^2 - 2x - 48}{x^2 - x - 6}$, with $x \neq -6, -2, 1, 3, 4$.
- L45.** $\frac{x^2 - 7x + 6}{x^2 - 3x - 10}$, with $x \neq -4, -2, 5, 6, 7$.

- L46.** $\frac{x^2 + 3x + 2}{x^2 - x - 56}$, with $x \neq -7, -6, -2, 4, 8$.
- L47.** $\frac{x^2 + 2x - 48}{x^2 - 6x + 5}$, with $x \neq -8, -6, -3, 1, 5$.
- L48.** $\frac{x^2 - 6x - 7}{x^2 - 9x + 18}$, with $x \neq -3, -1, 3, 6, 8$.
- L49.** $\frac{x^2 - 4x - 5}{x^2 - 8x + 12}$, with $x \neq -6, 1, 2, 5, 6$.
- L50.** $\frac{x^2 + 2x - 48}{x^2 + 2x - 3}$, with $x \neq -7, -3, 1, 6, 7$.

M. Addition and Subtraction of Rational Expressions

- M1.** $\frac{-3x + 3}{x^2 + x - 20}$, with $x \neq -5, 4$.
- M2.** $\frac{9x + 12}{x^2 - 16}$, with $x \neq -4, 4$.
- M3.** $\frac{-3x + 8}{x^2 - 8x}$, with $x \neq 0, 8$.
- M4.** $\frac{3x - 46}{x^2 - 5x - 24}$, with $x \neq -3, 8$.
- M5.** $\frac{-3x + 8}{x^2 - x - 42}$, with $x \neq -6, 7$.
- M6.** $\frac{5x + 1}{x^2 + 6x + 5}$, with $x \neq -5, -1$.
- M7.** $\frac{-10x + 48}{x^2 - 9x + 18}$, with $x \neq 3, 6$.
- M8.** $\frac{-5x + 19}{x^2 - 8x + 15}$, with $x \neq 3, 5$.
- M9.** $\frac{-3x + 54}{x^2 + 3x - 40}$, with $x \neq -8, 5$.
- M10.** $\frac{64}{x^2 - 64}$, with $x \neq -8, 8$.
- M11.** $\frac{4x - 16}{x^2 - 8x + 15}$, with $x \neq 3, 5$.
- M12.** $\frac{8x + 35}{x^2 + 7x}$, with $x \neq -7, 0$.
- M13.** $\frac{6x - 17}{x^2 + 3x - 40}$, with $x \neq -8, 5$.
- M14.** $\frac{-7x - 5}{x^2 - 1}$, with $x \neq -1, 1$.
- M15.** $\frac{-7x - 36}{x^2 + x - 42}$, with $x \neq -7, 6$.
- M16.** $\frac{8}{x^2 - 2x - 3}$, with $x \neq -1, 3$.

- M17. $\frac{2x - 24}{x^2 - 14x + 48}$, with $x \neq 6, 8$.
- M18. $\frac{2x - 2}{x^2 - 6x + 8}$, with $x \neq 2, 4$.
- M19. $\frac{5x - 17}{x^2 - 5x + 4}$, with $x \neq 1, 4$.
- M20. $\frac{-2x + 61}{x^2 - x - 56}$, with $x \neq -7, 8$.
- M21. $\frac{x + 28}{x^3 + 14x^2 + 49x}$, with $x \neq -7, 0$.
- M22. $\frac{11x + 78}{x^3 + 22x^2 + 160x + 384}$, with $x \neq -8, -6$.
- M23. $\frac{x - 62}{x^3 - 4x^2 - 60x + 288}$, with $x \neq -8, 6$.
- M24. $\frac{-x - 42}{x^3 + 6x^2 - 36x - 216}$, with $x \neq -6, 6$.
- M25. $-\frac{x}{x^3 + 7x^2 + 16x + 12}$, with $x \neq -3, -2$.
- M26. $\frac{-9x + 6}{x^3 - 13x^2 + 16x + 192}$, with $x \neq -3, 8$.
- M27. $\frac{8x - 14}{x^3 - 12x - 16}$, with $x \neq -2, 4$.
- M28. $\frac{-12x - 66}{x^3 + 18x^2 + 105x + 196}$, with $x \neq -7, -4$.
- M29. $\frac{-9x - 27}{x^3 + 9x^2 + 24x + 16}$, with $x \neq -4, -1$.
- M30. $\frac{-7x + 10}{x^3 - 4x^2 + 5x - 2}$, with $x \neq 1, 2$.
- M31. $\frac{5x - 22}{x^3 - 10x^2 + 28x - 24}$, with $x \neq 2, 6$.
- M32. $\frac{7x + 5}{x^3 + 5x^2 + 3x - 9}$, with $x \neq -3, 1$.
- M33. $\frac{11x + 41}{x^3 + 13x^2 + 48x + 36}$, with $x \neq -6, -1$.
- M34. $\frac{2x - 16}{x^3 + x^2 - 33x + 63}$, with $x \neq -7, 3$.
- M35. $\frac{9x - 69}{x^3 - 23x^2 + 176x - 448}$, with $x \neq 7, 8$.
- M36. $\frac{-6x - 16}{x^3 + 8x^2 + 16x}$, with $x \neq -4, 0$.
- M37. $\frac{x - 28}{x^3 + 4x^2 - 35x - 150}$, with $x \neq -5, 6$.
- M38. $-\frac{2x}{x^3 - 16x^2 + 84x - 144}$, with $x \neq 4, 6$.
- M39. $\frac{3x + 3}{x^3 + 9x^2 + 24x + 20}$, with $x \neq -5, -2$.

- M40.** $\frac{3x + 14}{x^3 + 14x^2 + 32x - 128}$, with $x \neq -8, 2$.
- M41.** $\frac{-x - 5}{x^4 + x^3 - x^2 + x - 2}$, with $x \neq -2, 1$.
- M42.** $\frac{-5x - 11}{x^4 - x^3 - 19x^2 - x - 20}$, with $x \neq -4, 5$.
- M43.** $\frac{-8x - 10}{x^4 + 4x^3 - 4x^2 + 4x - 5}$, with $x \neq -5, 1$.
- M44.** $\frac{10x - 8}{x^4 - 4x^3 - 31x^2 - 4x - 32}$, with $x \neq -4, 8$.
- M45.** $\frac{-5x - 32}{x^4 + 14x^3 + 49x^2 + 14x + 48}$, with $x \neq -8, -6$.
- M46.** $\frac{10x - 36}{x^4 - 6x^3 + x^2 - 6x}$, with $x \neq 0, 6$.
- M47.** $\frac{-4x + 18}{x^4 - 9x^3 + 19x^2 - 9x + 18}$, with $x \neq 3, 6$.
- M48.** $\frac{-4x + 13}{x^4 - 11x^3 + 29x^2 - 11x + 28}$, with $x \neq 4, 7$.
- M49.** $\frac{-6x - 31}{x^4 + 7x^3 + 7x^2 + 7x + 6}$, with $x \neq -6, -1$.
- M50.** $\frac{4x - 6}{x^4 - 3x^3 - 17x^2 - 3x - 18}$, with $x \neq -3, 6$.

N. Complex Rational Expressions

- N1.** $-\frac{x}{4} + \frac{1}{2}$, with $x \neq -2, 6$.
- N2.** $\frac{6x + 3}{x - 5}$, with $x \neq -6, 5$.
- N3.** $\frac{x^2 + 5x}{x + 6}$, with $x \neq -6, -5, 2$.
- N4.** $\frac{x^2 + 5x + 9}{x^2 - 1}$, with $x \neq -2, -1, 1$.
- N5.** $\frac{x}{3} - \frac{2}{3}$, with $x \neq -1, 5$.
- N6.** $\frac{-6x - 8}{x + 1}$, with $x \neq -3, -1$.
- N7.** $\frac{x^2 + 3x}{x + 4}$, with $x \neq -6, -4, -3$.
- N8.** $\frac{x^2 + 5x + 24}{x^2 - 1}$, with $x \neq -5, -1, 1$.
- N9.** $\frac{2x}{9} - \frac{1}{9}$, with $x \neq -4, 5$.
- N10.** $\frac{3x - 7}{x - 1}$, with $x \neq 1, 3$.

- N11.** $\frac{x^2 + 5x}{x + 4}$, with $x \neq -5, -4, -3$.
- N12.** $\frac{x^2 - 5x - 16}{x^2 - 1}$, with $x \neq -3, -1, 1$.
- N13.** $\frac{2x}{9} - \frac{1}{9}$, with $x \neq -4, 5$.
- N14.** $\frac{6}{x - 3}$, with $x \neq 0, 3$.
- N15.** $\frac{x^2 - 3x}{x - 2}$, with $x \neq -3, 2, 3$.
- N16.** $\frac{x^2 + 2x - 1}{x^2 - 1}$, with $x \neq -1, 0, 1$.
- N17.** $\frac{x}{4} - \frac{1}{4}$, with $x \neq -3, 5$.
- N18.** $\frac{6x + 16}{x}$, with $x \neq -4, 0$.
- N19.** $\frac{x^2 - 5x}{x - 4}$, with $x \neq 0, 4, 5$.
- N20.** $\frac{x + 3}{x - 1}$, with $x \neq -1, 1$.
- N21.** $-\frac{x}{4} - \frac{1}{2}$, with $x \neq -6, 2$.
- N22.** $\frac{x}{x - 4}$, with $x \neq 2, 4$.
- N23.** $\frac{x^2 + 4x}{x + 3}$, with $x \neq -4, -3, -2$.
- N24.** $\frac{x - 3}{x - 1}$, with $x \neq -1, 1$.
- N25.** $\frac{x}{4} + \frac{1}{4}$, with $x \neq -5, 3$.
- N26.** $\frac{5}{x - 4}$, with $x \neq 3, 4$.
- N27.** $\frac{x^2 + 4x}{x + 5}$, with $x \neq -5, -4, -2$.
- N28.** $\frac{x^2 - 3x - 1}{x^2 - 1}$, with $x \neq -1, 0, 1$.
- N29.** $\frac{x}{5} + \frac{1}{5}$, with $x \neq -6, 4$.
- N30.** $\frac{-2x + 24}{x + 6}$, with $x \neq -6, 3$.
- N31.** $\frac{x^2 + 5x}{x + 4}$, with $x \neq -5, -4, -3$.
- N32.** $\frac{x^2 - 3x + 8}{x^2 - 1}$, with $x \neq -1, 1, 3$.
- N33.** $-2x - 3$, with $x \neq -2, -1$.

- N34.** $\frac{3x-4}{x}$, with $x \neq 0, 2$.
- N35.** $\frac{x^2-7x}{x-6}$, with $x \neq -2, 6, 7$.
- N36.** $\frac{x^2-2x+5}{x^2-1}$, with $x \neq -1, 1, 3$.
- N37.** $2x+3$, with $x \neq -2, -1$.
- N38.** $\frac{-8x-18}{x+6}$, with $x \neq -6, 0$.
- N39.** $\frac{x^2-3x}{x-2}$, with $x \neq 2, 3, 5$.
- N40.** $\frac{x^2-5x-1}{x^2-1}$, with $x \neq -1, 0, 1$.
- N41.** $-\frac{x}{4} + \frac{1}{2}$, with $x \neq -2, 6$.
- N42.** $\frac{-4x-6}{x}$, with $x \neq 0, 6$.
- N43.** $x-1$, with $x \neq 0, 1, 2$.
- N44.** $\frac{x^2+2x-9}{x^2-1}$, with $x \neq -1, 1, 4$.
- N45.** $\frac{2x}{3} - \frac{1}{3}$, with $x \neq -1, 2$.
- N46.** $\frac{x-15}{x+1}$, with $x \neq -5, -1$.
- N47.** $\frac{x^2-6x}{x-5}$, with $x \neq 2, 5, 6$.
- N48.** $\frac{x+4}{x-1}$, with $x \neq -1, 1$.
- N49.** $-\frac{2x}{9} - \frac{1}{3}$, with $x \neq -6, 3$.
- N50.** $\frac{3x+12}{x+6}$, with $x \neq -6, 0$.

O. Polynomial Long Division

- O1.** $Q(x) = 4x - 1, R(x) = 0$.
- O2.** $Q(x) = -2x^2 + 4x + 1, R(x) = 1$.
- O3.** $Q(x) = -3x^2 - 2x + 1, R(x) = 1$.
- O4.** $Q(x) = -3x^2 - x + 1, R(x) = 4$.
- O5.** $Q(x) = -3x^2 - x + 1, R(x) = -3$.
- O6.** $Q(x) = 2x^2 - 4x + 1, R(x) = 0$.
- O7.** $Q(x) = -x - 1, R(x) = 1$.
- O8.** $Q(x) = -x^2, R(x) = 2$.

- O9.** $Q(x) = 2x^2 - 4x + 3$, $R(x) = 4$.
- O10.** $Q(x) = -4x + 4$, $R(x) = 0$.
- O11.** $Q(x) = -3x^2 - x + 3$, $R(x) = 0$.
- O12.** $Q(x) = 3x^3 - 3x^2 - 3x + 1$, $R(x) = 2$.
- O13.** $Q(x) = -3x^2 + 4x + 3$, $R(x) = 3$.
- O14.** $Q(x) = -2x + 4$, $R(x) = -5$.
- O15.** $Q(x) = -4x^3 - 3x^2 + 4x - 3$, $R(x) = -1$.
- O16.** $Q(x) = x^3 + 4x^2 - 4x - 1$, $R(x) = 0$.
- O17.** $Q(x) = x^2 - 4x + 1$, $R(x) = 5$.
- O18.** $Q(x) = -4x^2 - 2x - 1$, $R(x) = 5$.
- O19.** $Q(x) = x^3 + 2x^2 + 2x$, $R(x) = 2$.
- O20.** $Q(x) = -3x - 4$, $R(x) = 4$.
- O21.** $Q(x) = 2x^2 - x - 4$, $R(x) = 0$.
- O22.** $Q(x) = -4x + 2$, $R(x) = -x - 5$.
- O23.** $Q(x) = x^3 - 3x^2 - x + 1$, $R(x) = 3$.
- O24.** $Q(x) = -x^4 + 4x^3 + 3x^2 - 4x - 2$, $R(x) = -4x + 4$.
- O25.** $Q(x) = 4x^3 - 2x^2 - 4x$, $R(x) = 5$.
- O26.** $Q(x) = -4x^3 + 3x^2 + 4x - 3$, $R(x) = 0$.
- O27.** $Q(x) = -4x^2 - 4$, $R(x) = -2$.
- O28.** $Q(x) = 3x - 2$, $R(x) = 4$.
- O29.** $Q(x) = x^2 + 4x + 4$, $R(x) = 2x$.
- O30.** $Q(x) = x - 1$, $R(x) = -5$.
- O31.** $Q(x) = 2x^5 + 2x^4 + 3x^3 + 4x^2 + x + 2$, $R(x) = 0$.
- O32.** $Q(x) = -x^5 - 3x^4 + 4x^3 - 2x^2 - 3$, $R(x) = -3x - 4$.
- O33.** $Q(x) = -3x^5 + 4x^4 - x^3 - 2x^2 - 3x - 1$, $R(x) = 3x + 2$.
- O34.** $Q(x) = x^2 - 3x + 1$, $R(x) = 4$.
- O35.** $Q(x) = -4x^2 + 3x + 3$, $R(x) = -5$.
- O36.** $Q(x) = -2x^2 - 3x + 1$, $R(x) = 0$.
- O37.** $Q(x) = 4x^5 - x^4 - x^3 - 4x - 4$, $R(x) = -2x - 1$.
- O38.** $Q(x) = -4x^5 + 2x^4 - x^3 - 2x^2 - 3x + 4$, $R(x) = 4$.
- O39.** $Q(x) = -3x^5 + 2x^4 + 3x^3 + 4x^2 - 3$, $R(x) = 5$.
- O40.** $Q(x) = -3x^2 - 4x$, $R(x) = 2x - 3$.
- O41.** $Q(x) = 2x^2 + 2x + 2$, $R(x) = 0$.
- O42.** $Q(x) = 4x^4 + x^2 - 3x - 3$, $R(x) = 4x + 3$.
- O43.** $Q(x) = -3x^3 - x^2 + 2x + 1$, $R(x) = -5$.

- O44.** $Q(x) = -3x$, $R(x) = 3$.
- O45.** $Q(x) = 4x^4 - x^3 + 2x^2 - 2x + 4$, $R(x) = 3x + 3$.
- O46.** $Q(x) = 2x^5 - x^4 + 4x^3 + 4x^2 - 4x - 3$, $R(x) = 0$.
- O47.** $Q(x) = -2x^5 + 2x^4 + x^3 - 3x^2 + x + 4$, $R(x) = 2x + 1$.
- O48.** $Q(x) = 2x^6 - 2x^5 + 2x^4 - x^3 - 4x^2 - 4x - 3$, $R(x) = -4$.
- O49.** $Q(x) = -2x^4 + x^3 - 4x^2 - 2x + 3$, $R(x) = 2$.
- O50.** $Q(x) = x^6 + 2x^5 - 3x^4 + x^3 + 2x^2 + 2x + 3$, $R(x) = -4$.

P. Synthetic Division

- P1.** $Q(x) = 2x^3 + 5x^2 + 2x + 1$, $R = 0$.
- P2.** $Q(x) = -2x^2 + 2x - 4$, $R = -9$.
- P3.** $Q(x) = -2x^3 - 2x^2 - 5x + 3$, $R = 4$.
- P4.** $Q(x) = -4x^3 + 2x^2 + 3x + 5$, $R = 9$.
- P5.** $Q(x) = -3x^2 - 5x - 5$, $R = 0$.
- P6.** $Q(x) = -2x^3 - x^2 - 4x - 3$, $R = 7$.
- P7.** $Q(x) = -2x^3 + 4x^2 + 2x + 2$, $R = -1$.
- P8.** $Q(x) = 2x^3 + 2x^2 + 5x - 2$, $R = 5$.
- P9.** $Q(x) = -5x^2 - 4x - 1$, $R = 0$.
- P10.** $Q(x) = 2x^2 + 1$, $R = -3$.
- P11.** $Q(x) = -x^4 - 2x^3 + 3x + 3$, $R = 1$.
- P12.** $Q(x) = 4x^3 - x^2 + 3x - 4$, $R = 5$.
- P13.** $Q(x) = -4x^3 + x^2 - 2x - 1$, $R = 0$.
- P14.** $Q(x) = -3x^4 + 4x^3 - 4x^2 - 2$, $R = -1$.
- P15.** $Q(x) = -x^2 + 2$, $R = -9$.
- P16.** $Q(x) = x^4 + 2x^3 - 4x^2 + 3x + 1$, $R = -8$.
- P17.** $Q(x) = x^3 - x^2 + x + 4$, $R = 0$.
- P18.** $Q(x) = 5x^2 + 2x + 2$, $R = 0$.
- P19.** $Q(x) = -2x^3 - 4x^2 + x - 1$, $R = 3$.
- P20.** $Q(x) = -3x^2 - x - 4$, $R = -7$.
- P21.** $Q(x) = 4x^3 + 2x^2 + x + 5$, $R = 0$.
- P22.** $Q(x) = 4x^4 + 3x^3 + 3x^2 + 4x + 3$, $R = -1$.
- P23.** $Q(x) = 2x^5 - 3x^4 - 4x^3 - x^2 + x - 1$, $R = 8$.
- P24.** $Q(x) = 4x^4 + x^3 + x^2 + x - 1$, $R = 9$.
- P25.** $Q(x) = -4x^4 - 2x^3 + 3x^2 + x - 1$, $R = 0$.

- P26.** $Q(x) = -4x^2 - x - 3$, $R = -9$.
- P27.** $Q(x) = -x^2 + 3x + 5$, $R = -4$.
- P28.** $Q(x) = -2x^3 + 4x^2 + 4x - 1$, $R = 8$.
- P29.** $Q(x) = x^4 - 4x^3 + x^2 + 4x - 1$, $R = 0$.
- P30.** $Q(x) = -3x^5 - x^4 - 3x^3 + x^2 + 5x + 1$, $R = 8$.
- P31.** $Q(x) = 3x^4 - 3x^3 + x^2 + 5x - 5$, $R = 5$.
- P32.** $Q(x) = 4x^4 - 4x^3 + 4x^2 - 4x + 3$, $R = 1$.
- P33.** $Q(x) = -2x^3 + 5x^2 - 4x - 5$, $R = 0$.
- P34.** $Q(x) = 4x^3 - 4x^2 - 3x + 4$, $R = -2$.
- P35.** $Q(x) = -x^3 - x^2 + x + 2$, $R = -5$.
- P36.** $Q(x) = x^4 + 4x^3 - 4x^2 + 3x - 3$, $R = -3$.
- P37.** $Q(x) = -2x^4 + 2x^2 + 5$, $R = 0$.
- P38.** $Q(x) = -5x^6 + 4x^5 + x^4 - x^3 + 3x^2 + 4x - 5$, $R = 3$.
- P39.** $Q(x) = x^5 + 5x^4 - 5x^3 - x^2 + 5x + 2$, $R = 5$.
- P40.** $Q(x) = -4x^6 + 5x^5 + 5x^4 + 3x^3 + 4x - 4$, $R = -4$.
- P41.** $Q(x) = -4x^2 + 4x - 1$, $R = 0$.
- P42.** $Q(x) = -4x^2 + 5x + 2$, $R = -6$.
- P43.** $Q(x) = -3x^4 + 2x^3 - 2x^2 + 3x + 2$, $R = 8$.
- P44.** $Q(x) = 2x^2 - 3x + 3$, $R = -5$.
- P45.** $Q(x) = -5x^4 + 3x^3 - 2x^2 + 5x$, $R = 0$.
- P46.** $Q(x) = -4x^3 - 3x^2 + 5x + 4$, $R = -2$.
- P47.** $Q(x) = 5x^4 - x^3 + x^2 + 5x - 1$, $R = 1$.
- P48.** $Q(x) = 3x^5 - 5x^4 - x^3 + 5x - 2$, $R = -2$.
- P49.** $Q(x) = 5x^6 - 3x^5 - 3x^4 + 5x^3 - 2x^2 - 2x + 4$, $R = 0$.
- P50.** $Q(x) = -2x^2 + 2$, $R = -2$.

Q. Choosing Long or Synthetic Division

- Q1.** synthetic division (or long division); $Q(x) = -3x + 1$, $R(x) = -2$.
- Q2.** polynomial long division; $Q(x) = -4x^2 + 3x + 2$, $R(x) = -2$.
- Q3.** polynomial long division; $Q(x) = 3x^2 + x + 3$, $R(x) = 4$.
- Q4.** synthetic division (or long division); $Q(x) = -x^2 + 4x - 2$, $R(x) = -3$.
- Q5.** polynomial long division; $Q(x) = 3x + 2$, $R(x) = 4$.
- Q6.** polynomial long division; $Q(x) = -3x^2 + 4x - 1$, $R(x) = 1$.
- Q7.** synthetic division (or long division); $Q(x) = -x^2 + 3x - 3$, $R(x) = -1$.

- Q8.** polynomial long division; $Q(x) = -x$, $R(x) = -1$.
- Q9.** polynomial long division; $Q(x) = -3x - 4$, $R(x) = 3x + 2$.
- Q10.** synthetic division (or long division); $Q(x) = 2x^3 + 4x^2 - x$, $R(x) = -1$.
- Q11.** polynomial long division; $Q(x) = -2x^4 + 4x^2 + x - 3$, $R(x) = 3$.
- Q12.** polynomial long division; $Q(x) = 4x - 1$, $R(x) = -4$.
- Q13.** synthetic division (or long division); $Q(x) = -2x^3 + 2x^2 - 2$, $R(x) = -1$.
- Q14.** polynomial long division; $Q(x) = -3x^2 + 1$, $R(x) = -2$.
- Q15.** polynomial long division; $Q(x) = -3x^3 + 4x^2 + 4x + 2$, $R(x) = -2$.
- Q16.** synthetic division (or long division); $Q(x) = 4x^2 + 2x - 3$, $R(x) = 0$.
- Q17.** polynomial long division; $Q(x) = 2x^3 + 4x^2 + 2x + 4$, $R(x) = 4$.
- Q18.** polynomial long division; $Q(x) = 3x^3 - 3x^2 + 4$, $R(x) = -x + 4$.
- Q19.** synthetic division (or long division); $Q(x) = 2x + 2$, $R(x) = -1$.
- Q20.** polynomial long division; $Q(x) = 4x^4 + x^3 - x^2 + 4x - 4$, $R(x) = 1$.
- Q21.** polynomial long division; $Q(x) = -x^3 + 3x^2 - 4x$, $R(x) = -2$.
- Q22.** synthetic division (or long division); $Q(x) = x^5 + x^4 + x^3 - x^2 + x + 2$, $R(x) = 2$.
- Q23.** polynomial long division; $Q(x) = 3x^2 - 2x + 1$, $R(x) = -3$.
- Q24.** polynomial long division; $Q(x) = -x^2 + 3x + 1$, $R(x) = 3$.
- Q25.** synthetic division (or long division); $Q(x) = x^4 + 2x^3 - 2x^2 - 4x + 2$, $R(x) = 4$.
- Q26.** polynomial long division; $Q(x) = 4x^4 - 3x^3 - 4x^2 - 3$, $R(x) = 1$.
- Q27.** polynomial long division; $Q(x) = 4x^5 - 4x^4 + 2x^3 + 2x^2 - 3x - 3$, $R(x) = 3x + 1$.
- Q28.** synthetic division (or long division); $Q(x) = -x^2 + 4x + 4$, $R(x) = 1$.
- Q29.** polynomial long division; $Q(x) = -2x - 1$, $R(x) = -4$.
- Q30.** polynomial long division; $Q(x) = x - 1$, $R(x) = 0$.
- Q31.** synthetic division (or long division); $Q(x) = 4x^4 + 4x^3 - 3x^2 + x + 3$, $R(x) = 4$.
- Q32.** polynomial long division; $Q(x) = -3x^4 - 4x^3 + 4x + 3$, $R(x) = 0$.
- Q33.** polynomial long division; $Q(x) = -x^5 + 3x^4 - x^3 - 2x^2 + 3x + 1$, $R(x) = -4x + 3$.
- Q34.** synthetic division (or long division); $Q(x) = -x^4 + 2x^3 + 3x^2 - 2x - 3$, $R(x) = 1$.
- Q35.** polynomial long division; $Q(x) = -4x + 2$, $R(x) = -1$.
- Q36.** polynomial long division; $Q(x) = -2x^5 + 4x^4 + x^2 - 2x + 4$, $R(x) = -2x - 3$.
- Q37.** synthetic division (or long division); $Q(x) = 2x^5 + x^4 + 2x^3 + 2x^2 - 4x - 3$, $R(x) = -4$.
- Q38.** polynomial long division; $Q(x) = 2x^3 - 3x^2 + 3x + 3$, $R(x) = -4$.
- Q39.** polynomial long division; $Q(x) = -4x^4 + x^3 + 4x^2 - 2x - 4$, $R(x) = 0$.
- Q40.** synthetic division (or long division); $Q(x) = x^5 + 4x^4 - 2x^2 + 4x + 4$, $R(x) = -3$.
- Q41.** polynomial long division; $Q(x) = -2x^5 - 3x^4 + 4x^3 - 4x^2 + 4x - 1$, $R(x) = -2$.
- Q42.** polynomial long division; $Q(x) = 3x - 1$, $R(x) = 0$.

- Q43.** synthetic division (or long division); $Q(x) = -x^6 - x^5 + 2x^4 - 2x^3 + x^2 - 4x - 2$, $R(x) = 2$.
- Q44.** polynomial long division; $Q(x) = -x^5 + 4x^4 + 3x^3 - 2x^2 - 2x - 1$, $R(x) = -1$.
- Q45.** polynomial long division; $Q(x) = -x^3 + 3x^2 - 3x + 3$, $R(x) = -2$.
- Q46.** synthetic division (or long division); $Q(x) = 3x^4 + 4x^3 - 3x^2 + 2x + 4$, $R(x) = -1$.
- Q47.** polynomial long division; $Q(x) = -2x^4 + x^3 - 2x^2 + x - 2$, $R(x) = 0$.
- Q48.** polynomial long division; $Q(x) = 2x^5 - 2x^4 + 4x^2 + 3x + 2$, $R(x) = -x - 1$.
- Q49.** synthetic division (or long division); $Q(x) = -2x^7 - 4x^6 + 3x^5 - 4x^4 + 4x^3 + 4x^2 + x + 1$, $R(x) = -3$.
- Q50.** polynomial long division; $Q(x) = x^5 + 3x^4 - 4x^3 + 3x - 2$, $R(x) = -4$.

R. Distinct Linear Factors

- R1.** $-\frac{1}{x+7} + \frac{7}{x-5}$.
- R2.** $-\frac{6}{x+3} + \frac{3}{x-8}$.
- R3.** $\frac{5}{x+8} - \frac{7}{x+2}$.
- R4.** $\frac{2}{x+4} + \frac{5}{x-5}$.
- R5.** $\frac{2}{x-2} - \frac{7}{x-7}$.
- R6.** $-\frac{5}{x+2} + \frac{1}{x-8}$.
- R7.** $\frac{6}{x+8} - \frac{7}{x-1}$.
- R8.** $-\frac{8}{x-5} + \frac{6}{x-6}$.
- R9.** $\frac{6}{x+7} + \frac{2}{x}$.
- R10.** $\frac{6}{x+8} - \frac{6}{x-6}$.
- R11.** $-\frac{5}{x+6} + \frac{8}{x+4}$.
- R12.** $-\frac{2}{x+5} - \frac{5}{x-8}$.
- R13.** $\frac{1}{x+3} + \frac{8}{x-6}$.
- R14.** $\frac{4}{x-1} - \frac{7}{x-7}$.
- R15.** $-\frac{4}{x-1} - \frac{6}{x-2}$.
- R16.** $-\frac{1}{x+6} - \frac{5}{x}$.

$$\text{R17. } -\frac{3}{x+5} + \frac{5}{x+2}.$$

$$\text{R18. } \frac{7}{x+7} + \frac{8}{x+6}.$$

$$\text{R19. } -\frac{6}{x+2} + \frac{7}{x-1}.$$

$$\text{R20. } -\frac{6}{x-3} + \frac{4}{x-5}.$$

$$\text{R21. } -\frac{5}{x+8} - \frac{8}{x+3} + \frac{7}{x+1}.$$

$$\text{R22. } \frac{2}{x+6} - \frac{6}{x+5} - \frac{1}{x+4}.$$

$$\text{R23. } \frac{7}{x+4} + \frac{4}{x+2} - \frac{7}{x-6}.$$

$$\text{R24. } -\frac{4}{x+3} + \frac{6}{x-2} + \frac{5}{x-4}.$$

$$\text{R25. } \frac{6}{x+4} - \frac{4}{x+1} + \frac{6}{x}.$$

$$\text{R26. } -\frac{2}{x-1} - \frac{8}{x-5} - \frac{2}{x}.$$

$$\text{R27. } -\frac{4}{x+3} + \frac{7}{x+2} - \frac{8}{x-3}.$$

$$\text{R28. } \frac{6}{x+5} - \frac{3}{x+1} + \frac{1}{x}.$$

$$\text{R29. } \frac{1}{x+3} + \frac{4}{x-3} - \frac{4}{x-5}.$$

$$\text{R30. } -\frac{6}{x+7} + \frac{2}{x+2} - \frac{5}{x-3}.$$

$$\text{R31. } -\frac{2}{x+7} - \frac{3}{x+5} - \frac{5}{x-2}.$$

$$\text{R32. } \frac{4}{x+5} + \frac{2}{x-5} - \frac{6}{x}.$$

$$\text{R33. } \frac{7}{x+6} + \frac{6}{x-2} + \frac{2}{x-8}.$$

$$\text{R34. } \frac{1}{x+5} - \frac{8}{x-1} + \frac{7}{x}.$$

$$\text{R35. } \frac{1}{x-3} - \frac{4}{x-4} - \frac{7}{x}.$$

$$\text{R36. } -\frac{2}{x+4} + \frac{1}{x-3} - \frac{7}{x-4}.$$

$$\text{R37. } \frac{2}{x+4} + \frac{8}{x+3} + \frac{5}{x-2}.$$

$$\text{R38. } -\frac{4}{x+8} + \frac{7}{x+6} - \frac{5}{x}.$$

$$\text{R39. } -\frac{2}{x+8} - \frac{4}{x+7} + \frac{1}{x+5}.$$

$$\mathbf{R40.} \quad \frac{7}{x+3} + \frac{8}{x-3} + \frac{4}{x-8}.$$

$$\mathbf{R41.} \quad -\frac{5}{x+7} + \frac{4}{x+6} + \frac{3}{x-1} - \frac{1}{x-4}.$$

$$\mathbf{R42.} \quad -\frac{6}{x+8} - \frac{5}{x-4} + \frac{2}{x-6} - \frac{4}{x-8}.$$

$$\mathbf{R43.} \quad \frac{2}{x+5} - \frac{2}{x-1} - \frac{4}{x-6} + \frac{3}{x-8}.$$

$$\mathbf{R44.} \quad -\frac{6}{x+8} + \frac{2}{x+3} + \frac{3}{x-6} + \frac{8}{x-8}.$$

$$\mathbf{R45.} \quad \frac{1}{x-4} - \frac{6}{x-5} + \frac{3}{x-6} - \frac{1}{x}.$$

$$\mathbf{R46.} \quad \frac{2}{x+7} + \frac{8}{x+1} - \frac{7}{x-8} + \frac{1}{x}.$$

$$\mathbf{R47.} \quad \frac{8}{x+4} + \frac{5}{x-3} - \frac{5}{x-4} - \frac{7}{x-7}.$$

$$\mathbf{R48.} \quad \frac{5}{x-1} - \frac{3}{x-5} - \frac{4}{x-8} + \frac{7}{x}.$$

$$\mathbf{R49.} \quad \frac{7}{x+8} + \frac{8}{x+7} + \frac{4}{x+1} + \frac{6}{x-2}.$$

$$\mathbf{R50.} \quad \frac{6}{x+8} - \frac{2}{x+3} - \frac{4}{x+2} - \frac{8}{x-7}.$$

S. Heaviside Cover-Up Method

$$\mathbf{S1.} \quad \frac{4}{x+2} - \frac{6}{x-3}.$$

$$\mathbf{S2.} \quad \frac{6}{x+8} - \frac{2}{x-1}.$$

$$\mathbf{S3.} \quad \frac{3}{x+8} + \frac{5}{x-1}.$$

$$\mathbf{S4.} \quad \frac{1}{x+8} - \frac{6}{x-1}.$$

$$\mathbf{S5.} \quad -\frac{2}{x+5} + \frac{7}{x}.$$

$$\mathbf{S6.} \quad -\frac{2}{x+2} - \frac{5}{x-3}.$$

$$\mathbf{S7.} \quad -\frac{6}{x+8} + \frac{4}{x-7}.$$

$$\mathbf{S8.} \quad \frac{6}{x-3} + \frac{1}{x}.$$

$$\mathbf{S9.} \quad -\frac{8}{x+5} + \frac{2}{x-4}.$$

$$\mathbf{S10.} \quad -\frac{4}{x-2} - \frac{5}{x}.$$

$$\text{S11. } -\frac{3}{x+1} + \frac{8}{x-4}.$$

$$\text{S12. } \frac{1}{x-7} - \frac{6}{x}.$$

$$\text{S13. } -\frac{5}{x+5} + \frac{3}{x}.$$

$$\text{S14. } -\frac{1}{x+5} + \frac{1}{x}.$$

$$\text{S15. } -\frac{3}{x+3} + \frac{1}{x-6}.$$

$$\text{S16. } -\frac{3}{x-4} + \frac{3}{x-8}.$$

$$\text{S17. } \frac{7}{x+1} - \frac{1}{x-8}.$$

$$\text{S18. } -\frac{5}{x+6} + \frac{4}{x+1}.$$

$$\text{S19. } -\frac{5}{x+6} - \frac{2}{x-3}.$$

$$\text{S20. } -\frac{7}{x+7} - \frac{3}{x+1}.$$

$$\text{S21. } -\frac{4}{x+6} + \frac{6}{x+5} + \frac{8}{x+1}.$$

$$\text{S22. } \frac{4}{x+8} + \frac{6}{x+6} - \frac{5}{x-1}.$$

$$\text{S23. } \frac{4}{x+4} + \frac{7}{x+1} - \frac{8}{x-5}.$$

$$\text{S24. } \frac{6}{x-2} - \frac{8}{x-4} - \frac{1}{x-8}.$$

$$\text{S25. } \frac{7}{x+5} - \frac{8}{x-4} - \frac{8}{x-5}.$$

$$\text{S26. } -\frac{6}{x+8} - \frac{5}{x+4} - \frac{8}{x+2}.$$

$$\text{S27. } -\frac{4}{x+8} + \frac{8}{x+5} - \frac{1}{x}.$$

$$\text{S28. } \frac{2}{x+3} - \frac{3}{x-2} - \frac{2}{x}.$$

$$\text{S29. } -\frac{4}{x+3} - \frac{5}{x-1} + \frac{1}{x-7}.$$

$$\text{S30. } \frac{2}{x+8} + \frac{5}{x+7} + \frac{6}{x-5}.$$

$$\text{S31. } \frac{4}{x+3} - \frac{7}{x-1} + \frac{6}{x-5}.$$

$$\text{S32. } -\frac{4}{x+8} + \frac{2}{x+3} + \frac{2}{x+1}.$$

$$\text{S33. } -\frac{4}{x+5} + \frac{5}{x-7} + \frac{4}{x}.$$

$$\text{S34. } -\frac{6}{x-1} - \frac{1}{x-2} + \frac{3}{x-7}.$$

$$\text{S35. } \frac{5}{x+8} - \frac{7}{x-1} + \frac{8}{x}.$$

$$\text{S36. } \frac{7}{x+8} - \frac{1}{x-2} - \frac{3}{x-7}.$$

$$\text{S37. } \frac{4}{x-5} + \frac{3}{x-8} - \frac{2}{x}.$$

$$\text{S38. } \frac{2}{x-5} - \frac{3}{x-8} + \frac{1}{x}.$$

$$\text{S39. } \frac{5}{x+1} + \frac{7}{x-2} + \frac{2}{x-4}.$$

$$\text{S40. } -\frac{5}{x+7} + \frac{2}{x+4} + \frac{1}{x-2}.$$

$$\text{S41. } \frac{4}{x-2} + \frac{7}{x-4} - \frac{6}{x-8} + \frac{4}{x}.$$

$$\text{S42. } \frac{7}{x+6} - \frac{6}{x+4} + \frac{6}{x-5} - \frac{1}{x-7}.$$

$$\text{S43. } \frac{3}{x+7} - \frac{4}{x+6} + \frac{3}{x+5} - \frac{4}{x-8}.$$

$$\text{S44. } \frac{6}{x+4} + \frac{5}{x+3} + \frac{6}{x-1} + \frac{2}{x-8}.$$

$$\text{S45. } \frac{8}{x+1} + \frac{5}{x-1} + \frac{8}{x-3} - \frac{7}{x-5}.$$

$$\text{S46. } -\frac{8}{x+8} - \frac{7}{x+6} + \frac{3}{x-6} - \frac{7}{x-7}.$$

$$\text{S47. } -\frac{3}{x+6} - \frac{1}{x+1} + \frac{2}{x-4} + \frac{5}{x-7}.$$

$$\text{S48. } -\frac{7}{x+7} - \frac{1}{x+3} + \frac{4}{x-1} + \frac{2}{x-5}.$$

$$\text{S49. } -\frac{8}{x+4} - \frac{6}{x-1} + \frac{1}{x-3} - \frac{6}{x-6}.$$

$$\text{S50. } \frac{1}{x+7} - \frac{4}{x-3} + \frac{3}{x-7} - \frac{1}{x}.$$

T. Strategic Substitution

$$\text{T1. } -\frac{6}{x+6} - \frac{1}{x-5}.$$

$$\text{T2. } -\frac{4}{x-2} + \frac{1}{x-7}.$$

$$\text{T3. } \frac{5}{x+5} + \frac{6}{x-2}.$$

$$\text{T4. } -\frac{6}{x+7} + \frac{2}{x+4}.$$

$$\mathbf{T5.} \quad \frac{6}{x+7} - \frac{3}{x}.$$

$$\mathbf{T6.} \quad -\frac{7}{x+3} + \frac{2}{x-3}.$$

$$\mathbf{T7.} \quad \frac{3}{x-4} - \frac{6}{x-6}.$$

$$\mathbf{T8.} \quad \frac{1}{x+3} - \frac{1}{x-7}.$$

$$\mathbf{T9.} \quad \frac{2}{x+1} + \frac{6}{x-6}.$$

$$\mathbf{T10.} \quad -\frac{2}{x+2} - \frac{4}{x-1}.$$

$$\mathbf{T11.} \quad \frac{5}{x-3} + \frac{6}{x-4}.$$

$$\mathbf{T12.} \quad \frac{2}{x+2} - \frac{6}{x-5}.$$

$$\mathbf{T13.} \quad \frac{4}{x+3} - \frac{1}{x-6}.$$

$$\mathbf{T14.} \quad \frac{6}{x+2} + \frac{1}{x-1}.$$

$$\mathbf{T15.} \quad \frac{4}{x+6} - \frac{6}{x-2}.$$

$$\mathbf{T16.} \quad -\frac{2}{x+6} - \frac{5}{x-5}.$$

$$\mathbf{T17.} \quad \frac{2}{x+2} - \frac{1}{x}.$$

$$\mathbf{T18.} \quad -\frac{3}{x-1} + \frac{5}{x-7}.$$

$$\mathbf{T19.} \quad -\frac{2}{x+3} + \frac{6}{x-2}.$$

$$\mathbf{T20.} \quad -\frac{5}{x-3} - \frac{5}{x-6}.$$

$$\mathbf{T21.} \quad -\frac{1}{x-1} - \frac{1}{x-6} + \frac{2}{x}.$$

$$\mathbf{T22.} \quad -\frac{2}{x+4} + \frac{1}{x-6} + \frac{2}{x-7}.$$

$$\mathbf{T23.} \quad \frac{6}{x+6} + \frac{6}{x-3} + \frac{1}{x-7}.$$

$$\mathbf{T24.} \quad \frac{5}{x+3} + \frac{2}{x-1} - \frac{3}{x-5}.$$

$$\mathbf{T25.} \quad -\frac{5}{x+5} - \frac{2}{x+4} - \frac{5}{x+3}.$$

$$\mathbf{T26.} \quad \frac{4}{x+2} - \frac{7}{x-4} + \frac{1}{x-6}.$$

$$\mathbf{T27.} \quad \frac{5}{x+5} + \frac{3}{x-1} + \frac{3}{x}.$$

$$\mathbf{T28.} \quad -\frac{3}{x+5} - \frac{4}{x+4} - \frac{4}{x-1}.$$

$$\mathbf{T29.} \quad \frac{3}{x+3} - \frac{1}{x+1} + \frac{3}{x-7}.$$

$$\mathbf{T30.} \quad -\frac{4}{x+4} - \frac{3}{x-1} + \frac{2}{x-7}.$$

$$\mathbf{T31.} \quad -\frac{3}{x+5} - \frac{2}{x-2} - \frac{6}{x-3}.$$

$$\mathbf{T32.} \quad \frac{4}{x+3} - \frac{4}{x-5} + \frac{1}{x}.$$

$$\mathbf{T33.} \quad \frac{3}{x+4} - \frac{4}{x+2} - \frac{4}{x-5}.$$

$$\mathbf{T34.} \quad -\frac{5}{x-3} + \frac{1}{x-6} - \frac{6}{x}.$$

$$\mathbf{T35.} \quad \frac{1}{x+6} - \frac{7}{x+5} + \frac{3}{x+3}.$$

$$\mathbf{T36.} \quad \frac{6}{x+3} - \frac{6}{x+1} + \frac{2}{x}.$$

$$\mathbf{T37.} \quad -\frac{5}{x+6} + \frac{3}{x+4} + \frac{2}{x-1}.$$

$$\mathbf{T38.} \quad \frac{7}{x-3} - \frac{1}{x-4} - \frac{3}{x}.$$

$$\mathbf{T39.} \quad -\frac{4}{x+7} - \frac{6}{x+2} + \frac{4}{x-1}.$$

$$\mathbf{T40.} \quad \frac{1}{x+1} - \frac{6}{x-3} - \frac{1}{x}.$$

$$\mathbf{T41.} \quad \frac{7}{x+3} - \frac{4}{x-7} + \frac{4}{x}.$$

$$\mathbf{T42.} \quad -\frac{6}{x+3} + \frac{5}{x-5} - \frac{3}{x}.$$

$$\mathbf{T43.} \quad -\frac{4}{x-2} - \frac{4}{x-6} - \frac{1}{x}.$$

$$\mathbf{T44.} \quad -\frac{3}{x+5} - \frac{6}{x+1} + \frac{2}{x-2}.$$

$$\mathbf{T45.} \quad \frac{3}{x+3} - \frac{6}{x-1} - \frac{1}{x-6}.$$

$$\mathbf{T46.} \quad \frac{3}{x-4} + \frac{6}{x-5} + \frac{1}{x-7}.$$

$$\mathbf{T47.} \quad -\frac{7}{x-3} + \frac{3}{x-4} + \frac{3}{x-7}.$$

$$\mathbf{T48.} \quad \frac{3}{x+1} + \frac{6}{x-2} + \frac{2}{x-7}.$$

$$\mathbf{T49.} \quad \frac{6}{x+6} - \frac{3}{x-4} - \frac{3}{x-7}.$$

$$\mathbf{T50.} \quad -\frac{2}{x+5} + \frac{6}{x+3} - \frac{4}{x-4}.$$

U. Equating Coefficients

$$\text{U1. } \frac{3}{x+4} + \frac{4}{x-4}.$$

$$\text{U2. } \frac{1}{x+1} - \frac{1}{x-5}.$$

$$\text{U3. } -\frac{2}{x-3} - \frac{4}{x}.$$

$$\text{U4. } \frac{1}{x-6} - \frac{3}{x}.$$

$$\text{U5. } -\frac{6}{x-1} - \frac{2}{x-5}.$$

$$\text{U6. } \frac{2}{x+4} - \frac{3}{x-5}.$$

$$\text{U7. } -\frac{5}{x+5} - \frac{4}{x-4}.$$

$$\text{U8. } \frac{1}{x+3} - \frac{5}{x+2}.$$

$$\text{U9. } \frac{2}{x+6} - \frac{6}{x}.$$

$$\text{U10. } -\frac{5}{x+1} - \frac{3}{x-6}.$$

$$\text{U11. } \frac{2}{x-1} + \frac{3}{x-6}.$$

$$\text{U12. } \frac{2}{x+1} - \frac{3}{x}.$$

$$\text{U13. } \frac{6}{x-1} - \frac{4}{x-4}.$$

$$\text{U14. } -\frac{6}{x-3} - \frac{4}{x-6}.$$

$$\text{U15. } \frac{5}{x-1} + \frac{2}{x-3}.$$

$$\text{U16. } \frac{2}{x-1} + \frac{3}{x-6}.$$

$$\text{U17. } \frac{2}{x+5} + \frac{6}{x}.$$

$$\text{U18. } -\frac{6}{x+5} - \frac{5}{x}.$$

$$\text{U19. } -\frac{1}{x+6} - \frac{3}{x-4}.$$

$$\text{U20. } \frac{6}{x+4} - \frac{1}{x+2}.$$

$$\text{U21. } \frac{4x+3}{x^2+4x+5} + \frac{5}{x+4}.$$

$$\text{U22. } \frac{2x+4}{x^2+5x+7} + \frac{4}{x-6}.$$

$$\text{U23. } \frac{3-x}{x^2+4} - \frac{1}{x-1}.$$

$$\text{U24. } \frac{x+2}{x^2-4x+5} - \frac{5}{x-5}.$$

$$\text{U25. } \frac{2-5x}{x^2-x+3} - \frac{4}{x-1}.$$

$$\text{U26. } \frac{2x+1}{x^2+4x+8} - \frac{1}{x+5}.$$

$$\text{U27. } \frac{x-5}{x^2+2x+6} + \frac{3}{x+4}.$$

$$\text{U28. } \frac{3x-1}{x^2+2x+6} - \frac{4}{x-6}.$$

$$\text{U29. } \frac{x-2}{x^2-5x+7} - \frac{4}{x+5}.$$

$$\text{U30. } \frac{3x-5}{x^2+5x+7} + \frac{3}{x-4}.$$

$$\text{U31. } \frac{-x-3}{x^2-x+6} + \frac{4}{x+5}.$$

$$\text{U32. } \frac{1-3x}{x^2+x+1} - \frac{1}{x-3}.$$

$$\text{U33. } \frac{4x+5}{x^2-x+6} + \frac{4}{x+6}.$$

$$\text{U34. } \frac{3x+3}{x^2+x+5} - \frac{1}{x+6}.$$

$$\text{U35. } \frac{x-2}{x^2+3x+8} - \frac{3}{x+5}.$$

$$\text{U36. } \frac{-3x-5}{x^2+x+1} - \frac{5}{x-5}.$$

$$\text{U37. } \frac{5x+5}{x^2+3x+3} - \frac{1}{x+4}.$$

$$\text{U38. } \frac{5-3x}{x^2-4x+5} + \frac{2}{x-2}.$$

$$\text{U39. } \frac{-x-3}{x^2+x+5} - \frac{2}{x+1}.$$

$$\text{U40. } \frac{x+3}{x^2-2x+7} + \frac{1}{x+1}.$$

$$\text{U41. } \frac{2-2x}{x^2+5x+7} + \frac{4x-2}{x^2+7}.$$

$$\text{U42. } \frac{2-3x}{x^2-2x+8} + \frac{3x-1}{x^2+4x+6}.$$

$$\text{U43. } \frac{3-x}{x^2-3x+3} + \frac{-x-3}{x^2+4x+6}.$$

$$\text{U44. } \frac{3x+2}{x^2+4x+6} + \frac{3x+3}{x^2-3x+4}.$$

$$\text{U45. } \frac{1-4x}{x^2-x+4} + \frac{2x+4}{x^2+4x+8}.$$

$$\text{U46. } \frac{2x-1}{x^2-4x+8} + \frac{2x+3}{x^2-4x+5}.$$

$$\text{U47. } \frac{-2x-1}{x^2+x+4} + \frac{x+2}{x^2+x+6}.$$

$$\text{U48. } \frac{1-x}{x^2+6} + \frac{-3x-2}{x^2-2x+4}.$$

$$\text{U49. } \frac{-3x-3}{x^2+5x+7} + \frac{4x+1}{x^2+2x+4}.$$

$$\text{U50. } \frac{2-4x}{x^2+4} + \frac{-x-3}{x^2+x+6}.$$

V. Repeated Linear Factors

$$\text{V1. } \frac{3}{x-4} - \frac{6}{(x-4)^2}.$$

$$\text{V2. } \frac{6}{x+3} - \frac{2}{(x+3)^2}.$$

$$\text{V3. } -\frac{5}{x+1} - \frac{5}{(x+1)^2}.$$

$$\text{V4. } -\frac{1}{x+3} - \frac{1}{(x+3)^2}.$$

$$\text{V5. } -\frac{6}{x+2} - \frac{6}{(x+2)^2}.$$

$$\text{V6. } -\frac{6}{x+1} - \frac{6}{(x+1)^2}.$$

$$\text{V7. } -\frac{4}{x+4} - \frac{1}{(x+4)^2}.$$

$$\text{V8. } -\frac{2}{x+1} - \frac{3}{(x+1)^2}.$$

$$\text{V9. } -\frac{5}{x-1} + \frac{5}{(x-1)^2}.$$

$$\text{V10. } -\frac{1}{x+2} - \frac{4}{(x+2)^2}.$$

$$\text{V11. } -\frac{2}{x+1} + \frac{2}{(x+1)^2}.$$

$$\text{V12. } -\frac{6}{x+3} - \frac{3}{(x+3)^2}.$$

$$\text{V13. } -\frac{2}{x-2} - \frac{3}{(x-2)^2}.$$

$$\text{V14. } -\frac{4}{x-4} - \frac{1}{(x-4)^2}.$$

$$\text{V15. } \frac{3}{x-5} + \frac{5}{(x-5)^2}.$$

$$\mathbf{V16.} \quad -\frac{5}{x-3} + \frac{6}{(x-3)^2}.$$

$$\mathbf{V17.} \quad \frac{5}{x+6} + \frac{1}{(x+6)^2}.$$

$$\mathbf{V18.} \quad \frac{6}{x+3} + \frac{4}{(x+3)^2}.$$

$$\mathbf{V19.} \quad \frac{3}{x-6} + \frac{2}{(x-6)^2}.$$

$$\mathbf{V20.} \quad -\frac{2}{x-2} - \frac{5}{(x-2)^2}.$$

$$\mathbf{V21.} \quad -\frac{3}{x+1} + \frac{6}{(x+1)^2} + \frac{2}{(x+1)^3}.$$

$$\mathbf{V22.} \quad \frac{6}{x+4} + \frac{2}{(x+4)^2} + \frac{3}{(x+4)^3}.$$

$$\mathbf{V23.} \quad -\frac{6}{x+1} + \frac{4}{(x+1)^2} - \frac{4}{(x+1)^3}.$$

$$\mathbf{V24.} \quad \frac{3}{x+5} + \frac{6}{(x+5)^2} - \frac{6}{(x+5)^3}.$$

$$\mathbf{V25.} \quad -\frac{5}{x-1} + \frac{5}{(x-1)^2} - \frac{3}{(x-1)^3}.$$

$$\mathbf{V26.} \quad -\frac{5}{x-4} - \frac{6}{(x-4)^2} - \frac{5}{(x-4)^3}.$$

$$\mathbf{V27.} \quad -\frac{4}{x-1} - \frac{2}{(x-1)^2} - \frac{5}{(x-1)^3}.$$

$$\mathbf{V28.} \quad -\frac{6}{x-1} + \frac{2}{(x-1)^2} + \frac{6}{(x-1)^3}.$$

$$\mathbf{V29.} \quad -\frac{4}{x-6} + \frac{5}{(x-6)^2} + \frac{1}{(x-6)^3}.$$

$$\mathbf{V30.} \quad \frac{2}{x-6} - \frac{3}{(x-6)^2} + \frac{2}{(x-6)^3}.$$

$$\mathbf{V31.} \quad \frac{2}{x+2} + \frac{2}{x+1} + \frac{4}{(x+1)^2} + \frac{6}{(x+1)^3}.$$

$$\mathbf{V32.} \quad \frac{6}{x-1} - \frac{1}{(x-1)^2} - \frac{4}{(x-1)^3} + \frac{2}{x-2}.$$

$$\mathbf{V33.} \quad \frac{4}{x+1} + \frac{4}{x-2} - \frac{3}{(x-2)^2} - \frac{5}{(x-2)^3}.$$

$$\mathbf{V34.} \quad -\frac{4}{x+4} + \frac{2}{(x+4)^2} + \frac{3}{(x+4)^3} + \frac{2}{x-3}.$$

$$\mathbf{V35.} \quad \frac{6}{x+1} - \frac{5}{(x+1)^2} + \frac{1}{(x+1)^3} + \frac{4}{x-4}.$$

$$\mathbf{V36.} \quad \frac{3}{x+7} - \frac{1}{x-5} + \frac{2}{(x-5)^2} + \frac{6}{(x-5)^3}.$$

- V37.** $\frac{3}{x+4} - \frac{4}{x-3} + \frac{4}{(x-3)^2} + \frac{1}{(x-3)^3}.$
- V38.** $-\frac{3}{x+4} + \frac{1}{(x+4)^2} - \frac{6}{(x+4)^3} + \frac{6}{x+3}.$
- V39.** $\frac{6}{x+5} + \frac{1}{(x+5)^2} + \frac{3}{(x+5)^3} - \frac{5}{x-7}.$
- V40.** $\frac{6}{x+4} - \frac{6}{(x+4)^2} - \frac{1}{(x+4)^3} - \frac{4}{x-6}.$
- V41.** $-\frac{3}{x-1} - \frac{1}{x-4} + \frac{6}{(x-4)^2} - \frac{3}{(x-4)^3} - \frac{5}{(x-4)^4}.$
- V42.** $\frac{4}{x+6} + \frac{4}{(x+6)^2} + \frac{6}{(x+6)^3} - \frac{2}{(x+6)^4} - \frac{1}{x+2}.$
- V43.** $-\frac{4}{x+6} - \frac{4}{(x+6)^2} + \frac{4}{(x+6)^3} + \frac{4}{(x+6)^4} - \frac{3}{x+1}.$
- V44.** $-\frac{2}{x+2} + \frac{4}{(x+2)^2} - \frac{5}{(x+2)^3} + \frac{3}{(x+2)^4} + \frac{2}{x-3}.$
- V45.** $\frac{1}{x+3} + \frac{5}{x-3} + \frac{2}{(x-3)^2} - \frac{6}{(x-3)^3} - \frac{5}{(x-3)^4}.$
- V46.** $-\frac{3}{x-4} + \frac{4}{(x-4)^2} + \frac{1}{(x-4)^3} + \frac{3}{(x-4)^4} + \frac{5}{x-5}.$
- V47.** $-\frac{6}{x+4} - \frac{5}{x+2} - \frac{2}{(x+2)^2} - \frac{3}{(x+2)^3} + \frac{1}{(x+2)^4}.$
- V48.** $-\frac{1}{x-2} + \frac{4}{(x-2)^2} - \frac{6}{(x-2)^3} + \frac{4}{(x-2)^4} - \frac{5}{x-6}.$
- V49.** $-\frac{2}{x-1} - \frac{6}{x-2} - \frac{2}{(x-2)^2} + \frac{1}{(x-2)^3} + \frac{4}{(x-2)^4}.$
- V50.** $-\frac{3}{x-1} - \frac{4}{(x-1)^2} + \frac{6}{(x-1)^3} - \frac{1}{(x-1)^4} - \frac{5}{x-7}.$

W. Distinct Irreducible Quadratic Factors

- W1.** $\frac{-3x-2}{x^2-x+1} + \frac{-3x-1}{x^2-2x+3}.$
- W2.** $\frac{4-x}{x^2+x+2} + \frac{-2x-4}{x^2+2x+6}.$
- W3.** $\frac{-5x-2}{x^2-2x+6} + \frac{x+5}{x^2+2x+8}.$
- W4.** $\frac{3x-3}{x^2+x+6} + \frac{4x-5}{x^2+8}.$
- W5.** $\frac{3-2x}{x^2-x+5} + \frac{4x+4}{x^2+5x+7}.$
- W6.** $\frac{4x+2}{x^2+5x+8} + \frac{5x-3}{x^2-x+1}.$

$$\text{W7. } \frac{5-2x}{x^2+2x+7} + \frac{5x+4}{x^2+2x+2}.$$

$$\text{W8. } \frac{-4x-4}{x^2+2} + \frac{-3x-2}{x^2+2x+4}.$$

$$\text{W9. } \frac{3-5x}{x^2+3x+3} + \frac{-2x-1}{x^2+x+1}.$$

$$\text{W10. } \frac{-4x-5}{x^2+4} + \frac{4x+1}{x^2-x+2}.$$

$$\text{W11. } \frac{3x-1}{x^2-x+8} + \frac{3x+3}{x^2+4}.$$

$$\text{W12. } \frac{5-5x}{x^2-4x+7} + \frac{4x-2}{x^2-3x+4}.$$

$$\text{W13. } \frac{1-3x}{x^2+5} + \frac{3-4x}{x^2-x+3}.$$

$$\text{W14. } \frac{-3x-2}{x^2-x+4} + \frac{3x-3}{x^2-3x+5}.$$

$$\text{W15. } \frac{3x+1}{x^2+6} + \frac{3x+5}{x^2+2x+8}.$$

$$\text{W16. } \frac{5-3x}{x^2+2x+2} + \frac{3x-3}{x^2-x+8}.$$

$$\text{W17. } \frac{5x-2}{x^2+4x+6} + \frac{5x-2}{x^2+3}.$$

$$\text{W18. } \frac{5-x}{x^2+x+8} + \frac{5x+5}{x^2-3x+3}.$$

$$\text{W19. } \frac{-3x-5}{x^2+4x+5} + \frac{5x-4}{x^2-x+1}.$$

$$\text{W20. } \frac{-2x-3}{x^2-2x+3} + \frac{3x+3}{x^2+x+6}.$$

$$\text{W21. } \frac{-4x-1}{x^2-3x+3} + \frac{5x+1}{x^2-3x+4}.$$

$$\text{W22. } \frac{-2x-1}{x^2+3x+5} + \frac{x-1}{x^2-x+2}.$$

$$\text{W23. } \frac{5-3x}{x^2-5x+8} + \frac{-2x-3}{x^2-x+6}.$$

$$\text{W24. } \frac{1-3x}{x^2+8} + \frac{-4x-1}{x^2-2x+4}.$$

$$\text{W25. } \frac{3-4x}{x^2-4x+7} + \frac{2x-5}{x^2-4x+8}.$$

$$\text{W26. } \frac{3-x}{x^2+3x+6} + \frac{3-x}{x^2+x+2}.$$

$$\text{W27. } \frac{3-3x}{x^2+x+4} + \frac{2x+3}{x^2-5x+7}.$$

$$\text{W28. } \frac{-5x-1}{x^2+3x+5} + \frac{4x+5}{x^2+6}.$$

$$\text{W29. } \frac{-5x-1}{x^2+4x+5} + \frac{-4x-3}{x^2-4x+8}.$$

$$\mathbf{W30.} \quad \frac{x-5}{x^2+6} + \frac{4x+3}{x^2+2x+8}.$$

$$\mathbf{W31.} \quad \frac{-4x-1}{x^2+4x+6} + \frac{-3x-4}{x^2-x+5}.$$

$$\mathbf{W32.} \quad \frac{4-5x}{x^2-x+5} + \frac{x-4}{x^2-2x+8}.$$

$$\mathbf{W33.} \quad \frac{-4x-4}{x^2+2x+7} + \frac{4x+3}{x^2-x+6}.$$

$$\mathbf{W34.} \quad \frac{1-3x}{x^2+3x+8} + \frac{4x+3}{x^2+x+5}.$$

$$\mathbf{W35.} \quad \frac{3x+2}{x^2-3x+4} + \frac{5x-5}{x^2-3x+6}.$$

$$\mathbf{W36.} \quad \frac{-4x-4}{x^2-3x+5} + \frac{-4x-2}{x^2+4x+6}.$$

$$\mathbf{W37.} \quad \frac{x-4}{x^2+3x+7} + \frac{4x+5}{x^2-2x+8}.$$

$$\mathbf{W38.} \quad \frac{-x-2}{x^2-2x+4} + \frac{5x+3}{x^2-2x+2}.$$

$$\mathbf{W39.} \quad \frac{5x+1}{x^2+3x+3} + \frac{5x+2}{x^2+4x+7}.$$

$$\mathbf{W40.} \quad \frac{-x-3}{x^2+4x+7} + \frac{-x-2}{x^2+2x+6}.$$

$$\mathbf{W41.} \quad \frac{2-3x}{x^2-5x+7} + \frac{3x-1}{x^2+3x+4}.$$

$$\mathbf{W42.} \quad \frac{-5x-3}{x^2+2x+6} + \frac{-x-5}{x^2+7}.$$

$$\mathbf{W43.} \quad \frac{-2x-4}{x^2+x+3} + \frac{3x+1}{x^2-5x+7}.$$

$$\mathbf{W44.} \quad \frac{2x-5}{x^2+3x+8} + \frac{2x-2}{x^2+x+2}.$$

$$\mathbf{W45.} \quad \frac{-5x-3}{x^2+3x+4} + \frac{x+4}{x^2-2x+6}.$$

$$\mathbf{W46.} \quad \frac{x+3}{x^2+x+3} + \frac{4x+5}{x^2+x+6}.$$

$$\mathbf{W47.} \quad \frac{-2x-2}{x^2+5x+7} + \frac{2x-1}{x^2+x+3}.$$

$$\mathbf{W48.} \quad \frac{2-3x}{x^2+3x+3} + \frac{x+1}{x^2+3x+7}.$$

$$\mathbf{W49.} \quad \frac{1-2x}{x^2-5x+8} + \frac{4x-3}{x^2-3x+7}.$$

$$\mathbf{W50.} \quad \frac{-x-5}{x^2+2x+3} + \frac{4x-2}{x^2+8}.$$

X. Repeated Irreducible Quadratic Factors

$$\mathbf{X1.} \quad \frac{-x-2}{x^2+4x+7} + \frac{x-2}{(x^2+4x+7)^2}.$$

$$\mathbf{X2.} \quad \frac{-2x-4}{x^2+x+1} + \frac{x+4}{(x^2+x+1)^2}.$$

$$\mathbf{X3.} \quad \frac{2-2x}{x^2-3x+5} + \frac{-2x-3}{(x^2-3x+5)^2}.$$

$$\mathbf{X4.} \quad \frac{-4x-5}{x^2-3x+5} + \frac{-x-4}{(x^2-3x+5)^2}.$$

$$\mathbf{X5.} \quad \frac{-3x-1}{x^2-4x+7} + \frac{3x-2}{(x^2-4x+7)^2}.$$

$$\mathbf{X6.} \quad \frac{3x+3}{x^2+3x+4} + \frac{4x+3}{(x^2+3x+4)^2}.$$

$$\mathbf{X7.} \quad \frac{4-x}{(x^2-2x+5)^2} + \frac{4x+5}{x^2-2x+5}.$$

$$\mathbf{X8.} \quad \frac{4-2x}{x^2-x+5} + \frac{x-3}{(x^2-x+5)^2}.$$

$$\mathbf{X9.} \quad \frac{-4x-5}{(x^2-2x+5)^2} + \frac{4x+3}{x^2-2x+5}.$$

$$\mathbf{X10.} \quad \frac{5-3x}{x^2+2x+4} + \frac{2x-3}{(x^2+2x+4)^2}.$$

$$\mathbf{X11.} \quad \frac{-2x-1}{x^2+2x+3} + \frac{4x-1}{(x^2+2x+3)^2}.$$

$$\mathbf{X12.} \quad \frac{5-3x}{(x^2-4x+5)^2} + \frac{-2x-4}{x^2-4x+5}.$$

$$\mathbf{X13.} \quad \frac{-x-5}{x^2-3x+4} + \frac{3x+4}{(x^2-3x+4)^2}.$$

$$\mathbf{X14.} \quad \frac{1-3x}{(x^2+3x+3)^2} + \frac{2x+1}{x^2+3x+3}.$$

$$\mathbf{X15.} \quad \frac{x+1}{(x^2-3x+8)^2} + \frac{4x+5}{x^2-3x+8}.$$

$$\mathbf{X16.} \quad \frac{3-3x}{(x^2+2x+2)^2} + \frac{-2x-1}{x^2+2x+2}.$$

$$\mathbf{X17.} \quad \frac{3-x}{x^2+2x+4} + \frac{2x+4}{(x^2+2x+4)^2}.$$

$$\mathbf{X18.} \quad \frac{-2x-1}{x^2+2x+4} + \frac{x+1}{(x^2+2x+4)^2}.$$

$$\mathbf{X19.} \quad \frac{-x-3}{(x^2-2x+2)^2} + \frac{3x+4}{x^2-2x+2}.$$

$$\mathbf{X20.} \quad \frac{4-4x}{x^2-3x+4} + \frac{-3x-4}{(x^2-3x+4)^2}.$$

$$\mathbf{X21.} \quad \frac{1-x}{(x^2-x+7)^2} + \frac{2x-4}{x^2-x+7}.$$

$$\mathbf{X22.} \quad \frac{4x-3}{(x^2+8)^2} + \frac{4x-2}{x^2+8}.$$

$$\mathbf{X23.} \quad \frac{4-4x}{(x^2+3x+4)^2} + \frac{3x-5}{x^2+3x+4}.$$

$$\mathbf{X24.} \quad \frac{-4x-2}{x^2+2x+3} + \frac{4x+1}{(x^2+2x+3)^2}.$$

$$\mathbf{X25.} \quad \frac{-2x-4}{x^2-x+5} + \frac{x-3}{(x^2-x+5)^2}.$$

$$\mathbf{X26.} \quad \frac{-3x-4}{x^2+4x+7} + \frac{4x+3}{(x^2+4x+7)^2}.$$

$$\mathbf{X27.} \quad \frac{2x+5}{(x^2-2x+7)^2} + \frac{3x-3}{x^2-2x+7}.$$

$$\mathbf{X28.} \quad \frac{-2x-4}{x^2+4x+5} + \frac{-x-4}{(x^2+4x+5)^2}.$$

$$\mathbf{X29.} \quad \frac{-2x-4}{x^2-3x+3} + \frac{4x-3}{(x^2-3x+3)^2}.$$

$$\mathbf{X30.} \quad \frac{-3x-5}{(x^2+4x+7)^2} + \frac{2x+3}{x^2+4x+7}.$$

$$\mathbf{X31.} \quad \frac{4-4x}{x^2-2x+5} + \frac{-x-3}{(x^2-2x+5)^2} + \frac{4x+5}{(x^2-2x+5)^3}.$$

$$\mathbf{X32.} \quad \frac{4-x}{(x^2-4x+7)^2} + \frac{2x-2}{(x^2-4x+7)^3} + \frac{3x+1}{x^2-4x+7}.$$

$$\mathbf{X33.} \quad \frac{-4x-2}{(x^2-x+2)^2} + \frac{3x+2}{x^2-x+2} + \frac{4x-4}{(x^2-x+2)^3}.$$

$$\mathbf{X34.} \quad \frac{-2x-4}{x^2+3x+8} + \frac{2x-2}{(x^2+3x+8)^2} + \frac{4x+2}{(x^2+3x+8)^3}.$$

$$\mathbf{X35.} \quad \frac{1-3x}{(x^2+4x+5)^2} + \frac{-2x-5}{x^2+4x+5} + \frac{x+3}{(x^2+4x+5)^3}.$$

$$\mathbf{X36.} \quad \frac{3-3x}{(x^2-3x+3)^3} + \frac{3x+3}{x^2-3x+3} + \frac{4x-2}{(x^2-3x+3)^2}.$$

$$\mathbf{X37.} \quad \frac{-3x-4}{(x^2-3x+4)^3} + \frac{3x-5}{(x^2-3x+4)^2} + \frac{4x+3}{x^2-3x+4}.$$

$$\mathbf{X38.} \quad \frac{-3x-1}{x^2+2x+5} + \frac{-2x-3}{(x^2+2x+5)^3} + \frac{-2x-1}{(x^2+2x+5)^2}.$$

$$\mathbf{X39.} \quad \frac{-3x-5}{x^2-2x+5} + \frac{2x+5}{(x^2-2x+5)^2} + \frac{3x-4}{(x^2-2x+5)^3}.$$

$$\mathbf{X40.} \quad \frac{4-4x}{(x^2+8)^3} + \frac{2x+3}{(x^2+8)^2} + \frac{3x+3}{x^2+8}.$$

$$\mathbf{X41.} \quad \frac{3-4x}{(x^2+6)^2} + \frac{2x+5}{(x^2+6)^3} + \frac{3x+5}{x^2+6} + \frac{4}{x-3}.$$

$$\begin{aligned}
\text{X42. } & \frac{2x-4}{x^2+5x+7} + \frac{3x-1}{(x^2+5x+7)^3} + \frac{4x-1}{(x^2+5x+7)^2} - \frac{4}{x-4}. \\
\text{X43. } & \frac{-3x-2}{(x^2-4x+6)^3} + \frac{2x-1}{(x^2-4x+6)^2} + \frac{4x-1}{x^2-4x+6} + \frac{5}{x-1}. \\
\text{X44. } & \frac{x+3}{x^2+2} + \frac{2x+5}{(x^2+2)^2} + \frac{4x-5}{(x^2+2)^3} - \frac{2}{x-5}. \\
\text{X45. } & \frac{4-x}{(x^2-4x+5)^3} + \frac{-4x-1}{x^2-4x+5} + \frac{-x-2}{(x^2-4x+5)^2} + \frac{2}{x-1}. \\
\text{X46. } & \frac{2-2x}{(x^2+x+2)^2} + \frac{3x+3}{(x^2+x+2)^3} + \frac{4x-4}{x^2+x+2} + \frac{3}{x-3}. \\
\text{X47. } & \frac{4-x}{(x^2-3x+6)^2} + \frac{3x+2}{x^2-3x+6} + \frac{4x-2}{(x^2-3x+6)^3} + \frac{2}{x+3}. \\
\text{X48. } & \frac{x+1}{(x^2+3x+6)^2} + \frac{2x-2}{(x^2+3x+6)^3} + \frac{2x+5}{x^2+3x+6} + \frac{5}{x-1}. \\
\text{X49. } & \frac{5-4x}{(x^2+7)^2} + \frac{-4x-5}{(x^2+7)^3} + \frac{x-5}{x^2+7} - \frac{2}{x+2}. \\
\text{X50. } & \frac{-3x-2}{(x^2-5x+7)^2} + \frac{x+3}{x^2-5x+7} + \frac{4x+2}{(x^2-5x+7)^3} + \frac{4}{x+5}.
\end{aligned}$$

Y. Mixed Denominator Types

$$\begin{aligned}
\text{Y1. } & \frac{4-x}{x^2+3x+8} + \frac{5}{x+1} + \frac{1}{x}. \\
\text{Y2. } & \frac{5-3x}{x^2+2x+6} - \frac{5}{x+2} - \frac{2}{x-5}. \\
\text{Y3. } & \frac{3-4x}{x^2+4x+7} + \frac{2}{x+3} - \frac{1}{x-2}. \\
\text{Y4. } & \frac{4x-2}{x^2-x+7} - \frac{1}{x-1} - \frac{2}{x-6}. \\
\text{Y5. } & \frac{x+5}{x^2+5x+7} - \frac{5}{x-1} - \frac{5}{x-4}. \\
\text{Y6. } & \frac{2x+2}{x^2+x+3} + \frac{3}{x-3} - \frac{4}{x}. \\
\text{Y7. } & \frac{x-2}{x^2-4x+6} + \frac{2}{x-2} + \frac{4}{x-4}. \\
\text{Y8. } & \frac{2x-5}{x^2-x+5} - \frac{2}{x+6} - \frac{5}{x-5}. \\
\text{Y9. } & \frac{2-3x}{x^2-3x+6} - \frac{1}{x+6} - \frac{3}{x-4}. \\
\text{Y10. } & \frac{2x-1}{x^2-x+1} - \frac{1}{x+2} - \frac{1}{x-3}. \\
\text{Y11. } & \frac{2x+5}{x^2+4} + \frac{4}{x-2} + \frac{2}{(x-2)^2} - \frac{5}{x-3}.
\end{aligned}$$

- Y12. $\frac{-2x-2}{x^2-x+7} - \frac{4}{x+5} + \frac{1}{x-5} + \frac{3}{(x-5)^2}.$
- Y13. $\frac{x+3}{x^2-x+5} - \frac{3}{x+4} - \frac{2}{x+3} + \frac{1}{(x+3)^2}.$
- Y14. $\frac{2x-2}{x^2-3x+5} - \frac{1}{x+4} - \frac{3}{(x+4)^2} + \frac{3}{x-1}.$
- Y15. $\frac{4-4x}{x^2+2x+8} - \frac{3}{x+6} - \frac{5}{x+1} - \frac{5}{(x+1)^2}.$
- Y16. $\frac{-x-1}{x^2+4} - \frac{5}{x-3} - \frac{3}{x} + \frac{4}{x^2}.$
- Y17. $\frac{3-x}{x^2-2x+4} - \frac{1}{x+5} - \frac{5}{x-4} + \frac{3}{(x-4)^2}.$
- Y18. $\frac{-4x-2}{x^2+2x+4} - \frac{3}{x+3} + \frac{1}{x} - \frac{4}{x^2}.$
- Y19. $\frac{2x-1}{x^2-2x+8} - \frac{3}{x+3} + \frac{5}{(x+3)^2} + \frac{1}{x+1}.$
- Y20. $\frac{2x+4}{x^2-2x+8} + \frac{4}{x-2} - \frac{4}{x-6} - \frac{4}{(x-6)^2}.$
- Y21. $\frac{x-2}{x^2+4x+7} + \frac{2}{x-5} + \frac{3}{x-6} - \frac{2}{(x-6)^2}.$
- Y22. $\frac{x-5}{x^2+x+1} - \frac{3}{x+6} - \frac{4}{(x+6)^2} + \frac{3}{x-2}.$
- Y23. $\frac{4x-1}{x^2-3x+3} + \frac{1}{x-1} - \frac{1}{(x-1)^2} - \frac{5}{x-4}.$
- Y24. $\frac{-2x-4}{x^2+x+5} + \frac{1}{x-5} - \frac{2}{(x-5)^2} - \frac{4}{x-6}.$
- Y25. $\frac{4x-1}{x^2+8} - \frac{4}{x+6} - \frac{1}{(x+6)^2} + \frac{2}{x-5}.$
- Y26. $\frac{2-2x}{x^2+6} + \frac{5}{x-6} - \frac{5}{x} - \frac{3}{x^2}.$
- Y27. $\frac{x-1}{x^2+8} + \frac{5}{x+4} - \frac{5}{(x+4)^2} + \frac{2}{x-6}.$
- Y28. $\frac{5-4x}{x^2-2x+5} + \frac{4}{x+3} + \frac{1}{(x+3)^2} - \frac{5}{x-3}.$
- Y29. $\frac{4-x}{x^2-x+7} + \frac{5}{x-3} + \frac{5}{x-5} + \frac{1}{(x-5)^2}.$
- Y30. $\frac{4-3x}{x^2+3x+8} - \frac{2}{x+1} + \frac{5}{x-3} + \frac{3}{(x-3)^2}.$
- Y31. $\frac{-4x-2}{(x^2-x+7)^2} + \frac{2x+4}{x^2-x+7} - \frac{5}{x+2} + \frac{3}{(x+2)^2} - \frac{1}{x-5}.$
- Y32. $\frac{-2x-2}{x^2+4} + \frac{3x-3}{(x^2+4)^2} - \frac{5}{x-1} - \frac{1}{(x-1)^2} + \frac{5}{x-3}.$

$$\mathbf{Y33.} \quad \frac{5-4x}{(x^2-5x+8)^2} + \frac{3x+1}{x^2-5x+8} + \frac{2}{x+2} - \frac{1}{x-3} - \frac{3}{(x-3)^2}.$$

$$\mathbf{Y34.} \quad \frac{-2x-4}{x^2+4} + \frac{2x-1}{(x^2+4)^2} - \frac{5}{x+3} - \frac{5}{(x+3)^2} + \frac{2}{x-4}.$$

$$\mathbf{Y35.} \quad \frac{5-4x}{x^2-5x+8} + \frac{-2x-5}{(x^2-5x+8)^2} + \frac{2}{x+2} + \frac{2}{x-4} - \frac{2}{(x-4)^2}.$$

$$\mathbf{Y36.} \quad \frac{x+3}{x^2+5} + \frac{3x+5}{(x^2+5)^2} - \frac{4}{x+1} + \frac{5}{(x+1)^2} - \frac{2}{x-3}.$$

$$\mathbf{Y37.} \quad \frac{5-4x}{(x^2-2x+4)^2} + \frac{-3x-2}{x^2-2x+4} + \frac{1}{x+6} - \frac{1}{(x+6)^2} + \frac{1}{x-5}.$$

$$\mathbf{Y38.} \quad \frac{2x+4}{(x^2-3x+8)^2} + \frac{4x+2}{x^2-3x+8} + \frac{4}{x+3} - \frac{5}{(x+3)^2} - \frac{2}{x-3}.$$

$$\mathbf{Y39.} \quad \frac{-x-2}{x^2+7} + \frac{2x+5}{(x^2+7)^2} - \frac{5}{x+2} - \frac{2}{x-3} - \frac{3}{(x-3)^2}.$$

$$\mathbf{Y40.} \quad \frac{-2x-2}{x^2+5x+7} + \frac{3x+3}{(x^2+5x+7)^2} + \frac{3}{x+6} - \frac{1}{(x+6)^2} - \frac{4}{x+5}.$$

$$\mathbf{Y41.} \quad \frac{5-2x}{(x^2-4x+5)^2} + \frac{2x-5}{x^2-4x+5} - \frac{4}{x+1} - \frac{4}{(x+1)^2} - \frac{5}{x-3}.$$

$$\mathbf{Y42.} \quad \frac{4-2x}{(x^2+x+4)^2} + \frac{-3x-2}{x^2+x+4} + \frac{2}{x-1} + \frac{5}{x-5} + \frac{3}{(x-5)^2}.$$

$$\mathbf{Y43.} \quad \frac{5-4x}{x^2+4x+5} + \frac{4x-4}{(x^2+4x+5)^2} + \frac{2}{x-2} - \frac{4}{x-4} - \frac{2}{(x-4)^2}.$$

$$\mathbf{Y44.} \quad \frac{x+5}{(x^2-3x+6)^2} + \frac{3x-4}{x^2-3x+6} + \frac{5}{x+4} + \frac{1}{x+2} + \frac{1}{(x+2)^2}.$$

$$\mathbf{Y45.} \quad \frac{3-3x}{x^2-4x+7} + \frac{2x-4}{(x^2-4x+7)^2} + \frac{2}{x+5} - \frac{3}{x+3} + \frac{3}{(x+3)^2}.$$

$$\mathbf{Y46.} \quad \frac{-2x-5}{(x^2-2x+2)^2} + \frac{3x-3}{x^2-2x+2} + \frac{5}{x+4} + \frac{1}{(x+4)^2} - \frac{1}{x-6}.$$

$$\mathbf{Y47.} \quad \frac{5-2x}{(x^2+2x+5)^2} + \frac{4x-4}{x^2+2x+5} - \frac{1}{x+5} - \frac{4}{x+4} + \frac{1}{(x+4)^2}.$$

$$\mathbf{Y48.} \quad \frac{-4x-3}{(x^2-2x+4)^2} + \frac{3x+3}{x^2-2x+4} + \frac{5}{x+1} + \frac{2}{x-2} - \frac{1}{(x-2)^2}.$$

$$\mathbf{Y49.} \quad \frac{2x-2}{(x^2+2x+3)^2} + \frac{4x-5}{x^2+2x+3} + \frac{5}{x-4} + \frac{1}{x} + \frac{4}{x^2}.$$

$$\mathbf{Y50.} \quad \frac{-2x-2}{(x^2-x+2)^2} + \frac{-x-3}{x^2-x+2} + \frac{1}{x+1} + \frac{3}{x-4} + \frac{4}{(x-4)^2}.$$

Z. Improper Rational Expressions and Partial Fractions

$$\mathbf{Z1.} \quad (2-3x) - \frac{4}{x+6} + \frac{3}{x+5}.$$

$$\mathbf{Z2.} \quad (2-x) + \frac{1}{x+3} + \frac{1}{x-5}.$$

$$\mathbf{Z3.} \quad (3-3x) - \frac{2}{x+3} - \frac{3}{x}.$$

$$\mathbf{Z4.} \quad (-2x-3) + \frac{2}{x+2} - \frac{5}{x-5}.$$

$$\mathbf{Z5.} \quad -2x - \frac{2}{x+1} + \frac{3}{x-5}.$$

$$\mathbf{Z6.} \quad (3-3x) - \frac{2}{x+6} + \frac{1}{x-1}.$$

$$\mathbf{Z7.} \quad (x-2) + \frac{4}{x+4} + \frac{1}{x-6}.$$

$$\mathbf{Z8.} \quad (2-x) - \frac{5}{x+6} + \frac{4}{x-6}.$$

$$\mathbf{Z9.} \quad (-3x-2) - \frac{2}{x+1} - \frac{3}{x-3}.$$

$$\mathbf{Z10.} \quad (x-2) + \frac{4}{x+4} - \frac{2}{x+2}.$$

$$\mathbf{Z11.} \quad (-x-3) - \frac{5}{x-2} + \frac{5}{x}.$$

$$\mathbf{Z12.} \quad (3x-3) + \frac{4}{x+6} - \frac{3}{x-2}.$$

$$\mathbf{Z13.} \quad -x + \frac{2}{x+1} + \frac{4}{x-6}.$$

$$\mathbf{Z14.} \quad (2x-1) - \frac{2}{x-3} - \frac{2}{x-6}.$$

$$\mathbf{Z15.} \quad (-x-3) - \frac{4}{x+1} + \frac{3}{x-5}.$$

$$\mathbf{Z16.} \quad (-x-3) + \frac{3}{x+4} + \frac{2}{x+3}.$$

$$\mathbf{Z17.} \quad (x-2) + \frac{2}{x+5} - \frac{2}{x-4}.$$

$$\mathbf{Z18.} \quad 2x + \frac{3}{x-4} - \frac{1}{x-6}.$$

$$\mathbf{Z19.} \quad (x+1) - \frac{4}{x+5} + \frac{4}{x+1}.$$

$$\mathbf{Z20.} \quad 3x + \frac{2}{x+1} + \frac{5}{x}.$$

$$\mathbf{Z21.} \quad (x^2-x) - \frac{1}{x+5} - \frac{1}{x-5}.$$

$$\mathbf{Z22.} \quad (x^2-3) + \frac{4}{x+3} + \frac{5}{x-2}.$$

$$\mathbf{Z23.} \quad (-3x^2+2x+1) + \frac{3}{x+4} + \frac{3}{x-5}.$$

$$\mathbf{Z24.} \quad (2x^2+2x+1) + \frac{3}{x+3} - \frac{2}{x-3}.$$

$$\mathbf{Z25.} \quad (2x^2 - 3x) - \frac{1}{x-6} + \frac{5}{x}.$$

$$\mathbf{Z26.} \quad (x^2 + x - 1) + \frac{4}{x+3} - \frac{4}{x-2}.$$

$$\mathbf{Z27.} \quad (1 - x^2) - \frac{5}{x+4} - \frac{3}{x-1}.$$

$$\mathbf{Z28.} \quad (-2x^2 + 2x - 2) - \frac{4}{x+2} - \frac{1}{x-6}.$$

$$\mathbf{Z29.} \quad (-3x^2 + 3x - 2) + \frac{1}{x+4} - \frac{2}{x}.$$

$$\mathbf{Z30.} \quad 2x^2 + \frac{3}{x+2} + \frac{3}{x-6}.$$

$$\mathbf{Z31.} \quad (-2x^2 + x - 2) + \frac{3}{x+3} - \frac{5}{x+1} - \frac{5}{x-6}.$$

$$\mathbf{Z32.} \quad (-3x^2 + 2x + 3) + \frac{4}{x-1} + \frac{4}{x-3} + \frac{1}{x}.$$

$$\mathbf{Z33.} \quad (-3x^2 - x) + \frac{5}{x+4} + \frac{4}{x+3} + \frac{1}{x-3}.$$

$$\mathbf{Z34.} \quad (2x^2 - 3x + 2) - \frac{1}{x+4} - \frac{4}{x+3} - \frac{2}{x-4}.$$

$$\mathbf{Z35.} \quad (3x^2 + 2x - 1) + \frac{4}{x+2} + \frac{5}{x-4} + \frac{4}{x-6}.$$

$$\mathbf{Z36.} \quad -2x^2 - \frac{2}{x-1} + \frac{4}{x-2} + \frac{5}{x-5}.$$

$$\mathbf{Z37.} \quad (-2x^2 + 2x - 2) + \frac{4}{x+6} - \frac{5}{x+2} + \frac{1}{x-6}.$$

$$\mathbf{Z38.} \quad (-2x^2 - 2x + 1) + \frac{2}{x+3} + \frac{2}{x+1} - \frac{4}{x}.$$

$$\mathbf{Z39.} \quad (-2x^2 - 3x + 1) + \frac{1}{x+6} - \frac{4}{x-4} + \frac{2}{x-6}.$$

$$\mathbf{Z40.} \quad (x^2 - 3x + 2) - \frac{1}{x+5} + \frac{5}{x-5} + \frac{1}{x-6}.$$

$$\mathbf{Z41.} \quad (2x^3 + 3x + 1) - \frac{1}{x+3} - \frac{4}{x-2} + \frac{5}{x-5}.$$

$$\mathbf{Z42.} \quad (2x^3 + 2x^2 - 3) + \frac{4}{x-1} - \frac{2}{x-4} - \frac{5}{x}.$$

$$\mathbf{Z43.} \quad (-2x^3 - 2x^2 + 2x + 3) + \frac{2}{x-3} - \frac{4}{x-6} + \frac{4}{x}.$$

$$\mathbf{Z44.} \quad (-x^3 - 2x^2 + x) - \frac{3}{x+4} - \frac{1}{x-6} + \frac{4}{x}.$$

$$\mathbf{Z45.} \quad (x^3 - 3x^2 + 2x - 2) + \frac{4}{x+4} + \frac{3}{x+2} + \frac{5}{x-3}.$$

$$\mathbf{Z46.} \quad (-x^3 + x^2 + x - 3) + \frac{5}{x-3} - \frac{3}{x-5} + \frac{1}{x}.$$

$$\mathbf{Z47.} \quad (3x^3 - x^2 + x - 3) + \frac{5}{x+3} - \frac{5}{x-1} + \frac{5}{x-3}.$$

$$\mathbf{Z48.} \quad (2x^3 - x^2 - x + 3) + \frac{2}{x+1} + \frac{1}{x-4} - \frac{1}{x-5}.$$

$$\mathbf{Z49.} \quad (-2x^3 - 2x^2 + 2x + 3) - \frac{1}{x-3} + \frac{5}{x-4} + \frac{4}{x-5}.$$

$$\mathbf{Z50.} \quad (x^3 - x^2 + 3x) - \frac{2}{x+2} + \frac{2}{x-3} - \frac{3}{x-6}.$$

AA. Hybrid Coefficient Methods

$$\mathbf{AA1.} \quad \frac{-4x-4}{x^2+x+7} - \frac{1}{x+4} - \frac{4}{x-4}.$$

$$\mathbf{AA2.} \quad \frac{-2x-5}{x^2-x+5} + \frac{2}{x+3} - \frac{5}{x}.$$

$$\mathbf{AA3.} \quad \frac{5-2x}{x^2+2} + \frac{5}{x-3} + \frac{5}{x-4}.$$

$$\mathbf{AA4.} \quad \frac{3x+3}{x^2-2x+7} - \frac{4}{x-3} - \frac{2}{x}.$$

$$\mathbf{AA5.} \quad \frac{2x-3}{x^2+2x+7} + \frac{1}{x+3} + \frac{1}{x+2}.$$

$$\mathbf{AA6.} \quad \frac{x+3}{x^2+x+6} - \frac{3}{x+1} - \frac{2}{x}.$$

$$\mathbf{AA7.} \quad \frac{-2x-1}{x^2-3x+8} + \frac{2}{x+4} - \frac{2}{x-4}.$$

$$\mathbf{AA8.} \quad \frac{3x+5}{x^2+3x+7} - \frac{4}{x+6} - \frac{1}{x-6}.$$

$$\mathbf{AA9.} \quad \frac{3x-1}{x^2-2x+5} + \frac{5}{x+6} + \frac{4}{x+5}.$$

$$\mathbf{AA10.} \quad \frac{x+3}{x^2-4x+7} - \frac{4}{x-6} - \frac{3}{x}.$$

$$\mathbf{AA11.} \quad \frac{x+2}{x^2+4x+7} - \frac{4}{x+3} - \frac{2}{x}.$$

$$\mathbf{AA12.} \quad \frac{x-3}{x^2+4x+5} - \frac{2}{x-6} - \frac{1}{x}.$$

$$\mathbf{AA13.} \quad \frac{4x+2}{x^2+4x+6} - \frac{5}{x+4} + \frac{1}{x+3}.$$

$$\mathbf{AA14.} \quad \frac{2-2x}{x^2-x+4} + \frac{3}{x+1} - \frac{5}{x-2}.$$

$$\mathbf{AA15.} \quad \frac{x-3}{x^2+x+8} + \frac{1}{x-3} + \frac{5}{x-5}.$$

$$\mathbf{AA16.} \quad \frac{4-4x}{x^2+4x+5} + \frac{5}{x-4} + \frac{2}{x-6}.$$

$$\mathbf{AA17.} \quad \frac{4x-3}{x^2+3x+3} - \frac{3}{x-4} + \frac{2}{x-5}.$$

$$\mathbf{AA18.} \quad \frac{5-3x}{x^2-3x+3} + \frac{3}{x+3} - \frac{1}{x-3}.$$

- AA19. $\frac{4-2x}{x^2+1} - \frac{4}{x+6} - \frac{4}{x-1}.$
- AA20. $\frac{2x-2}{x^2-x+5} + \frac{5}{x-2} + \frac{3}{x-3}.$
- AA21. $\frac{-3x-5}{x^2-x+4} - \frac{2}{x+5} - \frac{4}{x-6} + \frac{2}{(x-6)^2}.$
- AA22. $\frac{-x-1}{x^2+2x+2} + \frac{3}{x+1} + \frac{3}{(x+1)^2} - \frac{3}{x-3}.$
- AA23. $\frac{5-4x}{x^2-x+5} - \frac{4}{x+4} + \frac{2}{(x+4)^2} + \frac{4}{x-4}.$
- AA24. $\frac{x+5}{x^2-2x+2} - \frac{2}{x+3} - \frac{4}{x-5} + \frac{1}{(x-5)^2}.$
- AA25. $\frac{-x-4}{x^2-4x+7} + \frac{4}{x+6} + \frac{2}{(x+6)^2} + \frac{5}{x-4}.$
- AA26. $\frac{4-2x}{x^2-5x+8} + \frac{2}{x-4} - \frac{5}{x} + \frac{2}{x^2}.$
- AA27. $\frac{2-4x}{x^2+x+1} + \frac{3}{x+3} + \frac{2}{(x+3)^2} - \frac{3}{x-3}.$
- AA28. $\frac{-3x-4}{x^2-3x+8} - \frac{3}{x+3} + \frac{3}{x+2} + \frac{3}{(x+2)^2}.$
- AA29. $\frac{2x-3}{x^2+x+8} - \frac{4}{x+5} + \frac{1}{x-2} + \frac{1}{(x-2)^2}.$
- AA30. $\frac{3-4x}{x^2-2x+5} + \frac{2}{x+2} - \frac{1}{(x+2)^2} + \frac{1}{x-4}.$
- AA31. $\frac{-3x-1}{x^2+4x+7} + \frac{3}{x-6} + \frac{3}{(x-6)^2} - \frac{2}{x}.$
- AA32. $\frac{2-3x}{x^2+3x+8} + \frac{5}{x+2} + \frac{5}{x-3} + \frac{3}{(x-3)^2}.$
- AA33. $\frac{4x+5}{x^2-x+5} - \frac{1}{x+4} - \frac{1}{x-5} - \frac{1}{(x-5)^2}.$
- AA34. $\frac{-2x-4}{x^2-3x+5} - \frac{5}{x+2} - \frac{4}{x+1} - \frac{3}{(x+1)^2}.$
- AA35. $\frac{-x-5}{x^2+x+7} + \frac{3}{x+1} - \frac{1}{x-1} + \frac{2}{(x-1)^2}.$
- AA36. $\frac{x-5}{x^2+3x+3} - \frac{5}{x+3} + \frac{3}{x-6} + \frac{1}{(x-6)^2}.$
- AA37. $\frac{x+1}{x^2-2x+8} + \frac{2}{x+5} - \frac{5}{x-6} - \frac{5}{(x-6)^2}.$
- AA38. $\frac{-2x-4}{x^2+1} - \frac{4}{x+2} - \frac{1}{x-5} + \frac{3}{(x-5)^2}.$
- AA39. $\frac{2x+2}{x^2-2x+4} - \frac{2}{x+3} - \frac{1}{(x+3)^2} + \frac{4}{x-4}.$

$$\mathbf{AA40.} \quad \frac{-3x-5}{x^2-x+5} - \frac{3}{x+6} + \frac{5}{x-5} + \frac{1}{(x-5)^2}.$$

$$\mathbf{AA41.} \quad \frac{2x-2}{x^2+2x+2} - \frac{1}{x+5} + \frac{3}{(x+5)^2} + \frac{2}{x-3}.$$

$$\mathbf{AA42.} \quad \frac{4-4x}{x^2+5x+8} + \frac{4}{x+6} + \frac{2}{(x+6)^2} - \frac{2}{x-1}.$$

$$\mathbf{AA43.} \quad \frac{3x+3}{x^2-3x+4} - \frac{5}{x-3} + \frac{3}{x-5} - \frac{1}{(x-5)^2}.$$

$$\mathbf{AA44.} \quad \frac{2x-5}{x^2+3x+5} + \frac{5}{x+1} - \frac{4}{(x+1)^2} - \frac{4}{x-2}.$$

$$\mathbf{AA45.} \quad \frac{x+1}{x^2+x+6} + \frac{3}{x+5} - \frac{2}{x+1} + \frac{2}{(x+1)^2}.$$

$$\mathbf{AA46.} \quad \frac{3x+3}{x^2-x+3} - \frac{3}{x-5} + \frac{3}{(x-5)^2} - \frac{5}{x-6}.$$

$$\mathbf{AA47.} \quad \frac{4x-4}{x^2-x+4} - \frac{3}{x-5} + \frac{1}{x-6} - \frac{5}{(x-6)^2}.$$

$$\mathbf{AA48.} \quad \frac{2x+2}{x^2-x+6} + \frac{5}{x+6} - \frac{3}{x+5} - \frac{2}{(x+5)^2}.$$

$$\mathbf{AA49.} \quad \frac{2x-3}{x^2+3} - \frac{2}{x-5} + \frac{1}{(x-5)^2} - \frac{3}{x}.$$

$$\mathbf{AA50.} \quad \frac{3-2x}{x^2-2x+3} - \frac{3}{x+4} - \frac{4}{x-1} + \frac{2}{(x-1)^2}.$$

AB. Verifying Decompositions

AB1. Valid.

AB2. Invalid. The corrected decomposition is $-\frac{3}{x+5} + \frac{3}{x-6}$.

AB3. Valid.

AB4. Invalid. The corrected decomposition is $-\frac{2}{x-3} - \frac{6}{x}$.

AB5. Valid.

AB6. Invalid. The corrected decomposition is $-\frac{2}{x+6} + \frac{2}{x+1}$.

AB7. Valid.

AB8. Invalid. The corrected decomposition is $-\frac{6}{x+7} - \frac{2}{x+3}$.

AB9. Valid.

AB10. Invalid. The corrected decomposition is $\frac{4}{x-4} + \frac{2}{x-7}$.

AB11. Valid.

AB12. Invalid. The corrected decomposition is $-\frac{6}{x+1} + \frac{4}{x-1}$.

AB13. Valid.

AB14. Invalid. The corrected decomposition is $\frac{5}{x+1} + \frac{2}{x}$.

AB15. Valid.

AB16. Invalid. The corrected decomposition is $\frac{6}{x+6} + \frac{1}{x+5}$.

AB17. Valid.

AB18. Invalid. The corrected decomposition is $-\frac{3}{x+4} + \frac{4}{x+3}$.

AB19. Valid.

AB20. Invalid. The corrected decomposition is $-\frac{3}{x+2} + \frac{5}{x+1}$.

AB21. Valid.

AB22. Invalid. The corrected decomposition is $\frac{1}{x+3} + \frac{4}{x-6}$.

AB23. Valid.

AB24. Invalid. The corrected decomposition is $\frac{2}{x+1} - \frac{3}{x-4}$.

AB25. Valid.

AB26. Invalid. The corrected decomposition is $-\frac{5}{x+5} - \frac{3}{x+3} + \frac{6}{x-5}$.

AB27. Valid.

AB28. Invalid. The corrected decomposition is $-\frac{3}{x+3} - \frac{3}{x+2} + \frac{4}{x-5}$.

AB29. Valid.

AB30. Invalid. The corrected decomposition is $\frac{1}{x+6} + \frac{4}{x+4} + \frac{2}{x-1}$.

AB31. Valid.

AB32. Invalid. The corrected decomposition is $\frac{2}{x+6} - \frac{3}{x-1} - \frac{2}{x-5}$.

AB33. Valid.

AB34. Invalid. The corrected decomposition is $\frac{3}{x+4} + \frac{3}{x-2} - \frac{3}{x-3}$.

AB35. Valid.

AB36. Invalid. The corrected decomposition is $\frac{1}{x+6} + \frac{4}{x+4} - \frac{6}{x-3}$.

AB37. Valid.

AB38. Invalid. The corrected decomposition is $-\frac{3}{x+5} - \frac{4}{x-2} + \frac{2}{x-6}$.

AB39. Valid.

AB40. Invalid. The corrected decomposition is $\frac{5}{x+5} - \frac{4}{x+2} - \frac{5}{x-1}$.

AB41. Valid.

AB42. Invalid. The corrected decomposition is $-\frac{4}{x+2} + \frac{3}{x-4} + \frac{2}{x}$.

AB43. Valid.

AB44. Invalid. The corrected decomposition is $-\frac{6}{x+5} + \frac{3}{x-5} - \frac{2}{x-6}$.

AB45. Valid.

AB46. Invalid. The corrected decomposition is $\frac{2}{x+4} - \frac{4}{x+1} - \frac{6}{x-5}$.

AB47. Valid.

AB48. Invalid. The corrected decomposition is $-\frac{6}{x+6} + \frac{6}{x-2} + \frac{6}{x-3}$.

AB49. Valid.

AB50. Invalid. The corrected decomposition is $\frac{6}{x+6} - \frac{1}{x+1} + \frac{6}{x-5}$.

CAP. Capstone

CAP1. Shape: common numerical/monomial factor. **Tools:** extract the greatest common factor. **Result:** $12x^2(2x+5)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP2. Shape: difference of two perfect squares. **Tools:** apply $A^2 - B^2 = (A - B)(A + B)$ repeatedly. **Result:** $9(2x-3)(2x+3)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP3. Shape: three-term perfect-square pattern. **Tools:** match $A^2 \pm 2AB + B^2$. **Result:** $(4x-1)^2$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP4. Shape: sum or difference of cubes. **Tools:** apply the cube identity, after any GCF. **Result:** $(2x+3)(4x^2-6x+9)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP5. Shape: monic quadratic trinomial. **Tools:** find two integers with sum $-b$ and product c . **Result:** $(x-4)(x-2)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP6. Shape: nonmonic quadratic trinomial. **Tools:** split the middle term or use an ac method, then group. **Result:** $(x+6)(3x+8)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP7. Shape: four or more terms with a repeated binomial after grouping. **Tools:** group terms and extract the common binomial. **Result:** $(x-5)(2x^2+2x-1)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP8. Shape: quadratic pattern in x^2 . **Tools:** substitute $u = x^2$, factor in u , then replace u . **Result:** $(x^2+1)(x^2+7)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP9. Shape: expression containing more than one factoring pattern. **Tools:** GCF first, then repeatedly factor until no listed tool applies. **Result:** $2x(x-2)^2(x+2)(x^2+2x+4)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP10. Shape: quotient with common polynomial factors. **Tools:** factor both parts, cancel factors only, preserve original restrictions. **Result:** $\frac{x+4}{x+5}$, with $x \neq -5, -3$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.

CAP11. Shape: product of rational expressions. **Tools:** factor, cancel across the product, then multiply. **Result:** $\frac{x^2-11x+28}{x^2+2x}$, with $x \neq -2, 0, 2, 8$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.

CAP12. Shape: quotient of rational expressions. **Tools:** record restrictions, multiply by the reciprocal, factor, and cancel. **Result:** $\frac{x^2+12x+32}{x^2+2x-24}$, with $x \neq -8, -6, -1, 0, 4$. **Check:** recombine or

substitute an allowed numerical value, while checking all original restrictions.

- CAP13. Shape:** sum or difference with unlike denominators. **Tools:** factor denominators, build the LCD, combine, and simplify. **Result:** $\frac{-10x + 48}{x^2 - 9x + 18}$, with $x \neq 3, 6$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP14. Shape:** fraction whose numerator or denominator contains fractions. **Tools:** identify the internal LCD and clear all small denominators. **Result:** $\frac{3x - 7}{x - 1}$, with $x \neq 1, 3$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP15. Shape:** polynomial dividend and nonzero polynomial divisor. **Tools:** polynomial long division with zero placeholders as needed. **Result:** $Q(x) = -3x^2 - 2x + 1$, $R(x) = 1$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP16. Shape:** linear divisor of the form $x - c$. **Tools:** synthetic division using c and every coefficient, including zeros. **Result:** $Q(x) = -2x^3 - x^2 - 4x - 3$, $R = 7$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP17. Shape:** inspect the divisor. **Tools:** use synthetic division only for $x - c$; otherwise use long division. **Result:** polynomial long division; $Q(x) = -3x - 4$, $R(x) = 3x + 2$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP18. Shape:** proper rational expression with distinct linear denominator factors. **Tools:** write the distinct-linear template and determine its constants. **Result:** $-\frac{6}{x + 3} + \frac{3}{x - 8}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP19. Shape:** one or more repeated linear factors. **Tools:** include a term for every power from 1 through the multiplicity. **Result:** $-\frac{6}{x + 2} - \frac{6}{(x + 2)^2}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP20. Shape:** distinct irreducible quadratic factors. **Tools:** place a general linear numerator over each quadratic factor. **Result:** $\frac{-4x - 4}{x^2 + 2} + \frac{-3x - 2}{x^2 + 2x + 4}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP21. Shape:** repeated irreducible quadratic factor. **Tools:** include a linear numerator over every power of the quadratic. **Result:** $\frac{5 - 3x}{(x^2 - 4x + 5)^2} + \frac{-2x - 4}{x^2 - 4x + 5}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP22. Shape:** mixed linear, repeated, and quadratic factors. **Tools:** classify each factor and write every required numerator and power. **Result:** $\frac{4 - 4x}{x^2 + 2x + 8} - \frac{3}{x + 6} - \frac{5}{x + 1} - \frac{5}{(x + 1)^2}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP23. Shape:** improper rational expression. **Tools:** perform polynomial division first, then decompose the proper remainder. **Result:** $2x + \frac{3}{x - 4} - \frac{1}{x - 6}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP24. Shape:** common numerical/monomial factor. **Tools:** extract the greatest common factor. **Result:** $12x^3(x^2 - 8)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP25. Shape:** difference of two perfect squares. **Tools:** apply $A^2 - B^2 = (A - B)(A + B)$ repeatedly. **Result:** $(2x^3 - 5)(2x^3 + 5)$ **Check:** multiply the proposed factors to recover the original polynomial.

- CAP26. Shape:** three-term perfect-square pattern. **Tools:** match $A^2 \pm 2AB + B^2$. **Result:** $(x - 5)^2$
Check: multiply the proposed factors to recover the original polynomial.
- CAP27. Shape:** sum or difference of cubes. **Tools:** apply the cube identity, after any GCF. **Result:** $(x - 6)(x^2 + 6x + 36)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP28. Shape:** monic quadratic trinomial. **Tools:** find two integers with sum $-b$ and product c . **Result:** $(x + 3)(x + 7)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP29. Shape:** nonmonic quadratic trinomial. **Tools:** split the middle term or use an ac method, then group. **Result:** $(3x - 1)(3x + 7)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP30. Shape:** four or more terms with a repeated binomial after grouping. **Tools:** group terms and extract the common binomial. **Result:** $-(4x + 1)(2x^2 - 7)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP31. Shape:** quadratic pattern in x^2 . **Tools:** substitute $u = x^2$, factor in u , then replace u . **Result:** $(x^2 - 7)(x^2 + 4)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP32. Shape:** expression containing more than one factoring pattern. **Tools:** GCF first, then repeatedly factor until no listed tool applies. **Result:** $5x(x - 1)^2(x + 1)(x^2 + x + 1)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP33. Shape:** quotient with common polynomial factors. **Tools:** factor both parts, cancel factors only, preserve original restrictions. **Result:** $\frac{x - 3}{x + 2}$, with $x \neq -7, -2$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP34. Shape:** product of rational expressions. **Tools:** factor, cancel across the product, then multiply. **Result:** $\frac{x^2 - 2x - 48}{x^2 - 8x + 7}$, with $x \neq -7, 1, 2, 7$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP35. Shape:** quotient of rational expressions. **Tools:** record restrictions, multiply by the reciprocal, factor, and cancel. **Result:** $\frac{x^2 + 9x + 8}{x^2 - 5x + 4}$, with $x \neq -8, 1, 3, 4, 8$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP36. Shape:** sum or difference with unlike denominators. **Tools:** factor denominators, build the LCD, combine, and simplify. **Result:** $\frac{2x - 24}{x^2 - 14x + 48}$, with $x \neq 6, 8$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP37. Shape:** fraction whose numerator or denominator contains fractions. **Tools:** identify the internal LCD and clear all small denominators. **Result:** $\frac{x + 3}{x - 1}$, with $x \neq -1, 1$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP38. Shape:** polynomial dividend and nonzero polynomial divisor. **Tools:** polynomial long division with zero placeholders as needed. **Result:** $Q(x) = -3x^2 + 4x + 3$, $R(x) = 3$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP39. Shape:** linear divisor of the form $x - c$. **Tools:** synthetic division using c and every coefficient, including zeros. **Result:** $Q(x) = x^4 + 2x^3 - 4x^2 + 3x + 1$, $R = -8$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP40. Shape:** inspect the divisor. **Tools:** use synthetic division only for $x - c$; otherwise use long division. **Result:** synthetic division (or long division); $Q(x) = 2x + 2$, $R(x) = -1$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.

- CAP41. Shape:** proper rational expression with distinct linear denominator factors. **Tools:** write the distinct-linear template and determine its constants. **Result:** $\frac{7}{x+4} + \frac{4}{x+2} - \frac{7}{x-6}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP42. Shape:** one or more repeated linear factors. **Tools:** include a term for every power from 1 through the multiplicity. **Result:** $-\frac{5}{x-4} - \frac{6}{(x-4)^2} - \frac{5}{(x-4)^3}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP43. Shape:** distinct irreducible quadratic factors. **Tools:** place a general linear numerator over each quadratic factor. **Result:** $\frac{-5x-1}{x^2+4x+5} + \frac{-4x-3}{x^2-4x+8}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP44. Shape:** repeated irreducible quadratic factor. **Tools:** include a linear numerator over every power of the quadratic. **Result:** $\frac{4x-3}{(x^2+8)^2} + \frac{4x-2}{x^2+8}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP45. Shape:** mixed linear, repeated, and quadratic factors. **Tools:** classify each factor and write every required numerator and power. **Result:** $\frac{4x-1}{x^2+8} - \frac{4}{x+6} - \frac{1}{(x+6)^2} + \frac{2}{x-5}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP46. Shape:** improper rational expression. **Tools:** perform polynomial division first, then decompose the proper remainder. **Result:** $\left(-2x^2 + 2x - 2\right) - \frac{4}{x+2} - \frac{1}{x-6}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP47. Shape:** common numerical/monomial factor. **Tools:** extract the greatest common factor. **Result:** $-5x^4(7x^4 + 8x^2 - 7)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP48. Shape:** difference of two perfect squares. **Tools:** apply $A^2 - B^2 = (A - B)(A + B)$ repeatedly. **Result:** $(x+2)(x+4)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP49. Shape:** three-term perfect-square pattern. **Tools:** match $A^2 \pm 2AB + B^2$. **Result:** $(4x^2 + 3)^2$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP50. Shape:** sum or difference of cubes. **Tools:** apply the cube identity, after any GCF. **Result:** $(4x-3)(16x^2 + 12x + 9)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP51. Shape:** monic quadratic trinomial. **Tools:** find two integers with sum $-b$ and product c . **Result:** $(x-6)(x-4)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP52. Shape:** nonmonic quadratic trinomial. **Tools:** split the middle term or use an ac method, then group. **Result:** $(2x+3)(3x+1)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP53. Shape:** four or more terms with a repeated binomial after grouping. **Tools:** group terms and extract the common binomial. **Result:** $2(2x+1)(2x^2 - 4x - 1)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP54. Shape:** quadratic pattern in x^3 . **Tools:** substitute $u = x^3$, factor in u , then replace u . **Result:** $(x+1)(x^3+5)(x^2-x+1)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP55. Shape:** expression containing more than one factoring pattern. **Tools:** GCF first, then repeatedly factor until no listed tool applies. **Result:** $-6x^3(x-2)^2(x+2)(x^2+2x+4)$ **Check:** multiply the proposed factors to recover the original polynomial.

- CAP56. Shape:** quotient with common polynomial factors. **Tools:** factor both parts, cancel factors only, preserve original restrictions. **Result:** $\frac{x-3}{x-5}$, with $x \neq -1, 5, 6$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP57. Shape:** product of rational expressions. **Tools:** factor, cancel across the product, then multiply. **Result:** $\frac{x^2+7x-8}{x^2+5x+6}$, with $x \neq -5, -3, -2, 4$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP58. Shape:** quotient of rational expressions. **Tools:** record restrictions, multiply by the reciprocal, factor, and cancel. **Result:** $\frac{x^2+10x+21}{x^2-2x-24}$, with $x \neq -7, -5, -4, 2, 6$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP59. Shape:** sum or difference with unlike denominators. **Tools:** factor denominators, build the LCD, combine, and simplify. **Result:** $\frac{8x-14}{x^3-12x-16}$, with $x \neq -2, 4$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP60. Shape:** fraction whose numerator or denominator contains fractions. **Tools:** identify the internal LCD and clear all small denominators. **Result:** $\frac{-2x+24}{x+6}$, with $x \neq -6, 3$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP61. Shape:** polynomial dividend and nonzero polynomial divisor. **Tools:** polynomial long division with zero placeholders as needed. **Result:** $Q(x) = x^2 - 3x + 1$, $R(x) = 4$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP62. Shape:** linear divisor of the form $x - c$. **Tools:** synthetic division using c and every coefficient, including zeros. **Result:** $Q(x) = -2x^4 + 2x^2 + 5$, $R = 0$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP63. Shape:** inspect the divisor. **Tools:** use synthetic division only for $x - c$; otherwise use long division. **Result:** synthetic division (or long division); $Q(x) = x^5 + 4x^4 - 2x^2 + 4x + 4$, $R(x) = -3$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP64. Shape:** proper rational expression with distinct linear denominator factors. **Tools:** write the distinct-linear template and determine its constants. **Result:** $\frac{7}{x+6} + \frac{6}{x-2} + \frac{2}{x-8}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP65. Shape:** one or more repeated linear factors. **Tools:** include a term for every power from 1 through the multiplicity. **Result:** $\frac{3}{x+7} - \frac{1}{x-5} + \frac{2}{(x-5)^2} + \frac{6}{(x-5)^3}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP66. Shape:** distinct irreducible quadratic factors. **Tools:** place a general linear numerator over each quadratic factor. **Result:** $\frac{5x+1}{x^2+3x+3} + \frac{5x+2}{x^2+4x+7}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP67. Shape:** repeated irreducible quadratic factor. **Tools:** include a linear numerator over every power of the quadratic. **Result:** $\frac{4-x}{(x^2-4x+7)^2} + \frac{2x-2}{(x^2-4x+7)^3} + \frac{3x+1}{x^2-4x+7}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP68. Shape:** mixed linear, repeated, and quadratic factors. **Tools:** classify each factor and write every required numerator and power. **Result:** $\frac{5-4x}{x^2-5x+8} + \frac{-2x-5}{(x^2-5x+8)^2} + \frac{2}{x+2} + \frac{2}{x-4} - \frac{2}{(x-4)^2}$. **Check:** recombine the decomposition over the original denominator and compare numerators.

- CAP69. Shape:** improper rational expression. **Tools:** perform polynomial division first, then decompose the proper remainder. **Result:** $(-2x^2 - 2x + 1) + \frac{2}{x+3} + \frac{2}{x+1} - \frac{4}{x}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP70. Shape:** common numerical/monomial factor. **Tools:** extract the greatest common factor. **Result:** $-12x^8(x-2)(2x+3)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP71. Shape:** difference of two perfect squares. **Tools:** apply $A^2 - B^2 = (A-B)(A+B)$ repeatedly. **Result:** $6x^3(x-2)(x+2)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP72. Shape:** three-term perfect-square pattern. **Tools:** match $A^2 \pm 2AB + B^2$. **Result:** $196x(x-1)^2$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP73. Shape:** sum or difference of cubes. **Tools:** apply the cube identity, after any GCF. **Result:** $2x^2(x+2)(x^2-2x+4)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP74. Shape:** monic quadratic trinomial. **Tools:** find two integers with sum $-b$ and product c . **Result:** $(x+3)(x+7)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP75. Shape:** nonmonic quadratic trinomial. **Tools:** split the middle term or use an ac method, then group. **Result:** $(3x-5)(3x+4)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP76. Shape:** four or more terms with a repeated binomial after grouping. **Tools:** group terms and extract the common binomial. **Result:** $(x-1)(6x^2+6x-1)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP77. Shape:** quadratic pattern in x^2 . **Tools:** substitute $u = x^2$, factor in u , then replace u . **Result:** $3x^2(x-1)(x+1)(x^2+4)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP78. Shape:** expression containing more than one factoring pattern. **Tools:** GCF first, then repeatedly factor until no listed tool applies. **Result:** $-3x(x-1)^2(x+1)(x^2+x+1)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP79. Shape:** quotient with common polynomial factors. **Tools:** factor both parts, cancel factors only, preserve original restrictions. **Result:** $\frac{x^3+3x^2+x+3}{x+5}$, with $x \neq -5, -1, 2$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP80. Shape:** product of rational expressions. **Tools:** factor, cancel across the product, then multiply. **Result:** $\frac{x^2+8x+12}{x^2+2x-48}$, with $x \neq -8, -5, 5, 6$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP81. Shape:** quotient of rational expressions. **Tools:** record restrictions, multiply by the reciprocal, factor, and cancel. **Result:** $\frac{x^2-7x+6}{x^2-3x-10}$, with $x \neq -4, -2, 5, 6, 7$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP82. Shape:** sum or difference with unlike denominators. **Tools:** factor denominators, build the LCD, combine, and simplify. **Result:** $\frac{-4x+13}{x^4-11x^3+29x^2-11x+28}$, with $x \neq 4, 7$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.
- CAP83. Shape:** fraction whose numerator or denominator contains fractions. **Tools:** identify the internal LCD and clear all small denominators. **Result:** $-\frac{x}{4} + \frac{1}{2}$, with $x \neq -2, 6$. **Check:** recombine or substitute an allowed numerical value, while checking all original restrictions.

- CAP84. Shape:** polynomial dividend and nonzero polynomial divisor. **Tools:** polynomial long division with zero placeholders as needed. **Result:** $Q(x) = -3x$, $R(x) = 3$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP85. Shape:** linear divisor of the form $x - c$. **Tools:** synthetic division using c and every coefficient, including zeros. **Result:** $Q(x) = 5x^4 - x^3 + x^2 + 5x - 1$, $R = 1$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP86. Shape:** inspect the divisor. **Tools:** use synthetic division only for $x - c$; otherwise use long division. **Result:** polynomial long division; $Q(x) = x^5 + 3x^4 - 4x^3 + 3x - 2$, $R(x) = -4$. **Check:** check $P(x) = D(x)Q(x) + R(x)$ and $\deg R < \deg D$.
- CAP87. Shape:** proper rational expression with distinct linear denominator factors. **Tools:** write the distinct-linear template and determine its constants. **Result:** $\frac{2}{x+5} - \frac{2}{x-1} - \frac{4}{x-6} + \frac{3}{x-8}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP88. Shape:** one or more repeated linear factors. **Tools:** include a term for every power from 1 through the multiplicity. **Result:** $-\frac{3}{x-4} + \frac{4}{(x-4)^2} + \frac{1}{(x-4)^3} + \frac{3}{(x-4)^4} + \frac{5}{x-5}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP89. Shape:** distinct irreducible quadratic factors. **Tools:** place a general linear numerator over each quadratic factor. **Result:** $\frac{1-2x}{x^2-5x+8} + \frac{4x-3}{x^2-3x+7}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP90. Shape:** repeated irreducible quadratic factor. **Tools:** include a linear numerator over every power of the quadratic. **Result:** $\frac{2x-4}{x^2+5x+7} + \frac{3x-1}{(x^2+5x+7)^3} + \frac{4x-1}{(x^2+5x+7)^2} - \frac{4}{x-4}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP91. Shape:** mixed linear, repeated, and quadratic factors. **Tools:** classify each factor and write every required numerator and power. **Result:** $\frac{3-3x}{x^2-4x+7} + \frac{2x-4}{(x^2-4x+7)^2} + \frac{2}{x+5} - \frac{3}{x+3} + \frac{3}{(x+3)^2}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP92. Shape:** improper rational expression. **Tools:** perform polynomial division first, then decompose the proper remainder. **Result:** $(2x^3 - x^2 - x + 3) + \frac{2}{x+1} + \frac{1}{x-4} - \frac{1}{x-5}$. **Check:** recombine the decomposition over the original denominator and compare numerators.
- CAP93. Shape:** common numerical/monomial factor. **Tools:** extract the greatest common factor. **Result:** $8x(x-9)(7x-9)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP94. Shape:** difference of two perfect squares. **Tools:** apply $A^2 - B^2 = (A-B)(A+B)$ repeatedly. **Result:** $(x-7)(x+7)(x^2+49)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP95. Shape:** three-term perfect-square pattern. **Tools:** match $A^2 \pm 2AB + B^2$. **Result:** $3(9x^3+1)^2$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP96. Shape:** sum or difference of cubes. **Tools:** apply the cube identity, after any GCF. **Result:** $2x(x-1)(x+1)(x^2-x+1)(x^2+x+1)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP97. Shape:** monic quadratic trinomial. **Tools:** find two integers with sum $-b$ and product c . **Result:** $(x-5)(x+2)$ **Check:** multiply the proposed factors to recover the original polynomial.
- CAP98. Shape:** nonmonic quadratic trinomial. **Tools:** split the middle term or use an ac method, then

group. **Result:** $(5x - 7)(7x + 8)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP99. Shape: four or more terms with a repeated binomial after grouping. **Tools:** group terms and extract the common binomial. **Result:** $(3x - 2)(4x^2 + 3x + 1)$ **Check:** multiply the proposed factors to recover the original polynomial.

CAP100. Shape: quadratic pattern in x^3 . **Tools:** substitute $u = x^3$, factor in u , then replace u . **Result:** $2x^2(x^3 - 5)(x^3 - 4)$ **Check:** multiply the proposed factors to recover the original polynomial.

Before Declaring the Algebra Complete

- I identified terms and factors correctly.
- I recorded restrictions from the original expression.
- I factored before cancelling.
- I cancelled only entire common factors.
- I used the least common denominator at its greatest required powers.
- I inserted zero coefficients in polynomial division when powers were missing.
- My remainder has degree smaller than the divisor.
- I made an improper rational expression proper before partial fractions.
- I factored the denominator completely and wrote every required partial-fraction term.
- I chose coefficient methods appropriate to the denominator shape.
- I verified the final result.